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YEAR 12 MATHEMATICS ADVANCED

Lyn Baker

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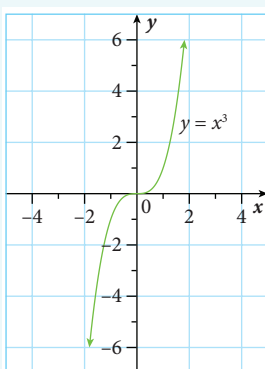
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1 Translations

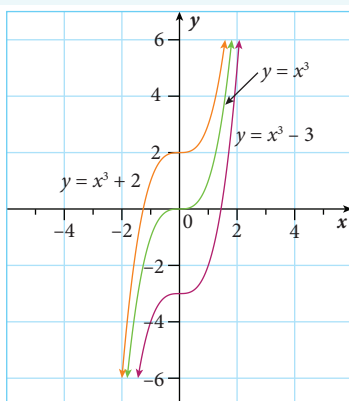
- It is important to be familiar with the shape of the basic curves. You need to be able to recognise straight lines, parabolas, absolute values and cubic, hyperbolic, exponential and logarithmic curves from their equations.
- Adding a constant to the equation of a function, $y = f(x) + c$, has the effect of moving the graph up, or down if the constant is negative.
- Adding a constant into the equation of a function, $y = f(x + c)$, has the effect of moving the graph left or right.

EXAMPLE 1

The diagram shows the graph of $y = x^3$.



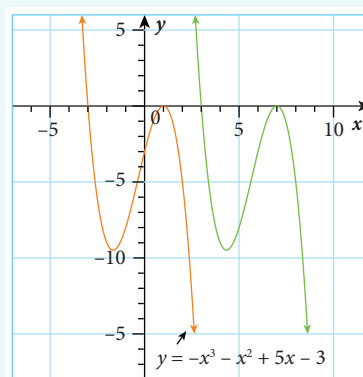
Copy the diagram and sketch $y = x^3 + 2$ and $y = x^3 - 3$ on the same diagram.



A translation is also known more informally as a slide.

EXAMPLE 2

Meg produced this diagram showing the curve $y = -x^3 - x^2 + 5x - 3$ and another curve that she forgot to label.



Find the equation of the second curve. Give the answer in the form $y = ax^3 + bx^2 + cx + d$.

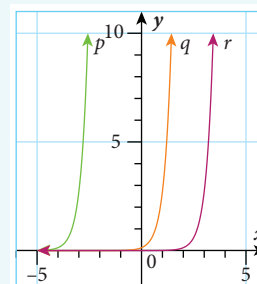
The second curve is a translation 6 units to the right. The equation is:

$$\begin{aligned} y &= -(x-6)^3 - (x-6)^2 + 5(x-6) - 3 \\ &= -(x^3 - 18x^2 + 108x - 216) - (x^2 - 12x + 36) + 5x - 30 - 3 \\ &= -x^3 + 18x^2 - 108x + 216 - x^2 + 12x - 36 + 5x - 33 \\ &= -x^3 + 17x^2 - 91x + 147 \end{aligned}$$

Because the translation is to the right the equation is of the form $y = f(x-6)$. If it was to the left it would be $y = f(x+6)$.

EXAMPLE 3

The diagram shows the graph of $y = e^{3x-2}$ and two translations of that graph. Determine the equations of p , q and r .



~

$$y = e^{3x-2}$$

When $x = 1$,

$$y = e^{3 \times 1 - 2}$$

$$= e \quad (\approx 2.718)$$

So q is the graph of $y = e^{3x-2}$.

Graph p has been translated 4 units to the left of q .

So p has equation:

$$y = e^{3(x+4)-2}$$

$$= e^{3x+10}$$

r has been translated 2 units to the right of q .

So r has equation:

$$y = e^{3(x-2)-2}$$

$$= e^{8x-8}$$

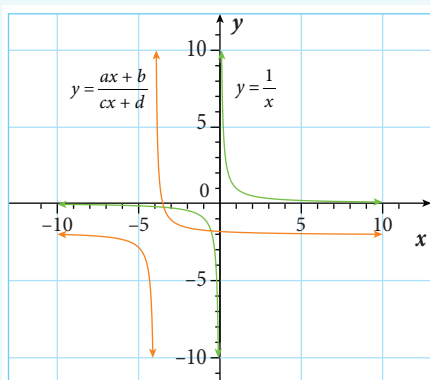
So p is $y = e^{3x+10}$, q is $y = e^{3x-2}$ and r is $y = e^{3x-8}$.

We know that one of the equations is $y = e^{3x-2}$ so we need to identify that first.

EXAMPLE 4

The diagram shows the graphs of $y = \frac{1}{x}$ and a

translation of $y = \frac{1}{x}$.



Determine the values of a , b , c and d .

The second curve has moved 4 units to the left and 2 units down.

Its equation is $y = \frac{1}{x+4} - 2$

$$y = \frac{1}{x+4} - \frac{2(x+4)}{x+4}$$

$$y = \frac{1-2x-8}{x+4}$$

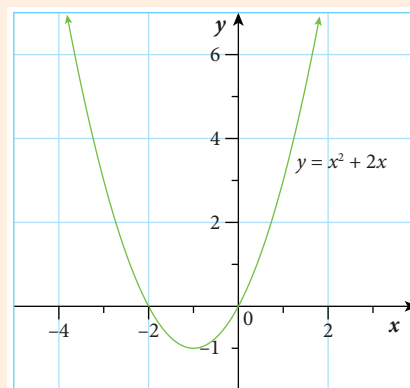
$$y = \frac{-2x-7}{x+4}$$

We find a common denominator and subtract fractions to change the equation into the desired form.

$\therefore a = -2$, $b = -7$, $c = 1$ and $d = 4$.

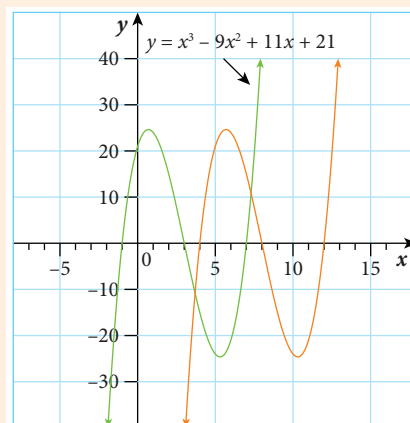
→ PRACTICE

- 1 The diagram shows the graph of $y = x^2 + 2x$.



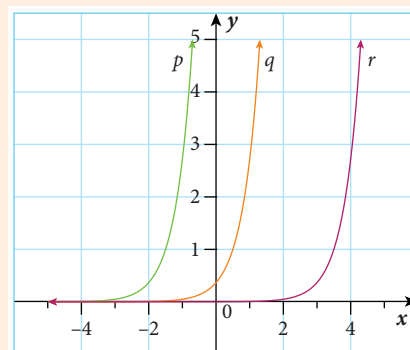
Copy the diagram and sketch $y = x^2 + 2x + 4$ and $y = x^2 + 2x - 2$ on the same diagram.

- 2 Ned produced this diagram showing the curve $y = x^3 - 9x^2 + 11x + 21$ and another curve that he forgot to label.

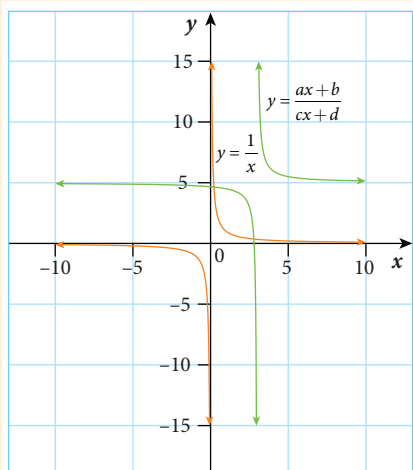


Find the equation of the second curve. Give the answer in the form $y = ax^3 + bx^2 + cx + d$.

- 3 The diagram shows the graph of $y = e^{2x-1}$ and two translations of that graph. Determine the equations of p , q and r .

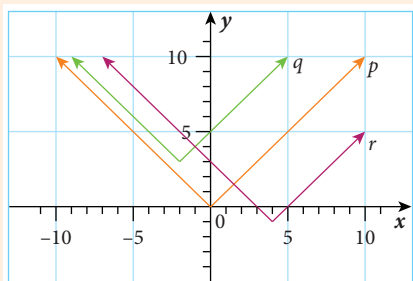


- 4 The diagram shows the graphs of $y = \frac{1}{x}$ and a translation of $y = \frac{1}{x}$.



Determine the values of a , b , c and d .

- 5* Determine the equations of the graphs labelled p , q and r .



Answers ➞ p. 24

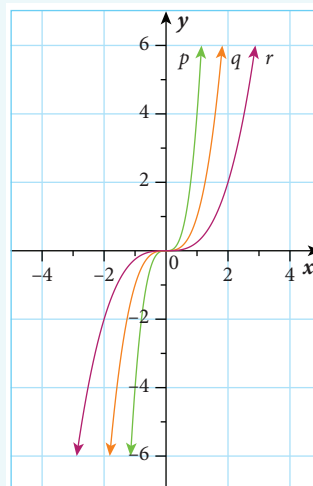
2 Dilations

- ➔ **Dilations are transformations** where size is not necessarily preserved.
- ➔ **Graphs of the form $y = kf(x)$ and $y = f(ax)$** , where k and a are constants, are dilations of the graph of $y = f(x)$. They are stretched (or reduced) or made wider or narrower but still have the basic features of $y = f(x)$.
- ➔ **The graph of $y = -kf(x)$** is the reflection of $y = kf(x)$ in the x -axis.

EXAMPLE 1

The graphs of $y = x^3$, $y = 4x^3$ and $y = \frac{x^3}{4}$ were produced with graphing software on the same diagram.

Identify each graph and comment on similarities and differences between the curves.



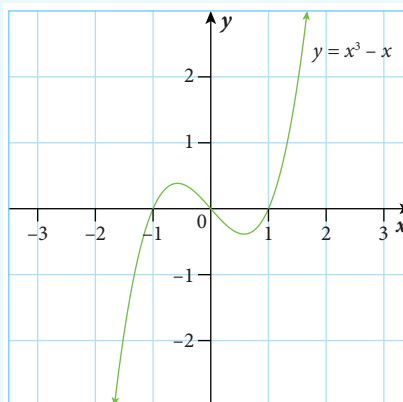
Graph q passes through $(1, 1)$ so it is the graph of $y = x^3$.
Graph p passes through $(1, 4)$ so it is the graph of $y = 4x^3$.
Graph r passes through $(2, 2)$ so it is the graph of $y = \frac{x^3}{4}$.

The graphs all have a similar cubic shape and all pass through the origin. They all have different widths, with $y = 4x^3$ being the narrowest and $y = \frac{x^3}{4}$ the widest.

The constant k in the equation $y = kx^3$ is the value of y when $x = 1$.

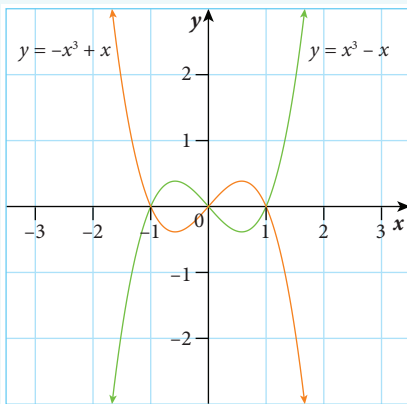
EXAMPLE 2

The graph of $y = x^3 - x$ is shown below.



Copy the graph and show the graph of $y = -x^3 + x$ on the same diagram.

$$y = -x^3 + x \\ = -(x^3 - x)$$

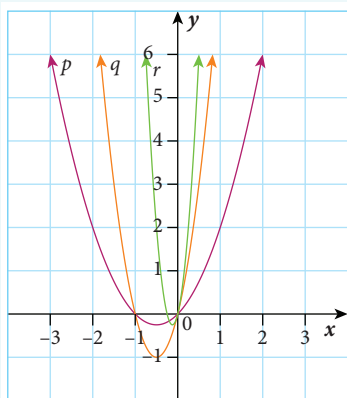


$-x^3 + x$ is the negative of $(x^3 - x)$ so the required curve is the same curve reflected in the x -axis.

EXAMPLE 3

Consider $f(x) = x^2 + x$.

- Find $f(4x)$.
- The graphs of $y = f(x)$, $y = 4f(x)$ and $y = f(4x)$ are shown below. Identify each curve and comment on the effect of multiplying the function by 4 or multiplying x in $f(x)$ by 4.



- $f(x) = x^2 + x$
 $f(4x) = (4x)^2 + 4x$
 $= 16x^2 + 4x$
- The curves are $y = x^2 + x$, $y = 4x^2 + 4x$ and $y = 16x^2 + 4x$.

Now both $y = x^2 + x$ and $y = 4x^2 + 4x$ have x -intercepts at -1 and 0 .

x -intercepts of $y = 16x^2 + 4x$ occur when

$$16x^2 + 4x = 0$$

$$4x(4x + 1) = 0$$

$$x = 0 \text{ or } x = -\frac{1}{4}$$

So r is $y = 16x^2 + 4x$

$$\text{When } x = -\frac{1}{2},$$

$$x^2 + x = \left(-\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)$$

$$= -\frac{1}{4}$$

$$4x^2 + 4x = 4\left(-\frac{1}{4}\right)$$

$$= -1$$

Think about the y -value at each point. For $y = 4f(x)$ each y -value is 4 times the corresponding y -value of $y = f(x)$.

So p is $y = x^2 + x$ and q is $y = 4x^2 + 4x$.

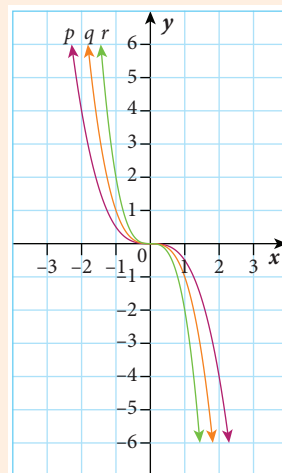
$\therefore p$ is $y = f(x)$, q is $y = 4f(x)$ and r is $y = f(4x)$.

Multiplying by 4 has the effect of narrowing the curve. $y = f(4x)$ is a narrower version of $y = f(x)$ without moving the curve up or down. $y = 4f(x)$ is also narrower but the vertex has also moved down.

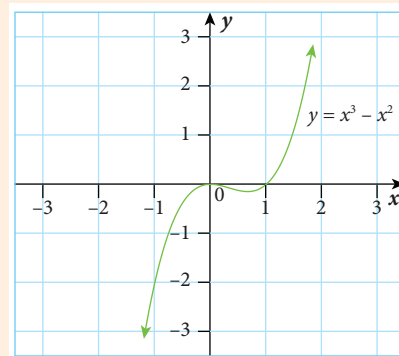
→ PRACTICE

- The graphs of $y = -x^3$, $y = -2x^3$ and $y = \frac{-x^3}{2}$ were produced with graphing software on the same diagram.

Identify each graph and comment on similarities and differences between the curves.



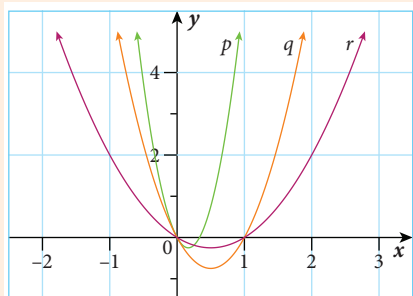
- The graph of $y = x^3 - x^2$ is shown below.



Copy the graph and show the graph of $y = -x^3 + x^2$ on the same diagram.

3 Consider $f(x) = x^2 - x$.

- a Find $f(3x)$.
- b The graphs of $y = f(x)$, $y = 3f(x)$ and $y = f(3x)$ are shown below. Identify each curve and comment on the effect of multiplying the function by 3 or multiplying x in $f(x)$ by 3.



4* Consider $f(x) = |3x - 8|$.

- a Sketch $y = f(x)$, $y = 4f(x)$ and $y = f(4x)$ on the same diagram.
- b Briefly comment on the similarities between the curves.

Answers ⇨ pp. 24–25

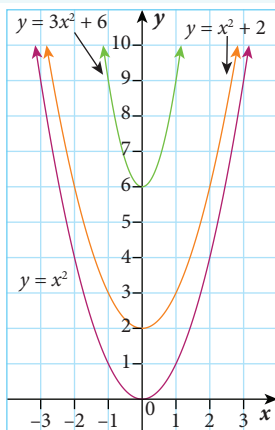
3 Mixed transformations

- ➔ Many transformations of graphs are a mixture of translations and dilations.
- ➔ The order in which transformations are applied is important.

EXAMPLE 1

Consider $f(x) = x^2$.

- a Sketch $y = f(x)$, $y = f(x) + 2$ and $y = 3(f(x) + 2)$ on the same diagram.
- b Sketch $y = f(x)$, $y = 3f(x)$ and $y = 3f(x) + 2$ on the same diagram.
- c Briefly comment on any differences the changed order of applying both transformations adding 2 and multiplying by 3 has on $y = f(x)$.

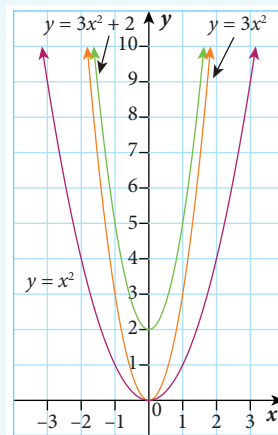


$$f(x) = x^2$$

$$3f(x) = 3x^2$$

$$3f(x) + 2 = 3x^2 + 2$$

- c In this case, the different order produces different graphs, one of which is a translation of the other.



By considering the equations we could determine the differences between the graphs without needing to sketch.

EXAMPLE 2

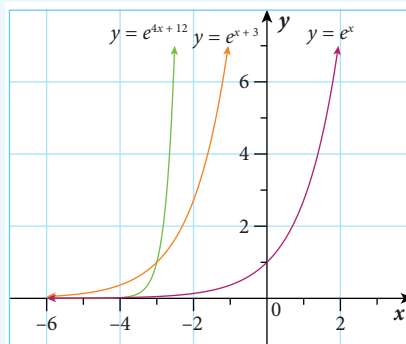
Consider $f(x) = e^x$.

- a Sketch $y = f(x)$, $y = f(x + 3)$ and $y = f(4(x + 3))$ on the same diagram.
- b Sketch $y = f(x)$, $y = f(4x)$ and $y = f(4x + 3)$ on the same diagram.
- c Briefly comment on any similarities or differences the changed order of applying both transformations has on $y = f(x)$.

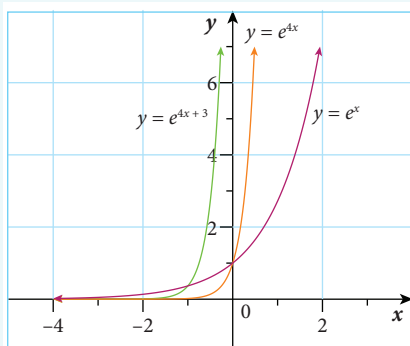
$$f(x) = e^x$$

$$f(x + 3) = e^{x+3}$$

$$f(4(x + 3)) = e^{4x+12}$$

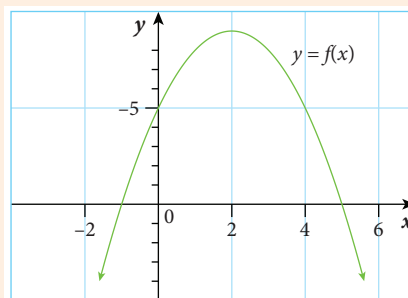


- b $f(x) = e^x$
- $f(4x) = e^{4x}$
- $f(4x + 3) = e^{4x+3}$



- c The curves $y = e^{4x+12}$ and $y = e^{4x+3}$ have the same shape. The position of the two graphs is the only difference; they are translations of each other. They are both translations of $y = e^{4x}$. The curve $y = e^{4x+3}$ has moved $\frac{3}{4}$ of a unit to the left and $y = e^{4x+12}$ has moved 3 units to the left.

$$e^{4x+3} = e^{4\left(x+\frac{3}{4}\right)}$$



- a Copy the diagram and show the graph of $y = \frac{1}{2}f(x) - 1$ on the same diagram.
- b Would $y = \frac{1}{2}f(x - 1)$ produce the same result? Justify your answer.

Answers ➞ p. 25

4 Sketching graphs

- ➔ A sketch of a curve does not need to be a strictly accurate graph.
- ➔ We make use of any easily available information to help determine (roughly) where the curve will go.
- Points can be plotted where appropriate.
 - The x and y -intercepts can often be found without too much difficulty.
 - The sign of the function could also be considered. It may be obvious that the curve is never negative, for example.
 - The value of the function $f(x)$ for very large (positive and negative) values of x can also be used.

EXAMPLE 1

Consider $y = 15x - 2x^2 - x^3$.

- a For what values of x does $y = 0$?
- b What happens to the y -values as x gets large positively?
- c Sketch the graph of $y = 15x - 2x^2 - x^3$.

a $y = 15x - 2x^2 - x^3$

When $y = 0$,

$$0 = 15x - 2x^2 - x^3$$

$$x^3 + 2x^2 - 15x = 0$$

$$x(x^2 + 2x - 15) = 0$$

$$x(x+5)(x-3) = 0$$

$$x = 0 \text{ or } x = -5 \text{ or } x = 3$$

So $y = 0$ when $x = -5, 0$ or 3 .

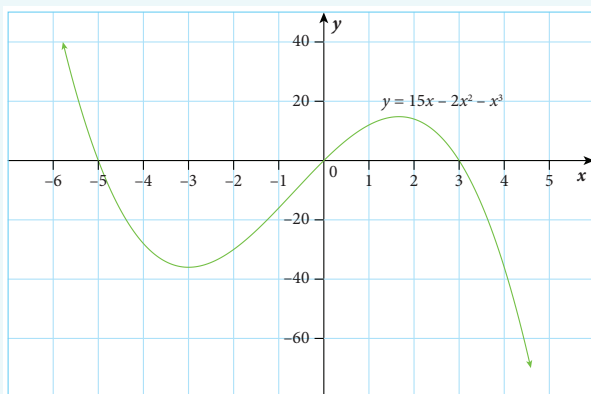
➔ PRACTICE

- 1 Consider $f(x) = |x|$.
- a Sketch $y = f(x)$, $y = f(x) + 3$ and $y = 2(f(x) + 3)$ on the same diagram.
- b Sketch $y = f(x)$, $y = 2f(x)$ and $y = 2f(x) + 3$ on the same diagram.
- c Briefly comment on any differences the changed order of applying both transformations adding 3 and multiplying by 2 has on $y = f(x)$.
- 2 Consider $f(x) = \ln x$.
- a Sketch $y = f(x)$, $y = f(x + 2)$ and $y = f(3(x + 2))$ on the same diagram.
- b Sketch $y = f(x)$, $y = f(3x)$ and $y = f(3x + 2)$ on the same diagram.
- c Briefly comment on any differences the changed order of applying both transformations has on $y = f(x)$.
- 3* The diagram shows the graph of $y = f(x)$ where $f(x) = 5 + 4x - x^2$.

b As $x \rightarrow \infty$, $y \rightarrow -\infty$

If x is a very large positive number then the highest power of x (x^3) will overpower the rest. As x^3 is negative, we say that as x tends to infinity y tends to negative infinity.

c



EXAMPLE 2

Sketch the curve $y = (x - 3)(x - 5)(x + 2)(x + 6)$.

$$y = (x - 3)(x - 5)(x + 2)(x + 6)$$

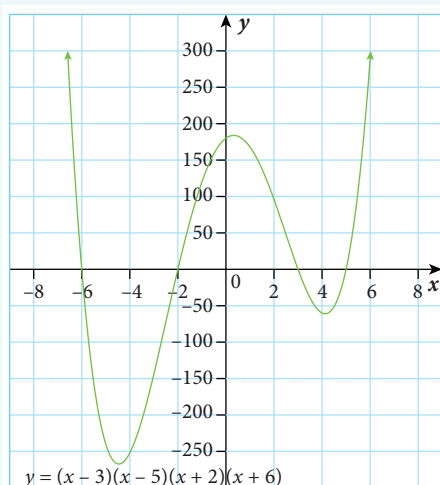
x -intercepts are 3, 5, -2 and -6.

$$y\text{-intercept} = -3 \times -5 \times 2 \times 6 = 180$$

As $x \rightarrow \infty$, $y \rightarrow \infty$

$$\text{When } x = 4, y = (4 - 3)(4 - 5)(4 + 2)(4 + 6) = -60$$

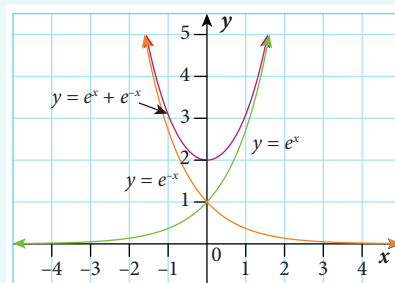
$$\text{When } x = -4, y = (-4 - 3)(-4 - 5)(-4 + 2)(-4 + 6) = -252$$



Finding the values midway between the intercepts gives us an idea of where the graph should go. It does not follow that the local maximum or minimum values will occur at those midway values. In the calculus section you will find how to determine the turning points of curves.

EXAMPLE 3

On the same diagram sketch the graphs of $y = e^x$, $y = e^{-x}$ and $y = e^x + e^{-x}$.



To sketch $y = e^x + e^{-x}$ we use the graphs of $y = e^x$ and $y = e^{-x}$ and add the ordinates.

→ PRACTICE

- Consider $y = x^3 + 2x^2 - 24x$.
 - For what values of x does $y = 0$?
 - What happens to the y -values as x gets large positively?
 - Sketch the graph of $y = x^3 + 2x^2 - 24x$.
- Sketch the curve $y = (x + 3)(x - 1)(x - 4)(x - 7)$.
- On the same diagram sketch $y = \log_2 x$, $y = \log_3 x$ and $y = \log_2 x + \log_3 x$.
- * Consider the curve $y = x^3 - 13x - 12$.
 - Where does the curve cut the y -axis?
 - Show that the curve cuts the x -axis at -3, -1 and 4.
 - Sketch the curve $y = x^3 - 13x - 12$.

Answers ➔ p. 26

5 Using symmetry

- ➔ If a function is **symmetrical**, then the working needed to sketch its graph can be simplified.
- ➔ An **even function** is **symmetrical** about the y -axis.
 $f(-x) = f(x)$
- ➔ An **odd function** has **point symmetry** about the origin.
 $f(-x) = -f(x)$

EXAMPLE 1

Consider $f(x) = 4x^2 + 28x + 52$.

- Rearrange the equation $y = 4x^2 + 28x + 52$ into the form $y = (ax + b)^2 + c$.

b What is the equation of the axis of symmetry of the parabola $y = f(x)$?

c Sketch the curve $y = f(x)$.

a $y = 4x^2 + 28x + 52$
 $= 4x^2 + 28x + 49 + 3$
 $= (2x + 7)^2 + 3$

Complete the square: $4x^2$ is $(2x)^2$, $28x$ is $2 \times 2x \times 7$ so we need to add 7^2 .

b Axis: $2x + 7 = 0$

$$2x = -7$$

$$x = -3.5$$

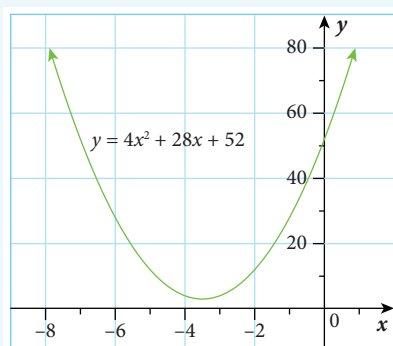
c When $x = -3.5$, $y = 3$

So the turning point of the parabola is at $(-3.5, 3)$ and, as the parabola is concave up, this is the minimum value.

The curve is always positive.

When $x = 0$, $y = 52$

So the y -intercept is 52.



We could plot a few points, $(-2, 12)$, $(-1, 28)$ and $(1, 84)$ perhaps, to see where the curve goes and then use symmetry to complete the sketch.

EXAMPLE 2

a Show that $f(x) = 4x^4 - 13x^2 + 9$ is an even function.

b Sketch $y = f(x)$.

a $f(x) = 4x^4 - 13x^2 + 9$
 $f(-x) = 4(-x)^4 - 13(-x)^2 + 9$
 $= 4x^4 - 13x^2 + 9$
 $= f(x)$

$\therefore f(x)$ is an even function.

b $f(0) = 9$

When $f(x) = 0$,

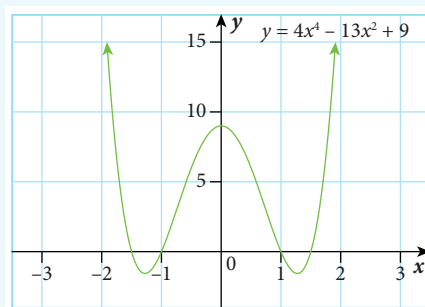
$$4x^4 - 13x^2 + 9 = 0$$

$$(4x^2 - 9)(x^2 - 1) = 0$$

$$x^2 = \frac{9}{4} \quad \text{or} \quad x^2 = 1$$

$$x = \pm \frac{3}{2} \quad \text{or} \quad x = \pm 1$$

As $x \rightarrow \infty$, $y \rightarrow \infty$



If the graph could be folded along the y -axis the two halves should match exactly.

EXAMPLE 3

a Show that $f(x) = 16x - x^3$ is an odd function.

b Sketch $y = f(x)$.

a $f(x) = 16x - x^3$
 $f(-x) = 16(-x) - (-x)^3$
 $= -16x + x^3$
 $= -(16x - x^3)$
 $= -f(x)$
 $\therefore f(x)$ is an odd function.

b If $f(x) = 0$

$$0 = 16x - x^3$$

$$= x(16 - x^2)$$

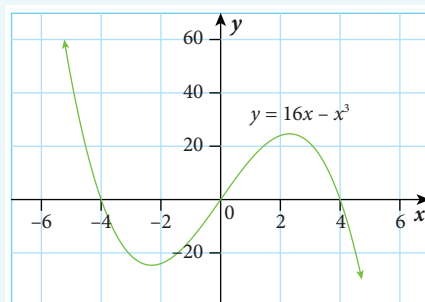
$$x = 0 \quad \text{or} \quad x^2 = 16$$

$$x = 0 \quad \text{or} \quad x = \pm 4$$

$$f(2) = 16(2) - 2^3$$

$$= 24$$

As $x \rightarrow \infty$, $y \rightarrow -\infty$



If a pin was placed at the origin and the graph spun through 180° it should be exactly the same.

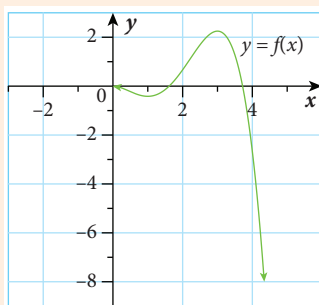
→ PRACTICE

- 1 Consider $f(x) = 9x^2 + 30x + 29$.
 - a Rearrange the equation $y = 9x^2 + 30x + 29$ into the form $y = (ax + b)^2 + c$.
 - b What is the equation of the axis of symmetry of the parabola $y = f(x)$?
 - c Sketch the curve $y = f(x)$.

- 2 a Show that $f(x) = -9x^4 + 37x^2 - 4$ is an even function.
 - b Sketch $y = f(x)$.

- 3 a Show that $f(x) = x^3 - 9x$ is an odd function.
 - b Sketch $y = f(x)$.

- 4* The graph of $y = f(x)$ is shown for $x \geq 0$.



On separate diagrams sketch the graph of $y = f(x)$ given that:

- a $f(x)$ is an even function,
- b $f(x)$ is an odd function.

Answers ➞ pp. 26–27

6 Discontinuities and asymptotes

- ➔ If a curve has a **gap or jump** it is said to be discontinuous at that point.
- ➔ As a curve **approaches a discontinuity** there might be an asymptote. An asymptote is a line that a curve tends to imitate.
- ➔ It is necessary to consider the domain over which a curve is sketched. The natural domain is the set of real numbers for which the function is defined.

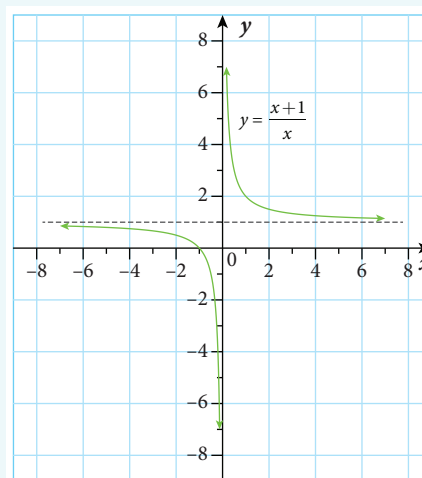
EXAMPLE 1

- a What are the equations of the asymptotes of the curve $y = \frac{x+1}{x}$?
- b Sketch the curve.

$$\begin{aligned} \text{a } y &= \frac{x+1}{x} \\ &= \frac{x}{x} + \frac{1}{x} \\ &= 1 + \frac{1}{x} \end{aligned}$$

So the asymptotes are $x = 0$ and $y = 1$.

b



By rearranging the equation we can see that the curve is a simple translation of the hyperbola $y = \frac{1}{x}$.

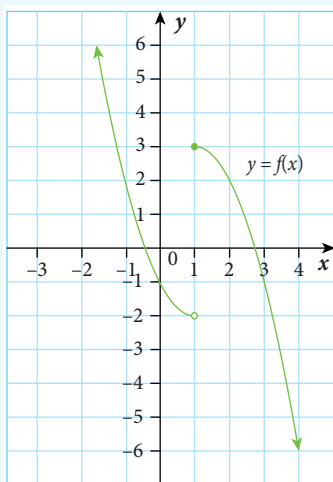
EXAMPLE 2

Consider the function $f(x) = \begin{cases} -x^2 + 2x + 2 & \text{for } x \geq 1 \\ x^2 - 2x - 1 & \text{for } x < 1 \end{cases}$.

- a Is the function continuous?
 - b Sketch $y = f(x)$.
- $$\text{a } f(x) = \begin{cases} -x^2 + 2x + 2 & \text{for } x \geq 1 \\ x^2 - 2x - 1 & \text{for } x < 1 \end{cases}$$
- $$\begin{aligned} f(1) &= -1^2 + 2 \times 1 + 2 \\ &= 3 \end{aligned}$$
- When $x = 1$,
- $$\begin{aligned} x^2 - 2x - 1 &= 1^2 - 2 \times 1 - 1 \\ &= -2 \\ &\neq f(1) \end{aligned}$$
- $\therefore$ the function is not continuous.
- b $-x^2 + 2x + 2 = -(x^2 - 2x) + 2$

$$\begin{aligned} &= -(x^2 - 2x + 1) + 3 \\ &= -(x - 1)^2 + 3 \end{aligned}$$

$$\begin{aligned} x^2 - 2x - 1 &= x^2 - 2x + 1 - 2 \\ &= (x - 1)^2 - 2 \end{aligned}$$



The solid circle tells us that the top part of the curve is the one when $x = 1$, while the open circle shows that $x = 1$ does not belong to the other part.

EXAMPLE 3

Consider the function $f(x) = \frac{1}{\sqrt{x^2 - 1}}$.

- a Is the function odd or even?
- b For what values of x is the function not defined?
- c Sketch $y = f(x)$.

$$\begin{aligned} \text{a } f(x) &= \frac{1}{\sqrt{x^2 - 1}} \\ f(-x) &= \frac{1}{\sqrt{(-x)^2 - 1}} \\ &= \frac{1}{\sqrt{x^2 - 1}} \\ &= f(x) \end{aligned}$$

So the function is even.

- b Natural domain:

$$x^2 - 1 > 0$$

$$x^2 > 1$$

$$x < -1 \text{ or } x > 1$$

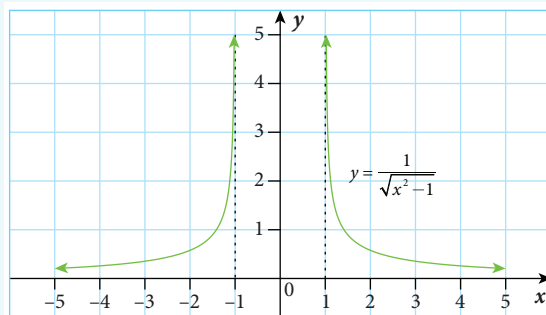
So the function is not defined when $-1 \leq x \leq 1$.

- c $y = \frac{1}{\sqrt{x^2 - 1}}$

Now $y > 0$ for all values of x .

$$\text{As } x \rightarrow \infty, y \rightarrow 0$$

$$\text{As } x \rightarrow 1^+, y \rightarrow \infty$$



The + notation in $x \rightarrow 1^+$, means as x approaches 1 from the positive side.

→ PRACTICE

- 1 a What are the equations of the asymptotes of the curve $y = \frac{2x-1}{x}$?
- b Sketch the curve.
- 2 Consider the function $f(x) = \begin{cases} x^2 - 4x + 2 & \text{for } x \geq 2 \\ -x^2 + 4x - 1 & \text{for } x < 2 \end{cases}$.
- a Is the function continuous?
- b Sketch $y = f(x)$.
- 3 Consider the function $f(x) = \frac{1}{\sqrt{1-x^2}}$.
- a Is the function odd or even?
- b For what values of x is the function not defined?
- c Sketch $y = f(x)$.
- 4* Sketch $y = \frac{5}{2x-1}$.

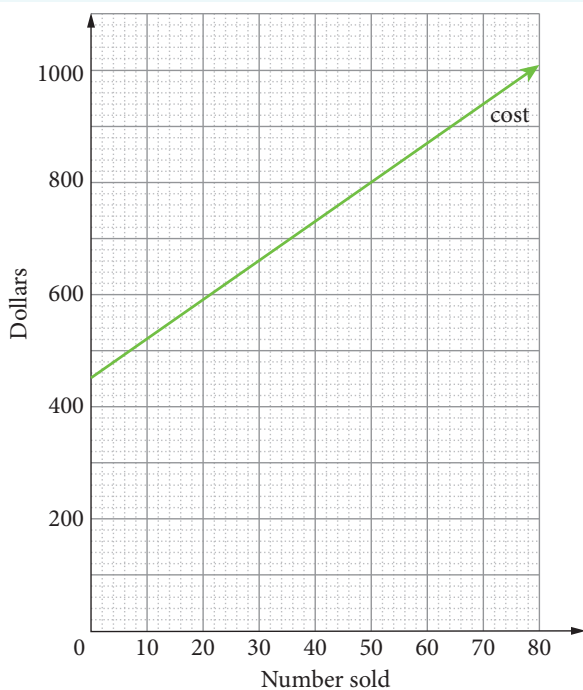
Answers ⇨ p. 27

7 Graphical solution of equations

- The simultaneous solution of two, or more, equations can easily be found if graphs are drawn. The points of intersection of two curves give the values that satisfy both equations at the same time.
- We can often determine the physical significance of a point of intersection; for example, in break-even analysis.
- Appropriate graphs can be drawn to determine the number of solutions of certain equations.

EXAMPLE 1

The graph shows the cost to make 'babybugs'.



- What is the gradient and what does it represent?
- What is the y -intercept and what does it represent?
- Each 'babybug' sells for \$22. Copy the graph and show the income on the same graph.
- What is the break-even point?
- How much is the profit when 50 'babybugs' are sold?
- If 20 'babybugs' are sold will there be a profit or a loss? What is the size of this profit or loss?

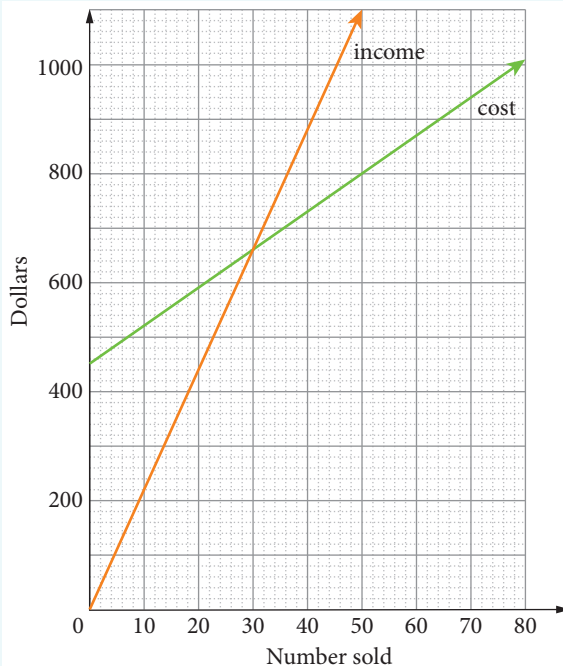
$$a \quad m = \frac{140}{20} = 7$$

The gradient is 7 and it represents the extra cost to make each 'babybug'.

- The y -intercept is 450. It represents the fixed costs.

Fixed costs are the costs for things like machinery which must be paid regardless of how many items are produced.

c



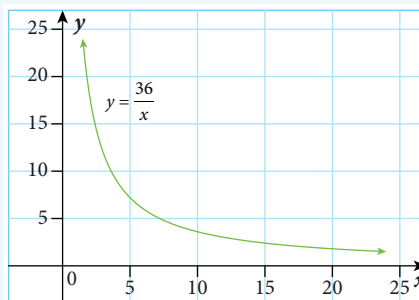
- The break-even point is when 30 'babybugs' are sold. Both the cost and income are \$660.
- The profit when 50 babybugs are sold is \$300.
- If 20 babybugs are sold there will be a loss of \$150.

There is no need to calculate either the cost or the income. The profit can be seen on the graph as the difference between them.

EXAMPLE 2

It is known that a certain rectangle has area of 36 cm^2 .

- If the length of the rectangle is $x \text{ cm}$ and the width $y \text{ cm}$, briefly explain why $y = \frac{36}{x}$.
- It is also known that the length of the rectangle is 5 cm longer than the width. Graphing software has been used to draw the graph of $y = \frac{36}{x}$ for positive values of x .



Find the dimensions of the rectangle.

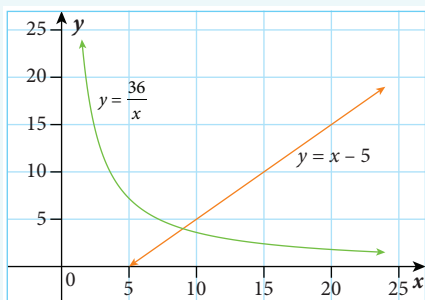
- a The area is equal to the length by the width.

$$\text{So } 36 = xy$$

$$y = \frac{36}{x}$$

- b Now the width is 5 cm shorter than the length.

$$\text{So } y = x - 5$$



From the graph the length is 9 cm and the width is 4 cm.

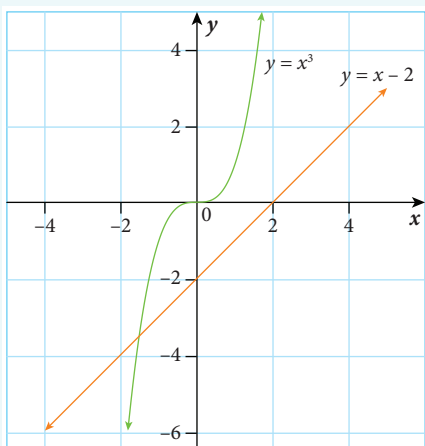
The point of intersection of the graphs gives us the required answer. It is not always necessary to draw the second graph; it might be possible to visualise where it must go and 'see' the answer.

EXAMPLE 3

Use a pair of graphs, drawn on the same diagram, to determine the number of solutions to the equation $x^3 - x + 2 = 0$.

$$x^3 - x + 2 = 0$$

$$x^3 = x - 2$$



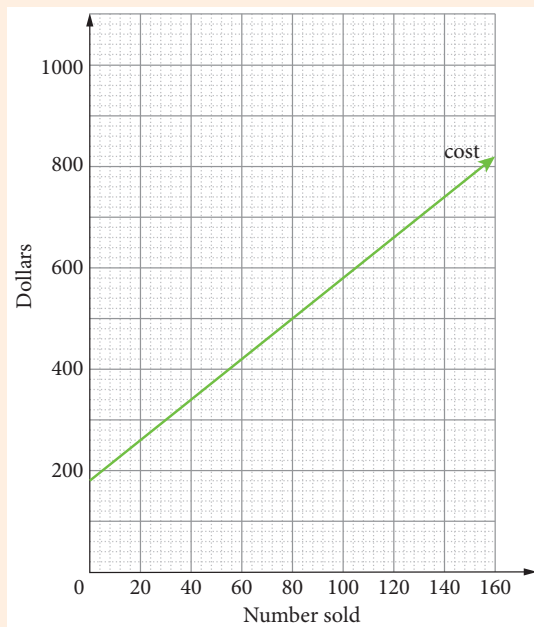
From the diagram $y = x^3$ and $y = x - 2$ only have one point of intersection.

So $x^3 - x + 2 = 0$ has one solution.

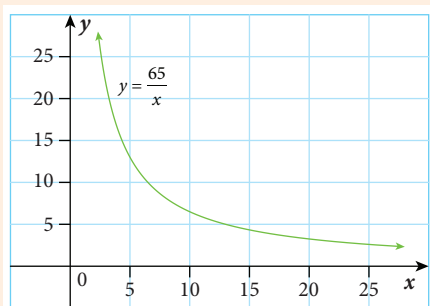
We can see from the graph that as x increases, in both the positive and negative directions, the curves are getting further apart, so there will be no other points of intersection.

→ PRACTICE

- 1 The graph shows the cost to make bags of 'sweetbeans'.



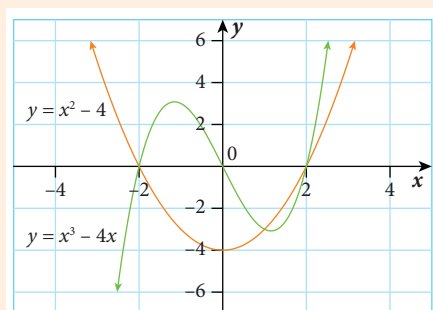
- What is the gradient and what does it represent?
 - What is the y -intercept and what does it represent?
 - Each bag of 'sweetbeans' sells for \$5.50. Copy the graph and show the income on the same graph.
 - What is the break-even point?
 - How much is the profit when 150 bags of 'sweetbeans' are sold?
 - If 100 bags of 'sweetbeans' are sold will there be a profit or a loss? What is the size of this profit or loss?
- 2 It is known that a particular rectangle has area of 65 cm^2 .
- If the length of the rectangle is $x \text{ cm}$ and the width $y \text{ cm}$, briefly explain why $y = \frac{65}{x}$.
 - It is also known that the length of the rectangle is 8 cm longer than the width. Graphing software has been used to draw the graph of $y = \frac{65}{x}$ for positive values of x .



Find the dimensions of the rectangle.

- 3 Use a pair of graphs, drawn on the same diagram, to determine the number of solutions to the equation $x^3 - 11x + 6 = 0$.

- 4* The diagram shows the graphs of $y = x^3 - 4x$ and $y = x^2 - 4$.



- a How many solutions are there to the equation $x^3 - 4x = x^2 - 4$?
- b How many times will the line $y = 4x - 4$ intersect the cubic curve $y = x^3 - x^2$? Justify your answer.

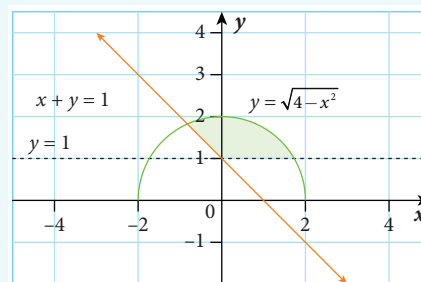
Answers ➞ pp. 27–28

8 Regions

- ➔ Any line, or curve, divides the number plane into different regions.
- ➔ A **broken line** means that the line forms the boundary of but is not part of the region.
- ➔ A **solid line** means that the boundary is part of the region.
- ➔ **Choosing a point either side of the boundary and testing** to see whether or not it satisfies the inequality allows us to determine the required region.

EXAMPLE 1

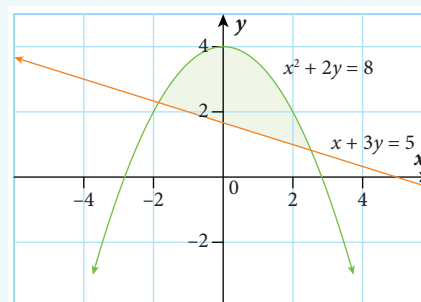
Draw a clear sketch of the region that satisfies all three inequalities: $y \leq \sqrt{4-x^2}$, $y > 1$, $x + y \geq 1$.



$y = 1$ forms the boundary of, but is not part of the region, so it is shown with a dotted line.

EXAMPLE 2

The graph shows the region bounded by the curve $x^2 + 2y = 8$ and the line $x + 3y = 5$.



Write down a pair of inequalities to describe the shaded region. Justify your answer.

Consider the point (0, 3) which lies in the shaded region.

When $x = 0$ and $y = 3$,

$$x^2 + 2y = 0^2 + 2 \times 3 = 6$$

Now $6 < 8$ so $x^2 + 2y \leq 8$

When $x = 0$ and $y = 3$,

$$x + 3y = 0 + 3 \times 3 = 9$$

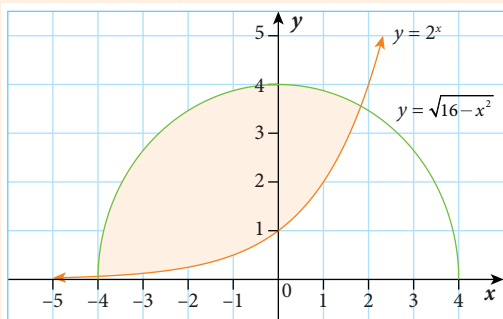
Now $9 > 5$ so $x + 3y \geq 5$.

The required region is $x^2 + 2y \leq 8$ and $x + 3y \geq 5$.

The boundary lines are solid so the symbols must include 'or equal to'.

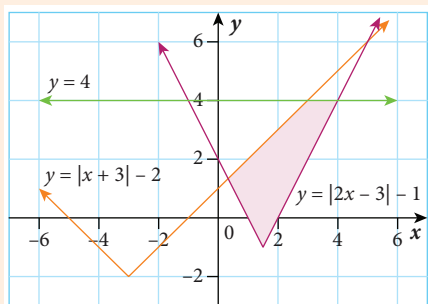
➔ PRACTICE

- 1 Draw a clear sketch of the region that satisfies all three inequalities: $y < 9 - x^2$, $y \leq 3 - 2x$, $y \geq -1$.
- 2 The graph shows the region bounded by the curves $y = 2^x$ and $y = \sqrt{16-x^2}$.



Write down a pair of inequalities to describe the shaded region. Justify your answer.

3* Write down a set of inequalities to describe the shaded region.



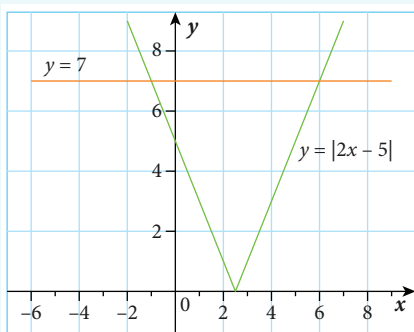
Answers ➞ p. 28

9 Solving inequalities

- ➔ Graphs can also be used to find the solution to inequalities.
- ➔ Quadratic inequalities can be solved by sketching a graph of the parabola represented by the equation and determining for what values of x it is positive or negative.

EXAMPLE 1

Use a graph to solve the inequality $|2x - 5| < 7$.



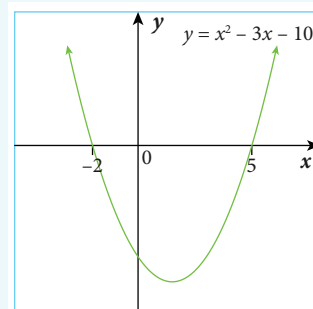
From the graph $|2x - 5| < 7$ when $-1 < x < 6$.

It is easy to see on the graph where the graph of the absolute value is below the line $y = 7$.

EXAMPLE 2

Solve $x^2 - 3x < 10$.

$$\begin{aligned} x^2 - 3x &< 10 \\ x^2 - 3x - 10 &< 0 \\ \text{Let } y &= x^2 - 3x - 10 \\ &= (x + 2)(x - 5) \end{aligned}$$



From the graph, $y < 0$ when $-2 < x < 5$.

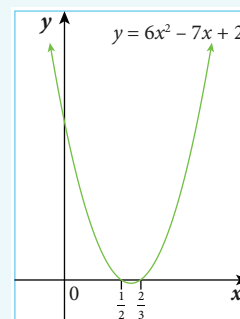
∴ $x^2 - 3x < 10$ when $-2 < x < 5$.

The sketch does not need to be accurate. It is just a tool to help us see where the parabola is; in this case, negative.

EXAMPLE 3

Solve $6x^2 \geq 7x - 2$.

$$\begin{aligned} 6x^2 &\geq 7x - 2 \\ 6x^2 - 7x + 2 &\geq 0 \\ \text{Let } y &= 6x^2 - 7x + 2 \\ &= (2x - 1)(3x - 2) \\ \text{So } y &= 0 \text{ when } x = \frac{1}{2} \text{ or } x = \frac{2}{3}. \end{aligned}$$



From the graph, $y \geq 0$ when $x \leq \frac{1}{2}$ or $x \geq \frac{2}{3}$.

∴ $6x^2 \geq 7x - 2$ when $x \leq \frac{1}{2}$ or $x \geq \frac{2}{3}$.

Alternatively, $y = 6x^2$ and $y = 7x - 2$ could both have been drawn in order to find the solution.

➔ PRACTICE

- 1 Use a graph to solve the inequality $|2x + 3| \geq 5$.
- 2 Solve $x^2 - 3x < 4$.
- 3 Solve $6x^2 \geq 19x - 10$.
- 4* Solve $8 + 2x - x^2 \geq 0$.

Answers ➞ pp. 28–29

10 Solving problems

- ➔ Graphs can be used to assist in solving problems.
- ➔ Algebraic working may also be required.

→ **Any problem needs to be carefully read and considered.** Take the time to stop and think. Use common sense. There is no need to produce pages of working if a simpler way exists. If stuck, check if there is some information that has not yet been used. Keep an eye on what you are required to achieve. Perhaps you can work backwards.

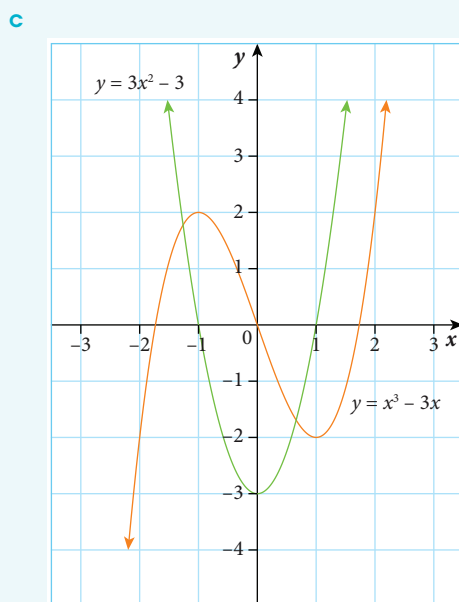
→ **Make sure you give the answer that is required.** For example, if asked for a point don't just give the x -value.

EXAMPLE 1

- Show that $f(x) = x^3 - 3x$ is an odd function.
- Show that $f'(x)$ is an even function.
- Sketch $y = f(x)$ and $y = f'(x)$ on the same diagram.
- When is $f'(x) < 0$?
- What conclusions can be made about $f(x)$ when $f'(x) < 0$?

a $f(x) = x^3 - 3x$
 $f(-x) = (-x)^3 - 3(-x)$
 $= -x^3 + 3x$
 $= -(x^3 - 3x)$
 $= -f(x)$
 $\therefore f(x)$ is an odd function.

b $f(x) = x^3 - 3x$
 $f'(x) = 3x^2 - 3$
 $f'(-x) = 3(-x)^2 - 3$
 $= 3x^2 - 3$
 $= f'(x)$
 $\therefore f'(x)$ is an even function.



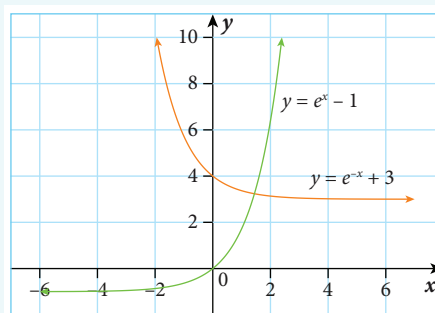
- $f'(x) < 0$ when $-1 < x < 1$
- When $f'(x) < 0$, $f(x)$ is decreasing.

This is covered in much greater detail in the calculus section.

EXAMPLE 2

- On the same set of axes sketch the graphs of $y = e^x - 1$ and $y = e^{-x} + 3$.
- Show that the point of intersection of the two curves occurs when $e^{2x} - 4e^x - 1 = 0$.
- Show that, at the point of intersection, $e^x = 2 + \sqrt{5}$.
- Hence show that the lines intersect when $x \approx 1.44$.

a



- b At the point of intersection:

$$e^x - 1 = e^{-x} + 3$$

$$e^x - e^{-x} - 4 = 0$$

$$e^{2x} - e^0 - 4e^x = 0$$

$$e^{2x} - 4e^x - 1 = 0$$

Multiplying every part of the equation by e^x .

c Let $u = e^x$
 $\therefore u^2 - 4u - 1 = 0$
 $u = \frac{-(-4) \pm \sqrt{(-4)^2 - 4 \times 1 \times -1}}{2 \times 1}$
 $= \frac{4 \pm \sqrt{20}}{2}$
 $= \frac{4 \pm 2\sqrt{5}}{2}$
 $= 2 \pm \sqrt{5}$

But $e^x > 0$ for all values of x .

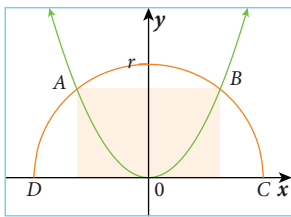
$$\therefore e^x = 2 + \sqrt{5}$$

- d $e^x = 2 + \sqrt{5}$
 $x = \ln(2 + \sqrt{5})$
 $= 1.443\,635\,47\dots$
 $= 1.44 \quad (2 \text{ d.p.})$
 $\therefore$ the lines intersect when $x \approx 1.44$.

EXAMPLE 3

The diagram shows the parabola $y = \frac{x^2}{2}$ and a semicircle of the form $y = \sqrt{r^2 - x^2}$. The graphs

intersect at A and B and the semicircle meets the x -axis at C and D, as shown.



- a Show that the equation of AB is given by $y = -1 + \sqrt{1+r^2}$.
- b If $r = 4$, show that the length of AB is about $\frac{5}{8}$ of the length of CD.
- c If $r = 4$, show that the area of the shaded rectangle is about $\frac{5}{8}$ of the area of the semicircle.

a $y = \sqrt{r^2 - x^2}$ (i)

$$y = \frac{x^2}{2}$$

$$x^2 = 2y \quad \text{(ii)}$$

Substituting (ii) into (i):

$$y = \sqrt{r^2 - 2y}$$

$$y^2 = r^2 - 2y$$

$$y^2 + 2y - r^2 = 0$$

$$y = \frac{-2 \pm \sqrt{2^2 - 4 \times 1 \times (-r^2)}}{2 \times 1}$$

$$= \frac{-2 \pm \sqrt{4 + 4r^2}}{2}$$

$$= \frac{-2 \pm 2\sqrt{1+r^2}}{2}$$

$$= -1 \pm \sqrt{1+r^2}$$

But $y > 0$,

$$\therefore y = -1 + \sqrt{1+r^2}$$

So the equation of AB is $y = -1 + \sqrt{1+r^2}$.

- b If $r = 4$,

$$\begin{aligned} \text{On AB, } y &= -1 + \sqrt{1+4^2} \\ &= -1 + \sqrt{17} \end{aligned}$$

Now at A and B, $x^2 = 2y$

$$\begin{aligned} &= -2 + 2\sqrt{17} \\ x &= \pm \sqrt{-2 + 2\sqrt{17}} \end{aligned}$$

$$\begin{aligned} \text{So the length of AB} &= 2\sqrt{-2 + 2\sqrt{17}} \\ &= 4.998\,484\,27\dots \\ &\approx 5 \text{ units} \end{aligned}$$

CD is the diameter of the semicircle.

So the length of CD is 8 units.

$\therefore$ the length of AB is approximately $\frac{5}{8}$ of the length of CD.

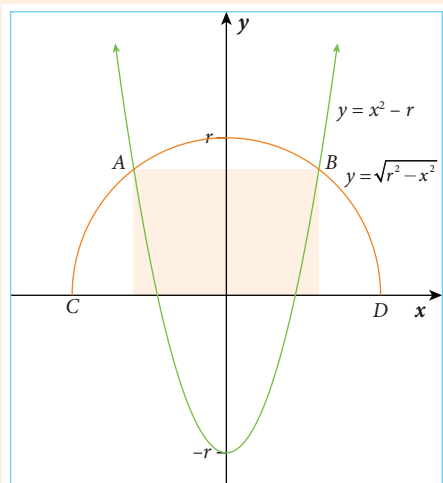
AB is parallel to the x-axis so its equation is of the form $y = c$.

- c Area of semicircle $= \frac{1}{2} \pi r^2$
- $$\begin{aligned} &= \frac{1}{2} \times \pi \times 4^2 \\ &= 25.132\,7412\dots \\ &= 25.1 \text{ units}^2 \quad (1 \text{ d.p.}) \end{aligned}$$
- So $\frac{5}{8}$ of the area is approximately 15.7 units^2 .
- Now $-1 + \sqrt{17} = 3.123\,105\,62\dots$
- Area of shaded rectangle
- $$\begin{aligned} &= 4.998\,484\,27\dots \times 3.123\,105\,62\dots \\ &= 15.610\,794\dots \text{ units}^2 \end{aligned}$$
- So the shaded area is about $\frac{5}{8}$ of the area of the semicircle.

→ PRACTICE

- 1
 - a Show that $f(x) = \frac{1}{3}x^3 - 4x$ is an odd function.
 - b Show that $f'(x)$ is an even function.
 - c Sketch $y = f(x)$ and $y = f'(x)$ on the same diagram.
 - d When is $f'(x) > 0$?
 - e What conclusions can be made about $f(x)$ when $f'(x) > 0$?
- 2
 - a On the same set of axes sketch the graphs of $y = e^x + 2$ and $y = e^{-x} - 2$.
 - b Show that the point of intersection of the two curves occurs when $e^{2x} + 4e^x - 1 = 0$.
 - c Show that, at the point of intersection, $e^x = -2 + \sqrt{5}$.
 - d Hence show that the lines intersect when $x \approx -1.44$.

- 3 The diagram shows the parabola $y = x^2 - r$ and a semicircle of the form $y = \sqrt{r^2 - x^2}$, ($r > 1$). The graphs intersect at A and B and the semicircle meets the x -axis at C and D , as shown.



- a Show that the equation of AB is given by $y = r - 1$.

- b If $r = 5$, show that the length of AB is $\frac{3}{5}$ of the length of CD .
- c If $r = 5$, show that the area of the shaded rectangle is about $\frac{3}{5}$ of the area of the semicircle.
- 4* a Use a pair of graphs to determine the number of solutions of the equation $|x| = 5 - x$.
- b Without drawing any further graphs determine the number of solution(s) of the equation $|x| = k - x$ where k is a constant. Justify your answer.

Answers ⇨ pp. 29–30

A QUICK NOTE! To help you study effectively we have made the **Examples** and **Practice** questions with the same numbers very similar to each other or you will require very similar skills to solve them.

- Please see the sample below: it shows that the **Example 1** question is similar to the **Practice 1** question and the **Example 2** question is similar to the **Practice 2** question.
 - » This is to enable you to **practise** what you have just read in an **Example**.
 - » You can also refer to the **Examples** when you are answering the **Practice** questions as their worked solutions and tips will help you answer the questions.
- The **Practice** questions that have an **asterisk (*)** are different to the other **Practice** questions. They do not match any particular **Example** but are designed to provide you with extra practice.
 - » You can see in the sample below that the **Practice 3** question is a different type to the other questions and is not directly related to any example. It is to give you further practice.

EXAMPLE 1

The present value of an annuity of \$700 per month for 3 years is \$22 207 when the interest rate is 8.4% p.a., compounded monthly. Find the future value.

EXAMPLE 2

The future value of an annuity is \$32 000 when the investment is over 10 years and earning 4.8% p.a. interest, compounded monthly. Find the present value.

is similar to

is similar to

This question is different from the other Practice questions.

→ PRACTICE

- The present value of an annuity of \$800 per month for 4 years is \$32 707 when the interest rate is 8.1% p.a., compounded monthly. Find the future value.
- The future value of an annuity is \$47 000 when the investment is over 12 years and earning 4.5% p.a. interest, compounded monthly. Find the present value.
- * What amount of money needs to be invested now at 6% p.a. interest, compounded quarterly, to be worth \$10 000 after 4 years?

HSC EXAM-TYPE QUESTIONS

Now for the real thing! The following questions are modelled on the types of questions you will face in the HSC Examination. If you get extensive practice at answering these sorts of questions, you will be more confident answering them in the HSC Examination.

Remember: The Answer section is located straight after this section so that you can check the answers readily. There are worked solutions and also ticks that indicate the mark allocation for all questions. Both these features will help you to strive for full marks.

Make sure also that when you check your work you highlight any questions you found difficult and earmark these areas for extra study.

Total marks: 60

PART A

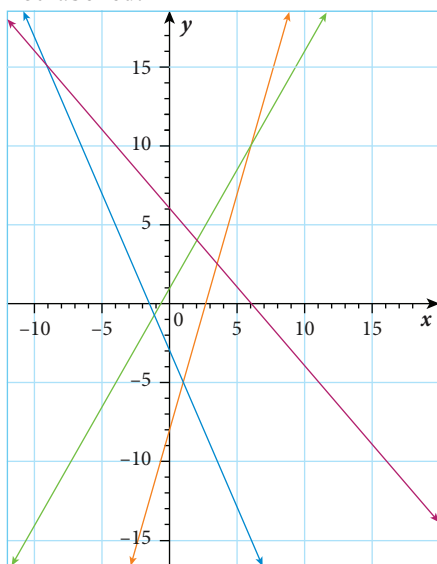
(Suggested time: 15 minutes)

(1 mark each)

- 1 What is the y -intercept of the curve $y = e^x + 2$?

A 0 B 1
C 2 D 3

- 2 A computer application drew four lines, but they were not labelled.



The lines are $y = -2x - 3$, $y = 3x - 8$, $y = \frac{3}{2}x + 1$ and $x + y = 6$.

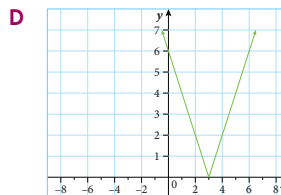
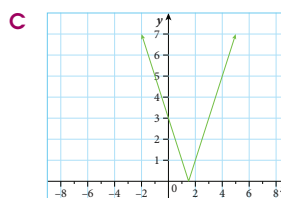
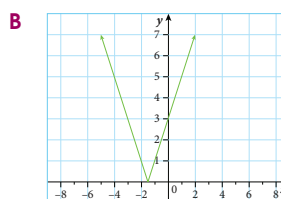
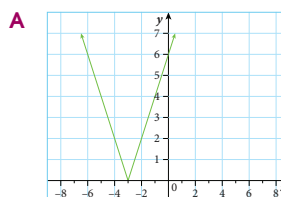
What is the point of intersection of $y = -2x - 3$ and $x + y = 6$?

A $(-9, 15)$ B $(1, -5)$
C $(2, 4)$ D $(3.5, 2.5)$

- 3 The curve $y = e^x - e^{-x}$ is:

A even.
B odd.
C neither even nor odd.
D both even and odd.

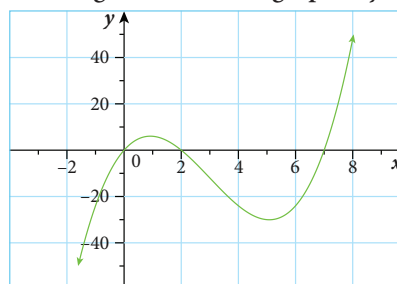
- 4 Which is the graph of $y = |2x + 3|$?



- 5 For what values of x is $x^2 > 5x + 24$?

A $x < -3$ and $x > 8$ B $x < -8$ and $x > 3$
C $-3 < x < 8$ D $-8 < x < 3$

- 6 The diagram shows the graph of $y = x^3 - 9x^2 + 14x$.



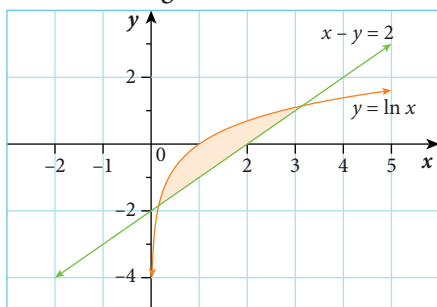
How many solutions are there to the equation $x^3 - 9x^2 + 14x = 10$?

A 3 B 2 C 1 D 0

7 Which function is continuous at $x = 4$?

- A $f(x) = \begin{cases} 6-x & \text{for } x < 4 \\ 2x-5 & \text{for } x \geq 4 \end{cases}$
- B $f(x) = \begin{cases} x^2-2x-6 & \text{for } x \leq 4 \\ \sqrt{x} & \text{for } x > 4 \end{cases}$
- C $f(x) = \frac{x}{x-4}$
- D $f(x) = \begin{cases} e^{\frac{x}{4}} & \text{for } x < 4 \\ e^{-4x} & \text{for } x \geq 4 \end{cases}$

8 The shaded region is where:

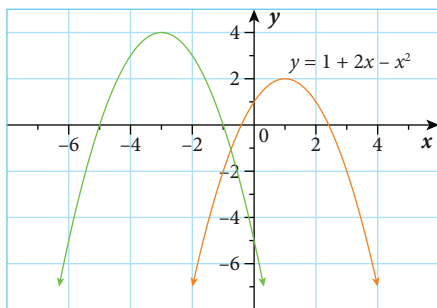


- A $y \leq \ln x, x - y \geq 2$ B $y \geq \ln x, x - y \leq 2$
- C $y \geq \ln x, x - y \geq 2$ D $y \leq \ln x, x - y \leq 2$

9 What are the equations of the asymptotes of $y = \frac{1}{x+2}$?

- A $x = 0$ and $y = 2$ B $x = 0$ and $y = -2$
- C $x = -2$ and $y = 0$ D $x = 2$ and $y = 0$

10 The graph shows $y = 1 + 2x - x^2$ and a translation of that graph.



What is the equation of the translated graph?

- A $y = -5 - 6x - x^2$ B $y = 1 - 4x - x^2$
- C $y = -21 + 10x - x^2$ D $y = 43 + 10x - x^2$

PART B

(Suggested time: 90 minutes)

Show all working.

11 Consider the curve $y = 4x^2 - 12x + 5$.

- a Where does the curve cut the y -axis? (1 mark)
- b Find the x -intercepts. (2 marks)
- c What is the equation of the axis of symmetry? (1 mark)
- d What are the coordinates of the vertex? (1 mark)
- e Sketch the curve. (1 mark)

12 a Show that $f(x) = 12x - x^3$ is an odd function. (1 mark)

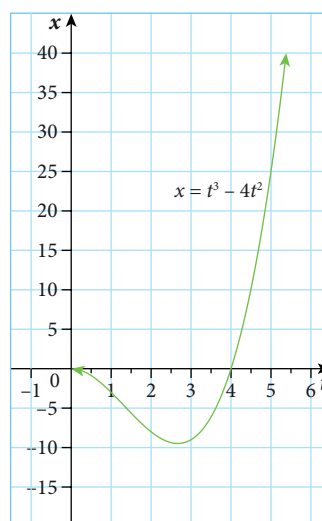
- b Given that $y = f(x)$ has a turning point at $(2, 16)$ and cuts the x -axis at $(2\sqrt{3}, 0)$, sketch the curve. (2 marks)

13 a Sketch the graph of $y = e^x + 1$. (1 mark)

- b On the same diagram show the line $y = x$ and the reflection of the graph of $y = e^x + 1$ in that line. (2 marks)
- c What is the equation of the reflected graph? (1 mark)

14 The displacements from the origin, x metres at time t seconds, of 2 particles A and B are given by $x_A = 4t + 5$ and $x_B = t^3 - 4t^2$.

The diagram shows the graph of $x = t^3 - 4t^2$ for $t \geq 0$.



When is particle B further from the origin than particle A? (2 marks)

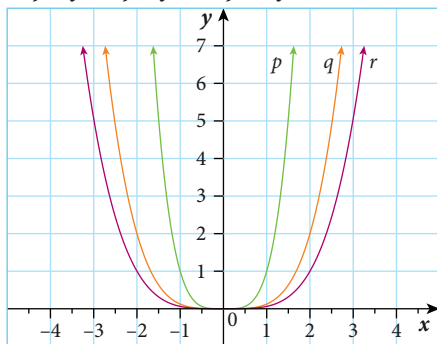
15 a Solve $x^4 - 25x^2 + 144 = 0$. (2 marks)

- b Explain why the solutions to the above equation give the x -values of the points of intersection of the circle $x^2 + y^2 = 25$ and the hyperbola $y = \frac{12}{x}$. (2 marks)

- c Write down the points of intersection of $xy = -12$ and $y = -\sqrt{25 - x^2}$. (2 marks)

- 16 Consider $f(x) = \frac{x^4}{16}$. The diagram shows the graphs

of $y = f(x)$, $y = f(2x)$, $y = 2f(x)$.

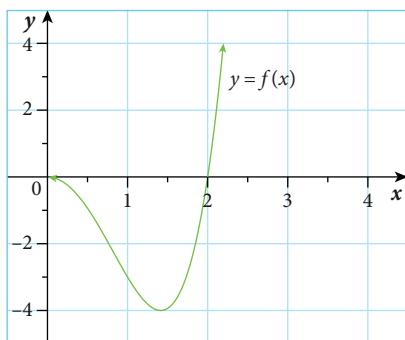


Identify the curves, justifying your answer.

(3 marks)

- 17 a On the same diagram sketch $y = \ln x$, $y = \ln(x+3)$, $y = \ln(x^3)$ and $y = \ln(3x)$, clearly labelling each curve. (4 marks)
- b Briefly comment on similarities and differences between the curves. (2 marks)

- 18 The graph shows $y = f(x)$ for $x \geq 0$, where $f(x) = x^4 - 4x^2$.



- a Copy and complete the graph for $x < 0$. (3 marks)
- b On a separate diagram sketch $y = f(x-3) + 1$. (2 marks)

- 19 The curve $y = \frac{ax+b}{cx+d}$ results from translating the curve

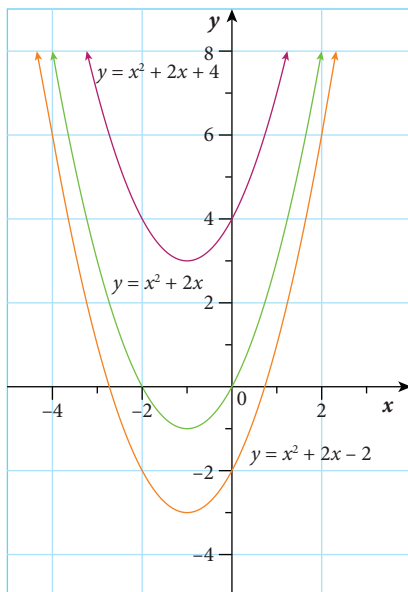
$y = \frac{1}{x}$ firstly 2 units to the right and then 3 units down.

- a Sketch the curves on the same diagram. (2 marks)
- b What are the equations of the asymptotes of $y = \frac{ax+b}{cx+d}$? (2 marks)
- c Would this be the same result if the curve was first translated 3 units down and then 2 units right? (1 mark)
- d Find the values of a , b , c and d . (3 marks)
- 20 a Sketch $y = e^x - 2$ and $y = e^{-x} + 1$ on the same diagram. (2 marks)
- b How many points of intersection are there? (1 mark)
- c Explain why the x -value(s) of the point(s) of intersection can be found by solving the equation $e^{2x} - 3e^x = 1$. (1 mark)
- d Find the point(s) of intersection, giving all values correct to one decimal place. (3 marks)

PRACTICE

1 Translations ⇨ pp. 5–6

1



- 2 The first curve is $y = x^3 - 9x^2 + 11x + 21$.

The second curve has moved 5 units to the right.

The equation is:

$$y = (x - 5)^3 - 9(x - 5)^2 + 11(x - 5) + 21$$

$$y = x^3 - 15x^2 + 75x - 125 - 9x^2 + 90x - 225 + 11x - 55 + 21$$

$$y = x^3 - 24x^2 + 176x - 384$$

- 3 $y = e^{2x-1}$

When $x = 0$,

$$y = e^{2 \times 0 - 1}$$

$$= e^{-1}$$

$$\approx 0.368$$

So q is the curve $y = e^{2x-1}$.

p has moved 2 units to the left.

$$p: y = e^{2(x+2)-1}$$

$$y = e^{2x+3}$$

r has moved 3 units to the right.

$$r: y = e^{2(x-3)-1}$$

$$y = e^{2x-7}$$

So p is the curve $y = e^{2x+3}$, q is $y = e^{2x-1}$ and r is $y = e^{2x-7}$.

- 4 $y = \frac{ax+b}{cx+d}$ is a translation of $y = \frac{1}{x}$ moving 3 units to the right and

5 units up.

$$\text{So } y = \frac{1}{x-3} + 5$$

$$= \frac{1+5(x-3)}{x-3}$$

$$= \frac{5x-14}{x-3}$$

$\therefore a = 5, b = -14, c = 1$ and $d = -3$.

- 5 p is $y = |x|$.

q is a translation of p , 2 units to the left and 3 units up.

$$q \text{ is } y = |x+2| + 3$$

r is a translation of p , 4 units to the right and 1 unit down.

$$r \text{ is } y = |x-4| - 1.$$

So p is $y = |x|$, q is $y = |x+2| + 3$ and r is $y = |x-4| - 1$.

2 Dilations ⇨ pp. 7–8

- 1 q passes through $(1, -1)$ and $(-1, 1)$ so it is $y = -x^3$.

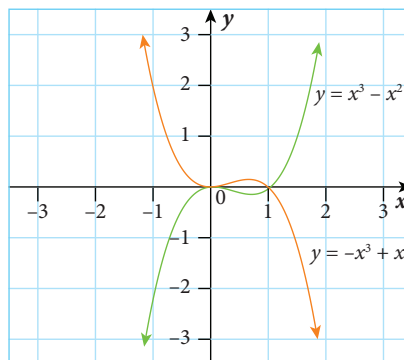
r passes through $(1, -2)$ so it is $y = -2x^3$.

$$p \text{ is } y = \frac{-x^3}{2}$$

The curves all have a basic cubic shape and all pass through the origin.

$y = -2x^3$ is narrower than $y = -x^3$ and $y = \frac{-x^3}{2}$ is wider.

2



- 3 a $f(x) = x^2 - x$

$$f(3x) = (3x)^2 - (3x)$$

$$= 9x^2 - 3x$$

- b The curves are $y = x^2 - x$, $y = 3x^2 - 3x$ and $y = 9x^2 - 3x$.

Now both $y = x^2 - x$ and $y = 3x^2 - 3x$ have x -intercepts at 0 and 1.

x -intercepts of $y = 9x^2 - 3x$ occur when

$$9x^2 - 3x = 0$$

$$3x(3x - 1) = 0$$

$$x = 0 \text{ or } x = \frac{1}{3}$$

So p is $y = 9x^2 - 3x$

$$\text{When } x = \frac{1}{2}, \quad x^2 - x = \left(\frac{1}{2}\right)^2 - \frac{1}{2} = -\frac{1}{4}$$

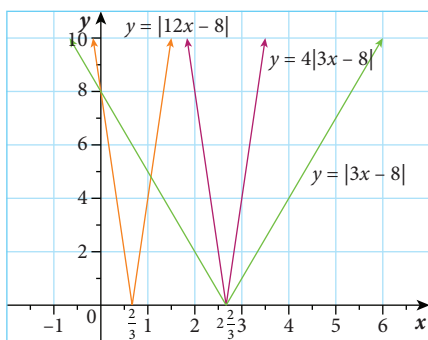
$$3x^2 - 3x = 3\left(-\frac{1}{4}\right) = -\frac{3}{4}$$

So r is $y = x^2 - x$ and q is $y = 3x^2 - 3x$.

Multiplying by 3 has the effect of narrowing the curve.

$y = f(3x)$ is a narrower version of $y = f(x)$ without moving the curve up or down. $y = 3f(x)$ is also narrower but the vertex has also moved down.

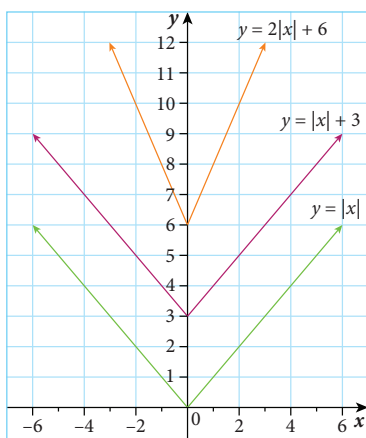
4 a



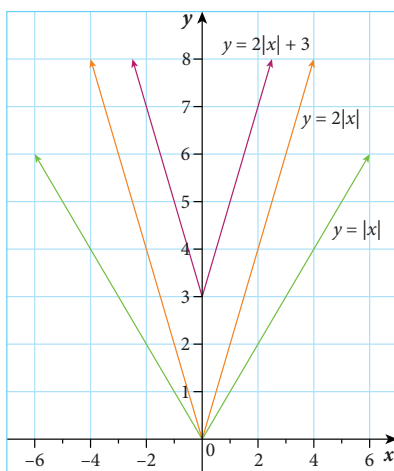
- b All curves are the V shape of absolute curves. Both $y = |3x - 8|$ and $y = 4|3x - 8|$ meet the x -axis at the same point $(\frac{2}{3}, 0)$, but $y = 4|3x - 8|$ is narrower than $y = |3x - 8|$. $y = |12x - 8|$ is a translation of $y = 4|3x - 8|$, 2 units to the left.

3 Mixed transformations ⇨ p. 9

1 a

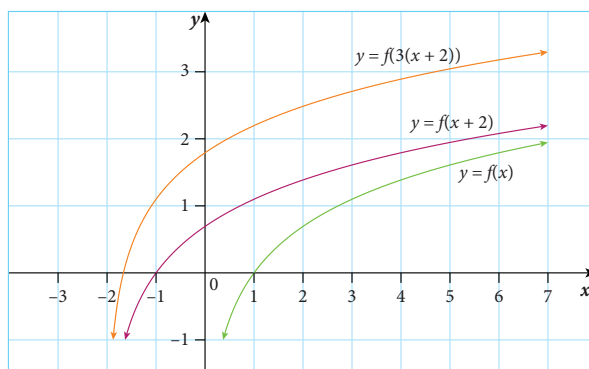


b

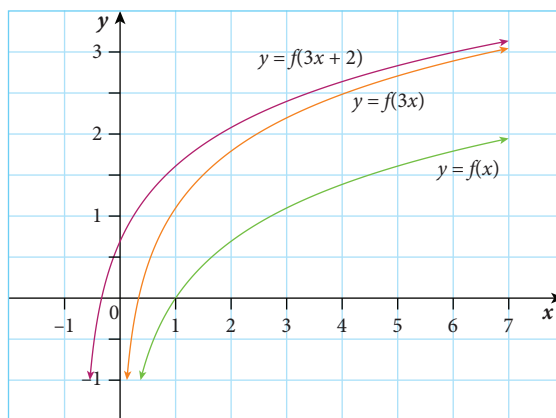


- c The final result in each case is two graphs that are translations of each other. By first adding 3 and then multiplying by 2 the curve moves up 2×3 or 6 units whereas when the function is first multiplied by 2 and then 3 added it moves up 3 units. **In both cases the multiplication by 2 means that the final curve is narrower than the original.**

2 a



b

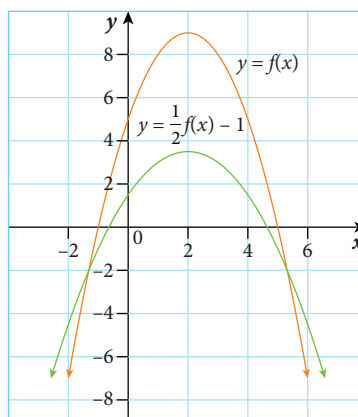


- c The different order of applying the transformations gives different results, but all the graphs are translations of each other. $\ln(3(x + 2)) = \ln 3 + \ln(x + 2)$ so in the first case, $y = \ln x$ first moves 2 units to the left and then up $\ln 3$ units. In the second case it moves up $\ln 3$ units first and then $\frac{2}{3}$ of a unit to the left.

3* a

$$f(x) = 5 + 4x - x^2$$

$$\begin{aligned} \frac{1}{2}f(x) - 1 &= \frac{1}{2}(5 + 4x - x^2) - 1 \\ &= 2\frac{1}{2} + 2x - \frac{1}{2}x^2 - 1 \\ &= 1\frac{1}{2} + 2x - \frac{1}{2}x^2 \end{aligned}$$



- b No, here each y -value is halved and then moved down 1 unit whereas $y = \frac{1}{2}f(x - 1)$ would have first moved right one unit and then each y -value would be halved.

4 Sketching graphs ⇨ p. 10

1 a $y = x^3 + 2x^2 - 24x$

When $y = 0$,

$$x^3 + 2x^2 - 24x = 0$$

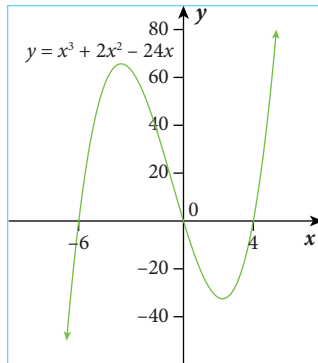
$$x(x^2 + 2x - 24) = 0$$

$$x(x + 6)(x - 4) = 0$$

$$x = 0 \text{ or } x = -6 \text{ or } x = 4$$

b As $x \rightarrow \infty, y \rightarrow \infty$

c

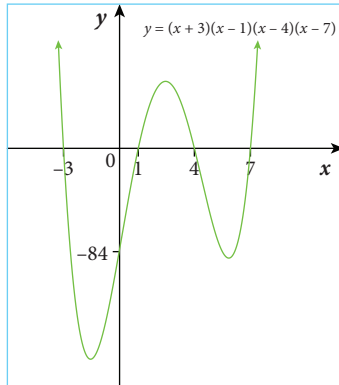


2 $y = (x + 3)(x - 1)(x - 4)(x - 7)$

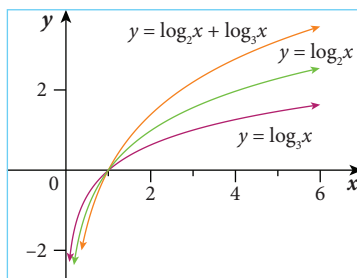
x -intercepts are $-3, 1, 4$ and 7

$$y\text{-intercept} = 3 \times -1 \times -4 \times -7 = -84$$

As $x \rightarrow \infty, y \rightarrow \infty$



3



4 $y = x^3 - 13x - 12$

a The curve cuts the y -axis at $(0, -12)$.

b When $x = -3$,

$$y = (-3)^3 - 13 \times -3 - 12 = 0$$

When $x = -1$,

$$y = (-1)^3 - 13 \times -1 - 12 = 0$$

When $x = 4$,

$$y = 4^3 - 13 \times 4 - 12 = 0$$

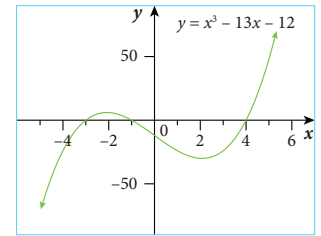
$\therefore$ the curve cuts the x -axis at $-3, -1$ and 4 .

c When $x = -2$,

$$y = (-2)^3 - 13 \times -2 - 12 = 6$$

When $x = 2$,

$$y = 2^3 - 13 \times 2 - 12 = -30$$



5 Using symmetry ⇨ p. 12

1 a $y = 9x^2 + 30x + 29$

$$= 9x^2 + 30x + 25 + 4$$

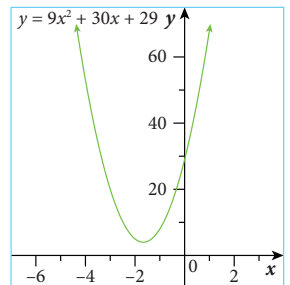
$$= (3x + 5)^2 + 4$$

b Axis: $3x + 5 = 0$

$$x = -1\frac{2}{3}$$

c Vertex $(-1\frac{2}{3}, 4)$

y -intercept = 29



2 a $f(x) = -9x^4 + 37x^2 - 4$

$$f(-x) = -9(-x)^4 + 37(-x)^2 - 4$$

$$= -9x^4 + 37x^2 - 4$$

$$= f(x)$$

$\therefore f(x)$ is an even function.

b $f(0) = -4$

When $f(x) = 0$,

$$-9x^4 + 37x^2 - 4 = 0$$

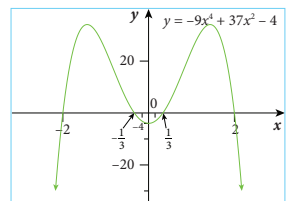
$$9x^4 - 37x^2 + 4 = 0$$

$$(9x^2 - 1)(x^2 - 4) = 0$$

$$9x^2 = 1 \text{ or } x^2 = 4$$

$$x = \pm\frac{1}{3} \text{ or } x = \pm 2$$

As $x \rightarrow \infty, y \rightarrow -\infty$



3 a $f(x) = x^3 - 9x$

$$f(-x) = (-x)^3 - 9(-x)$$

$$= -x^3 + 9x$$

$$= -(x^3 - 9x)$$

$$= -f(x)$$

$\therefore f(x)$ is an odd function.

b If $f(x) = 0$

$$0 = x^3 - 9x$$

$$= x(x^2 - 9)$$

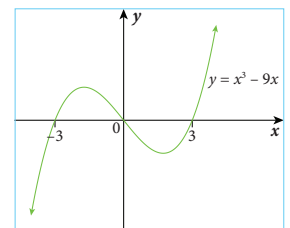
$$x = 0 \text{ or } x^2 = 9$$

$$x = 0 \text{ or } x = \pm 3$$

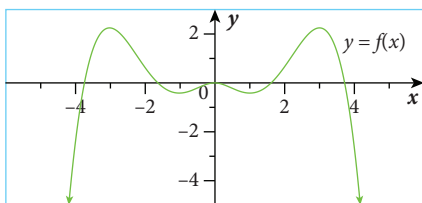
$$f(1.5) = (1.5)^3 - 9 \times 1.5$$

$$= -10.125$$

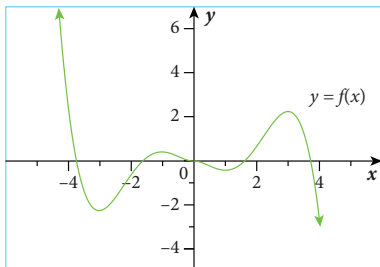
As $x \rightarrow \infty, y \rightarrow \infty$



4 a



b

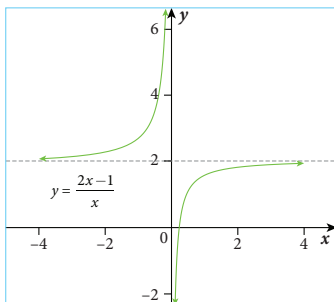


6 Discontinuities and asymptotes ⇨ p. 13

$$\begin{aligned}
 1 \quad a \quad y &= \frac{2x-1}{x} \\
 &= \frac{2x}{x} - \frac{1}{x} \\
 &= 2 - \frac{1}{x}
 \end{aligned}$$

So the asymptotes are $x = 0$ and $y = 2$.

b



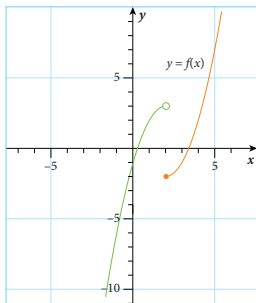
$$\begin{aligned}
 2 \quad a \quad f(x) &= \begin{cases} x^2 - 4x + 2 & \text{for } x \geq 2 \\ -x^2 + 4x - 1 & \text{for } x < 2 \end{cases} \\
 f(2) &= 2^2 - 4 \times 2 + 2 \\
 &= -2
 \end{aligned}$$

When $x = 2$,

$$\begin{aligned}
 -x^2 + 4x - 1 &= -2^2 + 4 \times 2 - 1 \\
 &= 3 \quad (\neq f(2))
 \end{aligned}$$

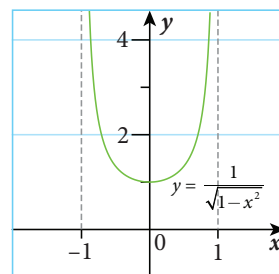
∴ the function is not continuous.

b



$$\begin{aligned}
 3 \quad a \quad f(x) &= \frac{1}{\sqrt{1-x^2}} \\
 f(-x) &= \frac{1}{\sqrt{1-(-x)^2}} \\
 &= \frac{1}{\sqrt{1-x^2}} \\
 &= f(x)
 \end{aligned}$$

∴ the function is even.

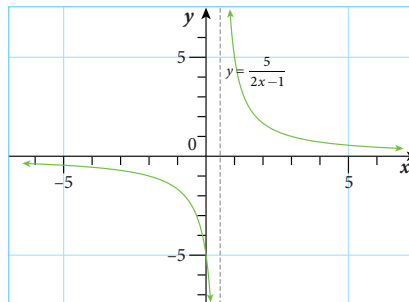
b The function is not defined when $1 - x^2 \leq 0$.So when $x \leq -1$ or $x \geq 1$ c $f(0) = 1$ As $x \rightarrow 1^-$, $f(x) \rightarrow \infty$ 

$$4 \quad y = \frac{5}{2x-1}$$

Now $2x - 1 \neq 0$

$$2x \neq 1$$

$$x \neq \frac{1}{2}$$

There is a vertical asymptote at $x = \frac{1}{2}$.When $x > \frac{1}{2}$, $2x - 1 > 0$ and so $y > 0$.When $x < \frac{1}{2}$, $2x - 1 < 0$ and so $y < 0$.There is a horizontal asymptote at $y = 0$.When $x = 0$, $y = -5$.

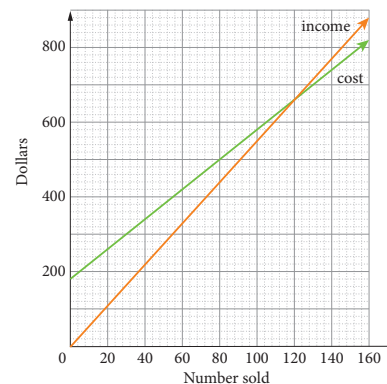
7 Graphical solution of equations ⇨ pp. 15–16

$$1 \quad a \quad m = \frac{80}{20} = 4$$

The gradient is 4 and it represents the extra cost to make each bag of 'sweetbeans'.

b The y-intercept is 180. It represents the fixed costs.

c



- d The break-even point is when 120 bags of 'sweetbeans' are sold. Both the cost and income are \$660.
 e The profit when 150 bags are sold is \$45.
 f If 100 bags are sold there will be a loss of \$30.

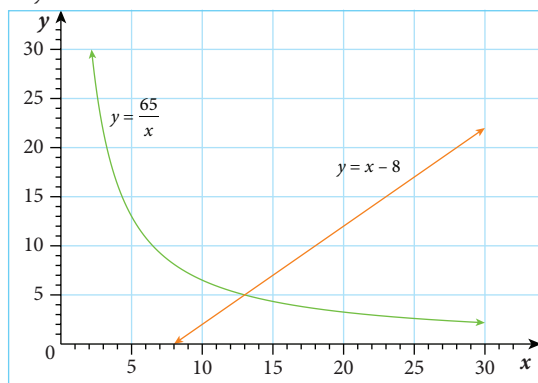
- 2 a The area is equal to the length by the width.

$$\text{So } 65 = xy$$

$$y = \frac{65}{x}$$

- b Now the width is 8 cm shorter than the length.

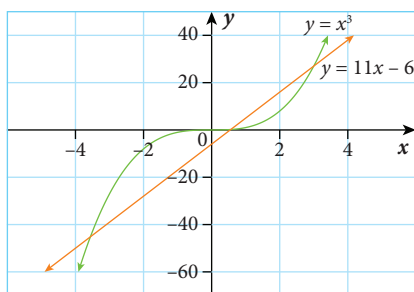
$$\text{So } y = x - 8$$



From the graph the length is 13 cm and the width is 5 cm.

- 3 $x^3 - 11x + 6 = 0$

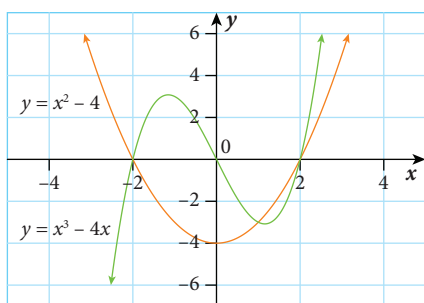
$$x^3 = 11x - 6$$



From the graph there are three points of intersection.

So there are three solutions to the equation $x^3 - 11x + 6 = 0$.

- 4 a There are three solutions to the equation $x^3 - 4x = x^2 - 4$.

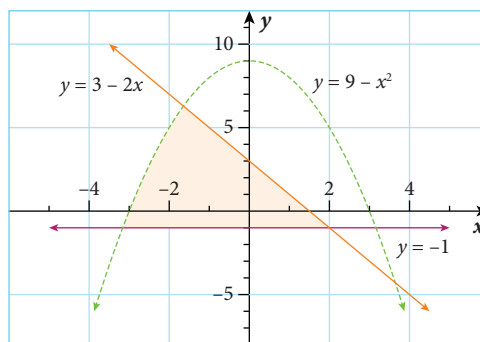


- b The line $y = 4x - 4$ will intersect the cubic curve $y = x^3 - x^2$ three times.

Rearranging the equation $x^3 - 4x = x^2 - 4$ gives $x^3 - x^2 = 4x - 4$ so there are the same number of solutions to both equations because they are in fact the same equation.

8 Regions ⇨ pp. 16–17

- 1 Consider (0, 2) in each equation: $2 > -1$; $2 < 9 - 0^2$; $2 < 3 - 2 \times 0$



- 2 Consider (0, 3) in each equation: $3 > 2^0$; $3 < \sqrt{16 - 0^2}$

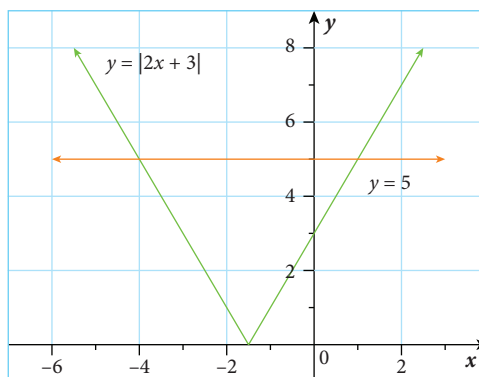
$$\text{So } y \geq 2^x \text{ and } y \leq \sqrt{16 - x^2}.$$

- 3 Consider (2, 2) in each equation: $2 < 4$; $2 < |2 + 3| - 2$;
 $2 > |2 \times 2 - 3| - 1$

$$\text{So } y \leq 4, y \leq |x + 3| - 2 \text{ and } y \geq |2x - 3| - 1.$$

9 Solving inequalities ⇨ p. 17

- 1 a



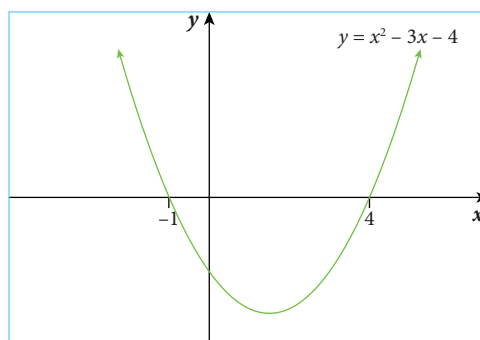
From the graph $|2x + 3| \geq 5$ when $x \leq -4$ and $x \geq 1$.

- 2 $x^2 - 3x < 4$

$$x^2 - 3x - 4 < 0$$

$$\text{Let } y = x^2 - 3x - 4$$

$$= (x + 1)(x - 4)$$



From the graph $y < 0$ when $-1 < x < 4$.

$$\therefore x^2 - 3x < 4 \text{ when } -1 < x < 4.$$

3 $6x^2 \geq 19x - 10$

$$6x^2 - 19x + 10 \geq 0$$

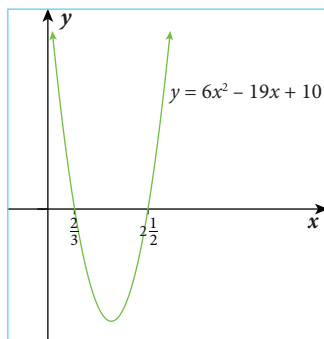
$$\text{Let } y = 6x^2 - 19x + 10 \\ = (3x - 2)(2x - 5)$$

From the graph $y \geq 0$ when

$$x \leq \frac{2}{3} \text{ or } x \geq 2\frac{1}{2}$$

$$\therefore 6x^2 \geq 19x - 10 \text{ when}$$

$$x \leq \frac{2}{3} \text{ or } x \geq 2\frac{1}{2}$$



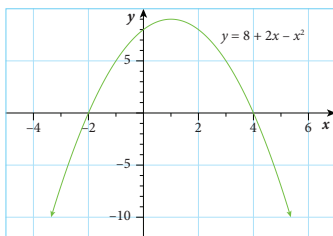
4 $8 + 2x - x^2 \geq 0$

$$\text{Let } y = 8 + 2x - x^2 \\ = (2 + x)(4 - x)$$

From the graph $y \geq 0$
when $-2 \leq x \leq 4$.

$$\therefore 8 + 2x - x^2 \geq 0$$

when $-2 \leq x \leq 4$.



10 Solving problems ⇨ pp. 19-20

1 a $f(x) = \frac{1}{3}x^3 - 4x$

$$f(-x) = \frac{1}{3}(-x)^3 - 4(-x)$$

$$= -\frac{1}{3}x^3 + 4x$$

$$= -\left(\frac{1}{3}x^3 - 4x\right)$$

$$= -f(x)$$

$\therefore f(x)$ is an odd function.

b $f'(x) = x^2 - 4$

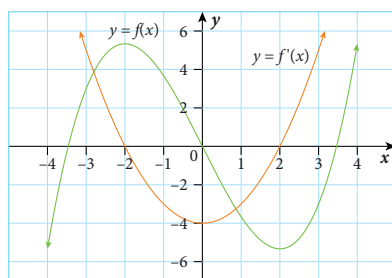
$$f'(-x) = (-x)^2 - 4$$

$$= x^2 - 4$$

$$= f'(x)$$

$\therefore f'(x)$ is an even function.

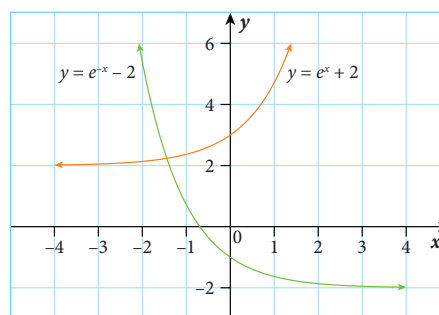
c



d $f'(x) > 0$ when $x < -2$ and $x > 2$.

e When $f'(x) > 0$, $f(x)$ is increasing.

2 a



b At the point of intersection:

$$e^x + 2 = e^{-x} - 2$$

$$e^x - e^{-x} + 4 = 0$$

$$e^{2x} - e^0 + 4e^x = 0$$

$$e^{2x} + 4e^x - 1 = 0$$

c Let $u = e^x$

$$\therefore u^2 + 4u - 1 = 0$$

$$u = \frac{-4 \pm \sqrt{4^2 - 4 \times 1 \times -1}}{2 \times 1}$$

$$= \frac{-4 \pm \sqrt{20}}{2}$$

$$= \frac{-4 \pm 2\sqrt{5}}{2}$$

$$= -2 \pm \sqrt{5}$$

But $e^x > 0$ for all values of x .

$$\therefore e^x = -2 + \sqrt{5}$$

d $e^x = -2 + \sqrt{5}$

$$x = \ln(-2 + \sqrt{5})$$

$$= -1.44363547\dots$$

$$= -1.44 \quad (2 \text{ d.p.})$$

$\therefore$ the lines intersect when $x \approx -1.44$.

3 a $y = \sqrt{r^2 - x^2}$ (i)

$$y = x^2 - r$$
 (ii)

At the points of intersection:

$$x^2 - r = \sqrt{r^2 - x^2}$$

$$x^4 - 2rx^2 + r^2 = r^2 - x^2$$

$$x^4 - (2r - 1)x^2 = 0$$

$$x^2(x^2 - (2r - 1)) = 0$$

$$x = 0 \text{ or } x^2 = 2r - 1$$

Substituting into (ii):

$$\text{If } x = 0, y = -r$$

But $y > 0$ at A and B.

$$\therefore x^2 = 2r - 1$$

$$y = 2r - 1 - r$$

$$= r - 1$$

So the equation of AB is $y = r - 1$.

b If $r = 5$,

$$\text{On AB, } y = 5 - 1$$

$$= 4$$

$$\text{Now at A and B, } x^2 = y + r$$

$$= 9$$

$$x = \pm 3$$

So the length of $AB = 2 \times 3$

$$= 6 \text{ units}$$

CD is the diameter of the semicircle.

So the length of CD is 10 units.

$\therefore$ the length of AB is $\frac{3}{5}$ of the length of CD .

c Area of semicircle $= \frac{1}{2} \pi r^2$

$$= \frac{1}{2} \times \pi \times 5^2$$

$$= 39.269\,908\,1\dots$$

$$= 39.3 \text{ units}^2 \quad (1 \text{ d.p.})$$

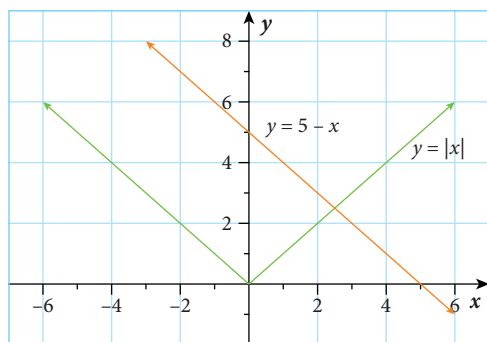
So $\frac{3}{5}$ of the area is approximately 23.6 units^2 .

Area of shaded rectangle $= 6 \times 4$

$$= 24 \text{ units}^2$$

So the shaded area is about $\frac{3}{5}$ of the area of the semicircle.

4 a The equation $|x| = 5 - x$ has one solution.



b If $k > 0$, the equation $|x| = k - x$ will have **one solution**.
If $k < 0$ the equation will have **no solutions** and if $k = 0$ there will be **infinite solutions**. The line $y = k - x$ will be parallel to one branch of $y = |x|$. When k is positive, the line $y = k - x$ will cut the other branch in one place, but if k is negative the line will not intersect $y = |x|$. If $k = 0$ the line will coincide with one branch of $y = |x|$.

HSC EXAM-TYPE QUESTIONS

Part A

1 D. $y = e^x + 2$

When $x = 0$,

$$y = e^0 + 2$$

$$= 1 + 2$$

$$= 3$$

2 A. The point of intersection of $y = -2x - 3$ and $x + y = 6$ is $(-9, 15)$.

3 B. $f(x) = e^x - e^{-x}$ is:

$$f(-x) = e^{-x} - e^{-(-x)}$$

$$= e^{-x} - e^x$$

$$= -(e^x - e^{-x})$$

$$= -f(x)$$

4 B. When $x = 0$, $y = 3$

When $y = 0$, $x = -1.5$

5 A. $x^2 > 5x + 24$

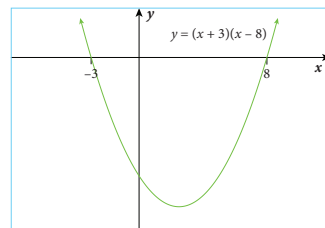
$$x^2 - 5x - 24 > 0$$

$$(x + 3)(x - 8) > 0$$

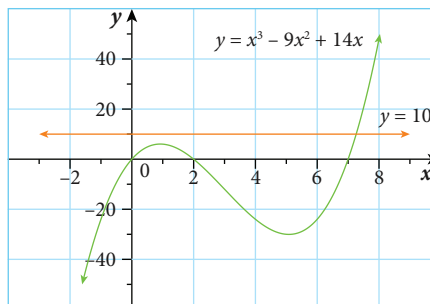
From the graph,

$$(x + 3)(x - 8) > 0 \text{ when}$$

$$x < -3 \text{ and } x > 8.$$



6 C. There is only one point of intersection between $y = 10$ and $y = x^3 - 9x^2 + 14x$.



7 B. When $x = 4$,

$$x^2 - 2x - 6 = 4^2 - 2 \times 4 - 6$$

$$= 2$$

$$\sqrt{4} = 2$$

$$\text{So } f(x) = \begin{cases} x^2 - 2x - 6 & \text{for } x \leq 4 \\ \sqrt{x} & \text{for } x > 4 \end{cases} \text{ is continuous at } x = 4.$$

8 D. When $x = 1.5$ and $y = 0$,

$$x - y = 1.5 \quad (< 2)$$

$$\ln 1.5 = 0.405\,465\dots \quad (> 0)$$

$$\text{So } y \leq \ln x, x - y \leq 2$$

9 C. $x + 2 \neq 0$

$$\text{So } x \neq -2$$

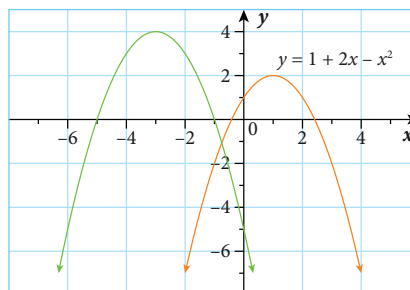
The asymptotes of $y = \frac{1}{x+2}$ are $x = -2$ and $y = 0$.

10 A. The graph has moved 4 units left and 2 units up.

$$\text{So } y = 1 + 2(x + 4) - (x + 4)^2 + 2$$

$$y = 1 + 2x + 8 - x^2 - 8x - 16 + 2$$

$$y = -5 - 6x - x^2$$



Part B

11 $y = 4x^2 - 12x + 5$.

a y -intercept $= 5$

The curve cuts the y -axis at $(0, 5)$. ✓

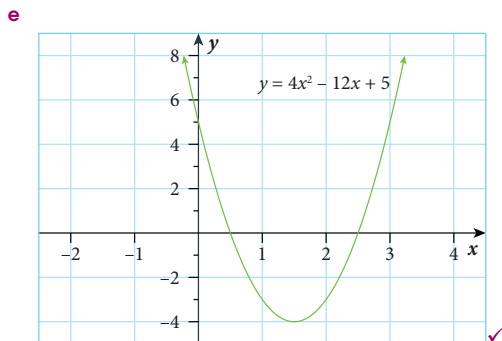
b $4x^2 - 12x + 5 = 0$
 $4x^2 - 2x - 10x + 5 = 0$
 $2x(2x - 1) - 5(2x - 1) = 0$
 $(2x - 1)(2x - 5) = 0$
 $x = \frac{1}{2}$ or $x = 2\frac{1}{2}$

x-intercepts are $\frac{1}{2}$ and $2\frac{1}{2}$. ✓✓

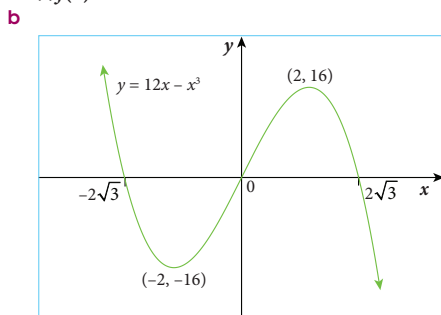
c Axis: $x = 1\frac{1}{2}$ ✓

d When $x = 1.5$,
 $y = 4(1.5)^2 - 12(1.5) + 5$
 $= -4$

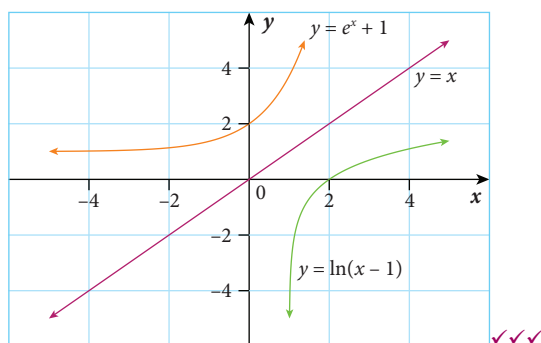
The vertex is $(1.5, -4)$. ✓



12 a $f(x) = 12x - x^3$
 $f(-x) = 12(-x) - (-x)^3$
 $= -12x + x^3$
 $= -(12x - x^3)$
 $= -f(x)$
 $\therefore f(x)$ is odd. ✓

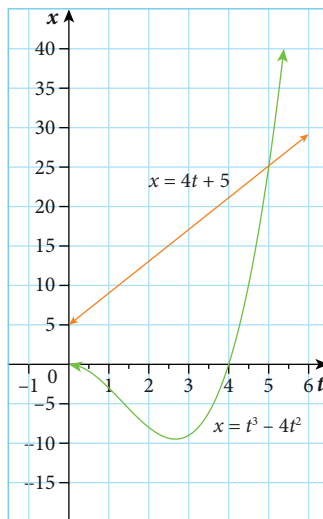


13 a, b



c The reflected graph is the graph of $y = \ln x$ moved 1 unit to the right. So it is $y = \ln(x - 1)$. ✓

14 From the graph, particle B is further from the origin than particle A after 5 seconds. ✓



15 a $x^4 - 25x^2 + 144 = 0$
 $(x^2 - 9)(x^2 - 16) = 0$
so $x^2 = 9$ or $x^2 = 16$
 $x = \pm 3$ or $x = \pm 4$ ✓✓

b $x^2 + y^2 = 25$ (i)
 $y = \frac{12}{x}$ (ii)

Substitute (ii) into (i).

$$x^2 + \left(\frac{12}{x}\right)^2 = 25$$

$$x^4 + 144 = 25x^2$$

$$x^4 - 25x^2 + 144 = 0$$
 ✓

The simultaneous solution of $x^2 + y^2 = 25$ and $y = \frac{12}{x}$ is given by

the equation $x^4 - 25x^2 + 144 = 0$ so the solutions to the equation give the x-values of the points of intersection of the circle

$$x^2 + y^2 = 25 \text{ and the hyperbola } y = \frac{12}{x}. \quad \checkmark$$

c The graphs would intersect at $(3, -4)$ and $(4, -3)$. ✓✓

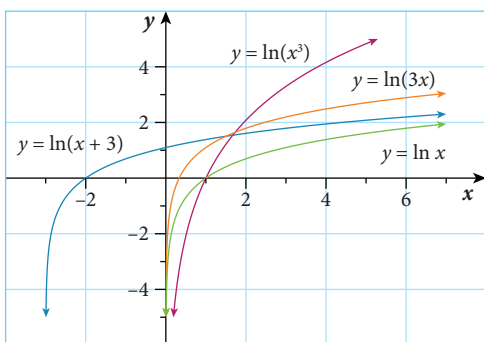
16 $f(x) = \frac{x^4}{16}$
 $f(2x) = \frac{(2x)^4}{16} = x^4$
 $2f(x) = \frac{2x^4}{16} = \frac{x^4}{8}$

Now the widest curve will be $y = \frac{x^4}{16}$ so graph r is $y = f(x)$. ✓

The narrowest curve is $y = x^4$ so graph p is $y = f(2x)$. ✓

So graph q will be $y = 2f(x)$. ✓

17 a

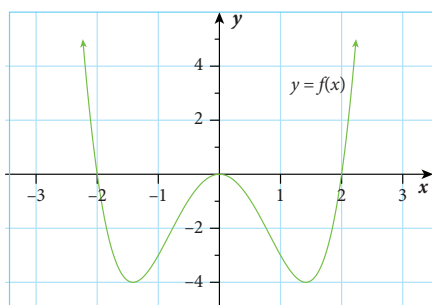


- b Three of the curves, $y = \ln x$, $y = \ln(x+3)$ and $y = \ln(3x)$, are the same shape and size. $y = \ln(x+3)$ is the curve $y = \ln x$ translated 3 units to the left. $\ln(3x) = \ln 3 + \ln x$ so $y = \ln(3x)$ is the curve $y = \ln x$ translated up $\ln 3$ units. $\ln x^3 = 3 \ln x$ so the graph of $y = \ln x^3$ has the same x -intercept as $y = \ln x$ but has been stretched. Each y -value is three times the corresponding y -value of $y = \ln x$.

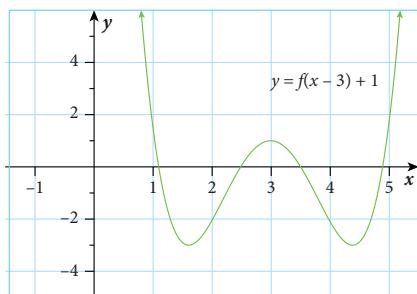
18 a

$$\begin{aligned} f(x) &= x^4 - 4x^2 \\ f(-x) &= (-x)^4 - 4(-x)^2 \\ &= x^4 - 4x^2 \\ &= f(x) \end{aligned}$$

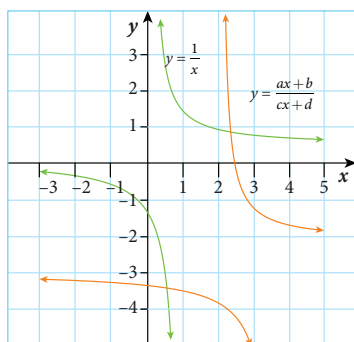
$\therefore f(x)$ is an even function.



b



19 a



b $x = 2$ and $y = -3$

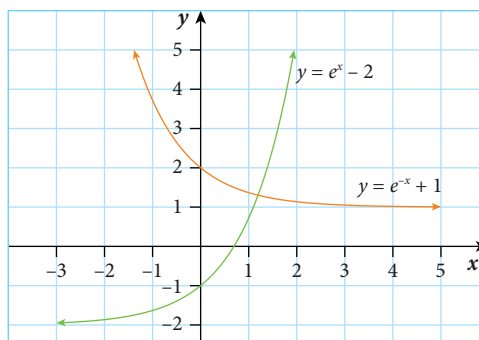
c Yes.

d Second curve:

$$\begin{aligned} y &= \frac{1}{x-2} - 3 \\ &= \frac{1-3(x-2)}{x-2} \\ &= \frac{-3x+7}{x-2} \end{aligned}$$

$\therefore a = -3, b = 7, c = 1$ and $d = -2$

20 a



b There is one point of intersection.

c At the point of intersection:

$$e^x - 2 = e^{-x} + 1$$

$$e^x - e^{-x} - 3 = 0$$

$$e^x(e^x - e^{-x} - 3) = 0$$

$$e^{2x} - 1 - 3e^x = 0$$

$$e^{2x} - 3e^x = 1$$

So solving that equation will give the x -value at the point of intersection.

d $e^{2x} - 3e^x = 1$

Let $u = e^x$

$$\text{So } u^2 - 3u - 1 = 0$$

$$\begin{aligned} u &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-(-3) \pm \sqrt{(-3)^2 - 4 \times 1 \times -1}}{2 \times 1} \\ &= \frac{3 \pm \sqrt{13}}{2} \end{aligned}$$

$$\therefore u = 3.30277563... \text{ or } u = -0.30277563...$$

But $e^x > 0$ for all real values of x .

$$\therefore e^x = 3.30277563...$$

$$\therefore x = 1.19476321...$$

$$x = 1.2 \text{ (1 d.p.)}$$

Now $y = e^x - 2$

$$= 1.30277563...$$

$$= 1.3 \text{ (1 d.p.)}$$

$\therefore$ the only point of intersection is $(1.2, 1.3)$, each value to one decimal place.

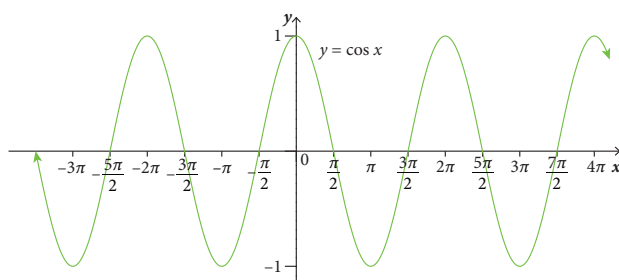
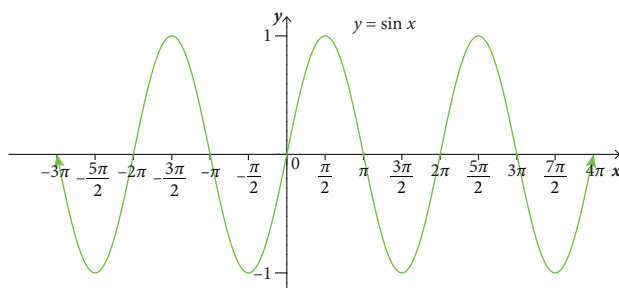
CHAPTER 2

TRIGONOMETRIC FUNCTIONS AND GRAPHS

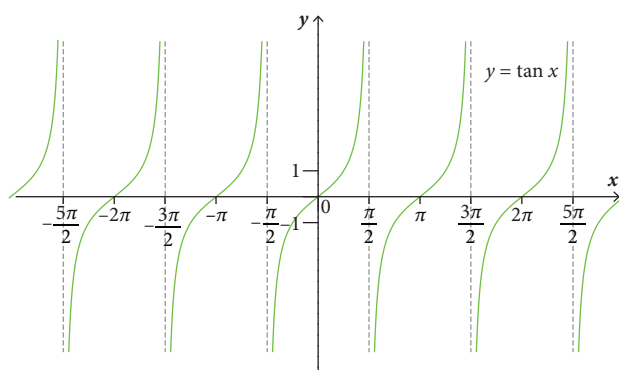
TOPIC
TRIGONOMETRIC
FUNCTIONS

1 Review of trig functions

- A radian is a measure of angle size. An angle measures one radian if the arc it subtends on a unit circle measures one unit in length. π radians = 180° .
- $y = \sin x$, $y = \cos x$ and $y = \tan x$ are functions of x (measured in radians).
- The graphs of the trig functions are periodic, meaning that they repeat after a certain interval.
- $y = \sin x$ and $y = \cos x$ have period 2π .



- $y = \tan x$ has period π . $\tan x = \frac{\sin x}{\cos x}$ so $\tan x$ is undefined whenever $\cos x = 0$.



EXAMPLE 1

Express 75° in radians in terms of π .

$$\begin{aligned} 75^\circ &= \frac{75\pi}{180} \\ &= \frac{5\pi}{12} \end{aligned}$$

It is important to be able to 'think in radians'. There should be no need to convert back and forth between degrees and radians unless specifically instructed.

EXAMPLE 2

Find $\cos \frac{2\pi}{5}$, giving the answer correct to four decimal places.

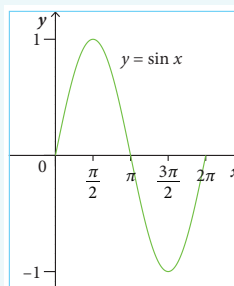
$$\begin{aligned} \cos \frac{2\pi}{5} &= 0.309016994\dots \\ &= 0.3090 \quad (4 \text{ d.p.}) \end{aligned}$$

The calculator must be in radians mode.

EXAMPLE 3

- Sketch $y = \sin x$ for $0 \leq x \leq 2\pi$.
- What is the range?

a



Remember: $\sin x$ and $\cos x$ always lie between (inclusively) -1 and 1 .

- Range: $-1 \leq y \leq 1$

→ PRACTICE

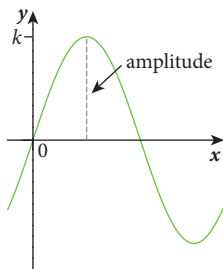
- Express 108° in radians in terms of π .
- Find $\sin \frac{3\pi}{8}$, giving the answer correct to four decimal places.

- 3 a Sketch $y = \cos x$ for $0 \leq x \leq 2\pi$.
 b What is the range?
-
- 4* a Sketch $y = \tan x$ for $0 \leq x \leq 2\pi$.
 b What is the range?
 c What is the significance of $y = 1$ to the graph?

Answers ➔ p. 46

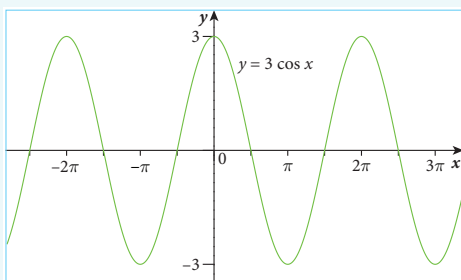
2 Amplitude and $y = kf(x)$

- ➔ If a trig function is multiplied by a constant the effect is to stretch the curve vertically. If the constant is negative the curve will be inverted.
- ➔ The amplitude is the distance from the central position to the maximum (or minimum) position.
- ➔ For $y = k \sin x$ and $y = k \cos x$ the amplitude is $|k|$.
- ➔ $y = k \tan x$ will pass through the point $\left(\frac{\pi}{4}, k\right)$.



EXAMPLE 1

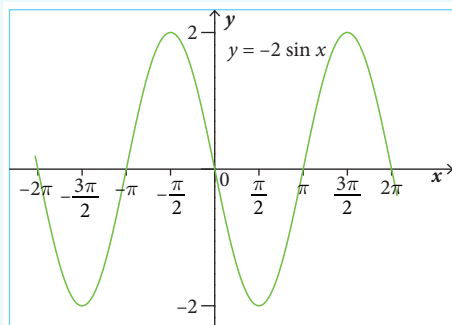
Sketch $y = 3 \cos x$.



The amplitude is 3. The range is $-3 \leq y \leq 3$.

EXAMPLE 2

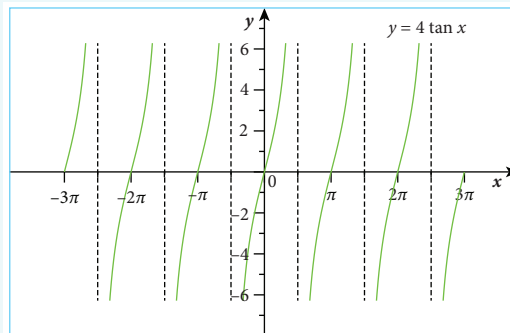
Sketch $y = -2 \sin x$ for $-2\pi \leq x \leq 2\pi$.



The negative has the effect of turning the curve upside down. The amplitude is always positive. It is 2.

EXAMPLE 3

Sketch $y = 4 \tan x$.



Because $\tan \frac{\pi}{4} = 1$ the curve will pass through the point $\left(\frac{\pi}{4}, 4\right)$.

➔ PRACTICE

- Sketch $y = 4 \sin x$.
- Sketch $y = -\cos x$ for $-2\pi \leq x \leq 2\pi$.
- Sketch $y = 2 \tan x$.
- * Sketch $y = \frac{1}{2} \sin x$ for $0 \leq x \leq \pi$.

Answers ➔ p. 46

3 Period and $y = f(ax)$

- ➔ All trigonometric functions are periodic, meaning they repeat after a certain interval.
- ➔ $\sin x$ and $\cos x$ have period 2π . $\tan x$ has period π .
- ➔ The value a in $y = k \sin ax$, $y = k \cos ax$ or $y = k \tan ax$ affects the period. $y = \sin ax$ and $y = \cos ax$ have period $\frac{2\pi}{a}$ and $y = \tan ax$ has period $\frac{\pi}{a}$.
- ➔ a is the frequency. It is the number of times the curve repeats in 2π for $\sin$ and $\cos$ and in π for $\tan$.

EXAMPLE 1

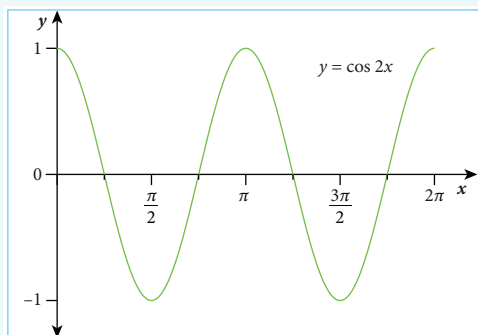
What is the period of the curve $y = -3 \sin \frac{x}{4}$?

$$\begin{aligned} \text{Period} &= \frac{2\pi}{\frac{1}{4}} \\ &= 8\pi \end{aligned}$$

The curve will just complete $\frac{1}{4}$ of its pattern in 2π .

EXAMPLE 2

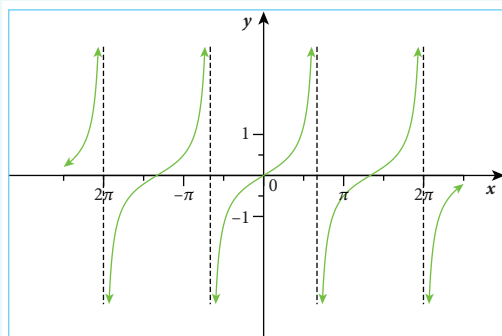
Sketch the curve $y = \cos 2x$ for $0 \leq x \leq 2\pi$.



Work out the period first.
 $\frac{2\pi}{2} = \pi$.

EXAMPLE 3

A computer application drew the graph of $y = \tan ax$.



What is the value of a ?

$$\begin{aligned} \text{Period} &= \frac{4\pi}{3} \\ \therefore \frac{\pi}{a} &= \frac{4\pi}{3} \\ 4a &= 3 \\ a &= \frac{3}{4} \end{aligned}$$

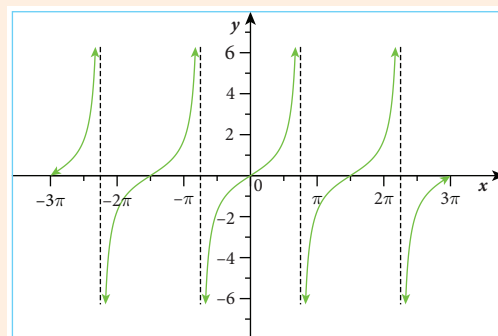
The graph repeats
every $\frac{4\pi}{3}$ units.

→ PRACTICE

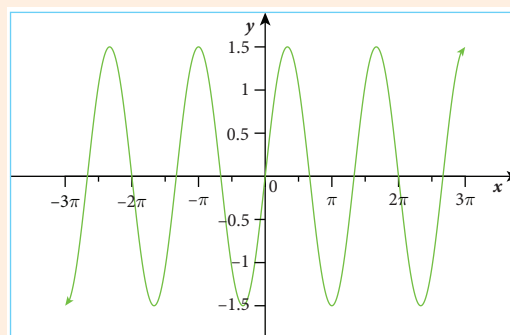
- What is the period of the curve $y = -2 \cos \frac{x}{3}$?
- Sketch the curve $y = \sin 4x$ for $0 \leq x \leq 2\pi$.

- A computer application drew the graph of $y = \tan ax$.

What is the value of a ?



- The graph of $y = k \sin ax$ is shown.



What are the values of k and a ?

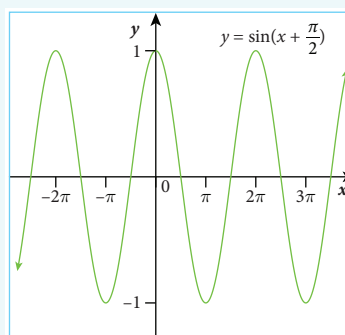
Answers ⇨ pp. 46–47

4 Phase and $y = f(x + b)$

- Adding a constant, b , in a trigonometric function (so $y = \sin a(x + b)$, $y = \cos a(x + b)$ or $y = \tan a(x + b)$) will move the curve left or right.
- The phase shift is how much the curve has moved from its original position along the x -axis.

EXAMPLE 1

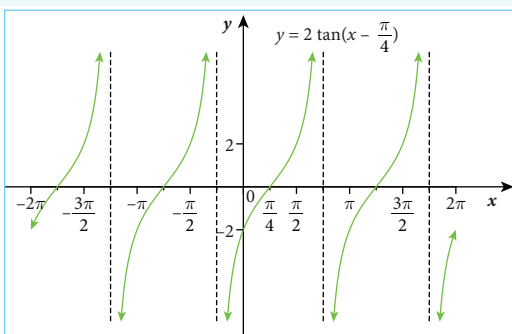
Sketch $y = \sin \left(x + \frac{\pi}{2} \right)$.



$y = \sin x$ has moved $\frac{\pi}{2}$ to the left. This is the same graph as $y = \cos x$. $y = \sin x$ and $y = \cos x$ are identical curves except for a phase shift.

EXAMPLE 2

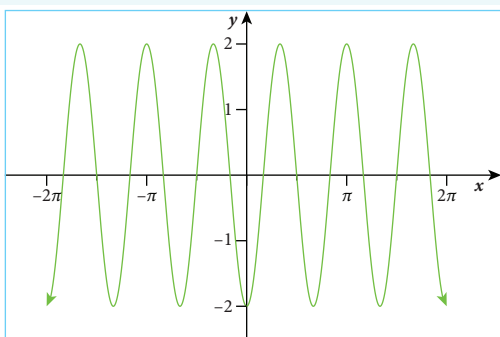
Sketch $y = 2 \tan \left(x - \frac{\pi}{4} \right)$ for $-2\pi \leq x \leq 2\pi$.



The curve, including asymptotes, moves $\frac{\pi}{4}$ units to the right.

EXAMPLE 3

A software application showed this graph of $y = k \cos(ax + b)$, $k > 0$, $0 \leq b \leq \pi$.



Find the values of k , a and b .

$$y = k \cos(ax + b)$$

The graph has amplitude 2, so $k = 2$. ($k > 0$)

The graph completes 3 cycles from 0 to 2π .

So $a = 3$.

When $x = 0$, $y = -2$

So $-2 = 2 \cos b$

$$\cos b = -1$$

$$b = \pi \quad (0 \leq b \leq \pi)$$

The phase shift is $\frac{\pi}{3}$ units to the left. $2 \cos(3x + \pi) = 2 \cos 3 \left(x + \frac{\pi}{3} \right)$.

EXAMPLE 4

Sketch $y = -3 \sin \left(2x - \frac{\pi}{6} \right)$.

Amplitude = 3

$$\text{Period} = \frac{2\pi}{2}$$

$$= \pi$$

$$y = -3 \sin \left(2x - \frac{\pi}{6} \right)$$

$$= -3 \sin 2 \left(x - \frac{\pi}{12} \right)$$

So the curve moves $\frac{\pi}{12}$ units to the right.

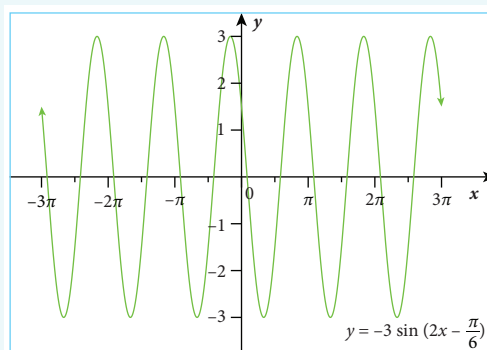
When $x = 0$,

$$y = -3 \sin \left(-\frac{\pi}{6} \right)$$

$$= -3 \times -\frac{1}{2}$$

$$= 1.5$$

The curve $y = 3 \sin 2x$ has been shifted $\frac{\pi}{12}$ to the right and inverted. This is sufficient information to sketch the curve; however, finding other values helps as a check.

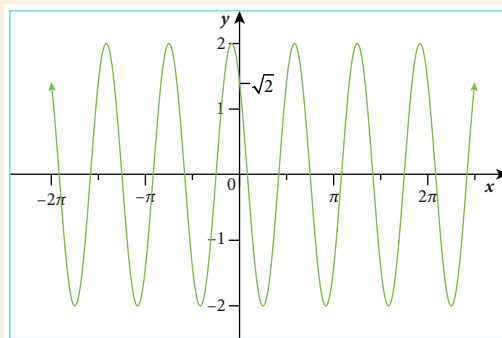


→ PRACTICE

1 Sketch $y = \cos \left(x - \frac{\pi}{2} \right)$.

2 Sketch $y = 4 \tan \left(x + \frac{\pi}{4} \right)$ for $-2\pi \leq x \leq 2\pi$.

3 A software application showed this graph of $y = k \cos(ax + b)$, $k > 0$, $0 \leq b \leq \frac{\pi}{2}$.



Find the values of k , a and b .

4 Sketch $y = -2 \sin \left(3x + \frac{\pi}{6} \right)$.

5* Sketch $y = 4 \cos \left(\frac{x}{2} - \frac{\pi}{3} \right)$ for $0 \leq x \leq 2\pi$.

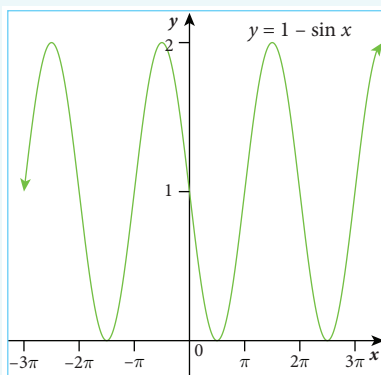
Answers $\Rightarrow$ p. 47

5 Vertical shift and $y = f(x) + c$

- Adding a constant, c , to the equation creates a vertical shift. The curve moves up, (or down if c is negative).
- The value c will not necessarily be the y -intercept, depending on the particular curve.
- For sine and cosine curves the range of values will change with the addition of a constant.

EXAMPLE 1

Sketch $y = 1 - \sin x$.



Think of the curve as $y = -\sin x + 1$. So invert $\sin x$ and move up one unit.

EXAMPLE 2

Consider the curve $y = 3 \cos \left(\frac{x + \pi}{2} \right) + 2$.

- a What is the amplitude?
- b What is the range?
- c What is the period?
- d Sketch the curve.

a $y = 3 \cos \left(\frac{x + \pi}{2} \right) + 2$

The amplitude is 3.

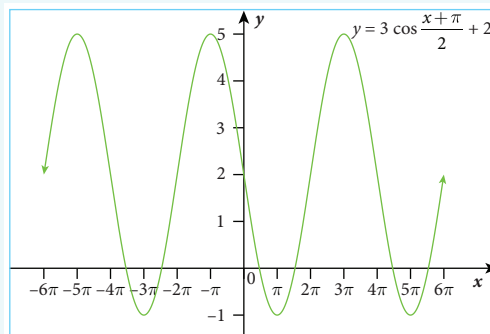
b Range: $-1 \leq y \leq 5$

c Period = $\frac{2\pi}{\frac{1}{2}}$
 $= 4\pi$

d $3 \cos \left(\frac{x + \pi}{2} \right) + 2 = 3 \cos \frac{1}{2}(x + \pi) + 2$

The range of $y = 3 \cos x$ is $-3 \leq y \leq 3$. If that curve is moved up 2 units then 2 must be added to both -3 and 3 .

The vertical shift is π units to the left.



→ PRACTICE

- 1 Sketch $y = 2 - \cos x$.
- 2 Consider the curve $y = 2 \sin \left(\frac{x + \pi}{4} \right) + 3$.
 - a What is the amplitude?
 - b What is the range?
 - c What is the period?
 - d Sketch the curve.
- 3* Sketch $y = 3 \sin (4x + 1) + 2$ for $0 \leq x \leq 6$.

Answers $\Rightarrow$ p. 48

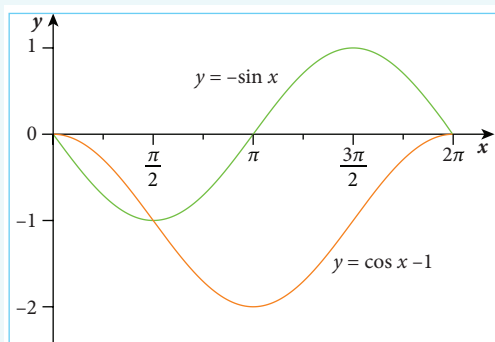
6 Solving equations graphically

- Equations involving the trigonometric functions can often be solved by sketching the graph of the function.
- The number of points of intersection within the specified domain will give the number of solutions to the equation.
- Solutions might be given in terms of π or as decimal approximations.

EXAMPLE 1

- a On the same diagram sketch $y = -\sin x$ and $y = \cos x - 1$ for $0 \leq x \leq 2\pi$.
- b For what values of x does $\sin x + \cos x = 1$, $0 \leq x \leq 2\pi$?

a



b $\sin x + \cos x = 1$

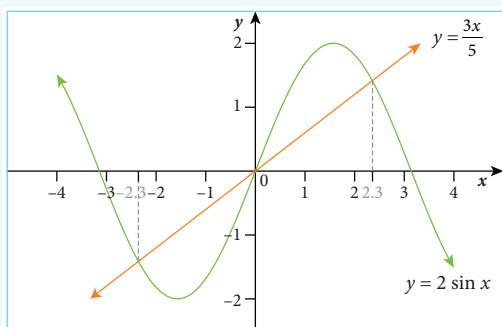
$\therefore \cos x - 1 = -\sin x$

From the graph this occurs when $x = 0, \frac{\pi}{2}$ and 2π .

A quick substitution into the equation can show that these values are correct.

EXAMPLE 2

On the same diagram sketch $y = 2 \sin x$ and $y = \frac{3x}{5}$ and hence write down the approximate values of x for which $2 \sin x = \frac{3x}{5}$.



From the diagram $2 \sin x = \frac{3x}{5}$ when $x = 0$ and $x \approx \pm 2.3$

π is about 3.14 so both numbers and values in terms of π can be shown on the x -axis at the same time.

EXAMPLE 3

a Show that $\frac{\pi}{6}$ is a solution of the equation

$\sin 2x = \cos x.$

b On the same diagram sketch $y = \sin 2x$ and $y = \cos x$ for $-2\pi \leq x \leq 2\pi$.

c Hence write down all solutions of $\sin 2x = \cos x, -2\pi \leq x \leq 2\pi$.

a When $x = \frac{\pi}{6}$,

$$\sin 2x = \sin \frac{2\pi}{6}$$

$$= \sin \frac{\pi}{3}$$

$$= \frac{\sqrt{3}}{2}$$

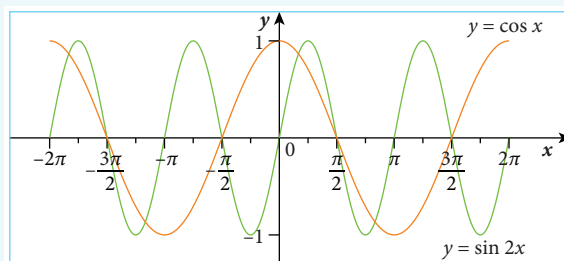
$$\cos x = \cos \frac{\pi}{6}$$

$$= \frac{\sqrt{3}}{2}$$

$$= \sin 2x$$

$\therefore \frac{\pi}{6}$ is a solution of $\cos 2x = \sin x$.

b



c From the diagram, $\sin 2x = \cos x$ when:

$$x = -\frac{11\pi}{6}, -\frac{3\pi}{2}, -\frac{7\pi}{6}, -\frac{\pi}{2}, \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6} \text{ and } \frac{3\pi}{2}.$$

Once it is established that $\frac{\pi}{6}$ is a solution we can see that $\pi - \frac{\pi}{6}, -\pi - \frac{\pi}{6}$ and $-2\pi + \frac{\pi}{6}$ are also solutions.

→ PRACTICE

1 a On the same diagram sketch $y = \cos x$ and $y = 1 - \cos x$ for $0 \leq x \leq 2\pi$.

b For what values of x does $2 \cos x = 1, 0 \leq x \leq 2\pi$?

2 On the same diagram sketch $y = \sin 2x$ and $y = \frac{3x}{4}$ and hence write down the approximate values of x for which $\sin 2x = \frac{3x}{4}$.

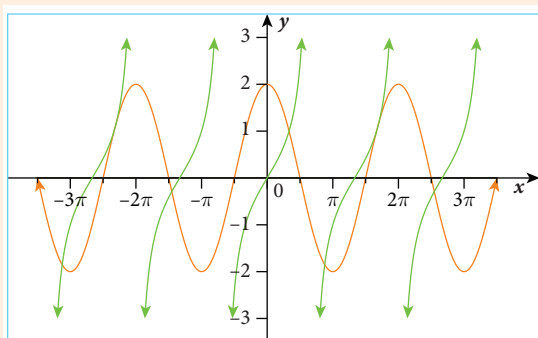
3 a Show that $\frac{\pi}{6}$ is a solution of the equation

$\cos 2x = \sin x.$

b On the same diagram sketch $y = \cos 2x$ and $y = \sin x$ for $-2\pi \leq x \leq 2\pi$.

c Hence write down all solutions of $\cos 2x = \sin x, -2\pi \leq x \leq 2\pi$.

- 4* A software application was used to graph $y = 2 \cos x$ and $y = \tan \frac{3x}{4}$ on the same diagram.



- How many solutions are there of $2 \cos x = \tan \frac{3x}{4}$ for $0 \leq x \leq \pi$?
- How many solutions are there of $2 \cos x = \tan \frac{3x}{4}$ for $-\pi \leq x \leq 0$?
- Lincoln said 'Both $y = \cos x$ and $y = \tan x$ are symmetrical curves so there should be the same number of solutions for $-\pi \leq x \leq 0$ as for $0 \leq x \leq \pi$.' Briefly explain why Lincoln's reasoning is wrong.
- Show that $\frac{\pi}{3}$ is a solution of the equation $2 \cos x = \tan \frac{3x}{4}$.
- Solve $2 \cos x \geq \tan \frac{3x}{4}$ for $-\pi \leq x \leq \frac{\pi}{2}$.

Answers ➞ pp. 48–49

7 Solving trigonometric equations algebraically

- ➔ Trigonometric equations can be solved algebraically as well as graphically.
- ➔ The solution should be found over the required domain, which will also determine whether the solution should be found in radians, in terms of π or as a decimal, or degrees.
- ➔ The exact solutions should be given wherever possible.

EXAMPLE 1

Solve $\tan x = 3$ for $0 \leq x \leq 2\pi$, giving the answers to two decimal places.

$$\tan x = 3$$

Now $\tan^{-1} 3 = 1.249\,045\,77\dots$

$\tan$ is positive in the first and third quadrants.

$$\therefore x = 1.249\,045\,77\dots \text{ or } \pi + 1.249\,045\,77\dots$$

$$x = 1.249\,045\,77\dots \text{ or } x = 4.390\,638\,42\dots$$

$$x = 1.25 \text{ or } x = 4.39 \quad (2 \text{ d.p.})$$

Because the required domain is between 0 and 2π we know that the answer must be in radians.

EXAMPLE 2

Solve $4 \cos^2 \theta - 3 = 0$ for $0 \leq \theta \leq 2\pi$.

$$4 \cos^2 \theta - 3 = 0$$

$$4 \cos^2 \theta = 3$$

$$\cos^2 \theta = \frac{3}{4}$$

$$\cos \theta = \pm \sqrt{\frac{3}{4}}$$

$$= \pm \frac{\sqrt{3}}{2}$$

Because both the positive and negative square roots satisfy the equation the solution can lie in any of the four quadrants.

$$\theta = \frac{\pi}{6}, \pi - \frac{\pi}{6}, \pi + \frac{\pi}{6} \text{ or } 2\pi - \frac{\pi}{6}$$

$$= \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6} \text{ or } \frac{11\pi}{6}$$

EXAMPLE 3

Explain why any solution to the equation $\frac{x}{2} = 1 - 3 \cos x$ must lie between $x = -4$ and $x = 8$.

$$\frac{x}{2} = 1 - 3 \cos x$$

$$\therefore 3 \cos x = 1 - \frac{x}{2}$$

$$\text{Now } -1 \leq \cos x \leq 1$$

$$\therefore -3 \leq 3 \cos x \leq 3$$

$$\text{So } -3 \leq 1 - \frac{x}{2} \leq 3$$

$$-4 \leq -\frac{x}{2} \leq 2$$

$$8 \geq x \geq -4$$

The inequality signs are reversed because we multiply by a negative number.

So any solution to $\frac{x}{2} = 1 - 3 \cos x$ lies between $x = -4$ and $x = 8$.

EXAMPLE 4

Solve $2 \sin \left(3x - \frac{\pi}{6} \right) = \sqrt{3}$ for $0 \leq x \leq \pi$.

$$0 \leq x \leq \pi$$

$$0 \leq 3x \leq 3\pi$$

$$-\frac{\pi}{6} \leq 3x - \frac{\pi}{6} \leq \frac{17\pi}{6}$$

$$2 \sin \left(3x - \frac{\pi}{6} \right) = \sqrt{3}$$

$$\sin \left(3x - \frac{\pi}{6} \right) = \frac{\sqrt{3}}{2}$$

$$\text{Now } \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$\sin$ is positive in the first and second quadrants

$$\text{So } 3x - \frac{\pi}{6} = \frac{\pi}{3} \text{ or } \pi - \frac{\pi}{3} \text{ or } 2\pi + \frac{\pi}{3} \text{ or } 3\pi - \frac{\pi}{3}$$

$$3x - \frac{\pi}{6} = \frac{\pi}{3} \text{ or } \frac{2\pi}{3} \text{ or } \frac{7\pi}{3} \text{ or } \frac{8\pi}{3}$$

$$3x = \frac{\pi}{2} \text{ or } \frac{5\pi}{6} \text{ or } \frac{5\pi}{2} \text{ or } \frac{17\pi}{6}$$

$$x = \frac{\pi}{6} \text{ or } \frac{5\pi}{18} \text{ or } \frac{5\pi}{6} \text{ or } \frac{17\pi}{18}$$

Take care to find all the values over the domain.

→ PRACTICE

1 Solve $\tan x = 5$ for $0 \leq x \leq 2\pi$, giving the answers to three decimal places.

2 Solve $2 \sin^2 \theta - 1 = 0$ for $0 \leq \theta \leq 2\pi$.

3 Explain why any solution to the equation $3x = 2 - 5 \sin x$ must lie between $x = -1$ and $x = 2\frac{1}{3}$.

4 Solve $10 \sin \left(2t + \frac{\pi}{4} \right) - 5 = 0$ for $0 \leq t \leq 2\pi$.

5* Solve $2 \cos \left(3x + \frac{\pi}{2} \right) = 1$ for $0 \leq x \leq \frac{\pi}{2}$.

Answers → p. 49

8 Further trigonometric equations and expressions

→ Trigonometric equations might involve the reciprocal

functions: $\operatorname{cosec} x = \frac{1}{\sin x}$, $\sec x = \frac{1}{\cos x}$ and $\cot x = \frac{1}{\tan x}$.

→ $\tan x = \frac{\sin x}{\cos x}$ and $\cot x = \frac{\cos x}{\sin x}$.

→ The Pythagorean identities: $\sin^2 x + \cos^2 x = 1$;
 $1 + \cot^2 x = \operatorname{cosec}^2 x$ and $\tan^2 x + 1 = \sec^2 x$ can be used to simplify expressions.

→ Quadratic equations involving trigonometric functions can be simplified by substituting u (or some other variable) for the trig function.

EXAMPLE 1

Solve $2 \cos^2 x + \cos x = 1$ for $0 \leq x \leq 2\pi$.

$$2 \cos^2 x + \cos x = 1$$

Let $u = \cos x$

$$\therefore 2u^2 + u - 1 = 0$$

$$(2u - 1)(u + 1) = 0$$

$$2u - 1 = 0 \text{ or } u + 1 = 0$$

$$u = \frac{1}{2} \text{ or } u = -1$$

$$\text{So } \cos x = \frac{1}{2} \text{ or } \cos x = -1$$

$$x = \frac{\pi}{3} \text{ or } 2\pi - \frac{\pi}{3} \text{ or } x = \pi$$

$$\therefore x = \frac{\pi}{3}, \pi \text{ or } \frac{5\pi}{3}$$

It is not necessary to use u (or another variable) but it is often less cumbersome to do so.

EXAMPLE 2

Solve $\frac{1}{\sec \theta - \tan \theta} - \frac{1}{\sec \theta + \tan \theta} = 2\sqrt{3}$ for $0 \leq \theta \leq 2\pi$.

$$\begin{aligned} & \frac{1}{\sec \theta - \tan \theta} - \frac{1}{\sec \theta + \tan \theta} \\ &= \frac{\sec \theta + \tan \theta - (\sec \theta - \tan \theta)}{(\sec \theta - \tan \theta)(\sec \theta + \tan \theta)} \\ &= \frac{\sec \theta + \tan \theta - \sec \theta + \tan \theta}{\sec^2 \theta - \tan^2 \theta} \\ &= \frac{2 \tan \theta}{1} \end{aligned}$$

$$= 2 \tan \theta$$

$$\text{So } 2 \tan \theta = 2\sqrt{3}$$

$$\tan \theta = \sqrt{3}$$

$$\text{Now } \tan \frac{\pi}{3} = \sqrt{3}$$

$$\theta = \frac{\pi}{3} \text{ or } \pi + \frac{\pi}{3}$$

$$\therefore \theta = \frac{\pi}{3} \text{ or } \theta = \frac{4\pi}{3}$$

There is no need to remember all the Pythagorean identities because as long as you know that $\sin^2 x + \cos^2 x = 1$ the other two can be found by simply dividing each part by either $\sin^2 x$ or $\cos^2 x$.

EXAMPLE 3

Find all values of θ ($0 \leq \theta \leq 2\pi$) for which $\cos^2 \theta = \sec \theta$.

$$\begin{aligned}\cos^2\theta &= \sec\theta \\ &= \frac{1}{\cos\theta} \\ \cos^3\theta &= 1 \\ \cos\theta &= 1 \\ \theta &= 0 \text{ or } 2\pi\end{aligned}$$

Picturing a trig curve in your mind often helps when solving simple equations.

EXAMPLE 4

a Show that $\cot x \operatorname{cosec} x + \operatorname{cosec}^2 x = \frac{1}{1 - \cos x}$.

b Hence solve $\cot x \operatorname{cosec} x + \operatorname{cosec}^2 x - 2 = 0$ for $0 \leq x \leq \frac{\pi}{2}$.

a $\cot x \operatorname{cosec} x + \operatorname{cosec}^2 x$

$$\begin{aligned}&= \frac{\cos x}{\sin x} \times \frac{1}{\sin x} + \frac{1}{\sin^2 x} \\ &= \frac{\cos x}{\sin^2 x} + \frac{1}{\sin^2 x} \\ &= \frac{\cos x + 1}{1 - \cos^2 x} \\ &= \frac{1 + \cos x}{(1 - \cos x)(1 + \cos x)} \\ &= \frac{1}{1 - \cos x}\end{aligned}$$

b $\cot x \operatorname{cosec} x + \operatorname{cosec}^2 x - 2 = 0$

$$\frac{1}{1 - \cos x} - 2 = 0$$

$$\frac{1}{1 - \cos x} = 2$$

$$1 - \cos x = \frac{1}{2}$$

$$\cos x = \frac{1}{2}$$

$$\text{Now } 0 \leq x \leq \frac{\pi}{2},$$

$$\therefore x = \frac{\pi}{3}$$

Because the question says 'hence' we know to start by using the result from part a.

EXAMPLE 5

Solve $3 \tan x = 2 \cos x$ for $0 \leq x \leq 2\pi$.

$$3 \tan x = 2 \cos x$$

$$\frac{3 \sin x}{\cos x} = 2 \cos x$$

$$3 \sin x = 2 \cos^2 x$$

$$3 \sin x = 2(1 - \sin^2 x)$$

$$3 \sin x = 2 - 2 \sin^2 x$$

$$\begin{aligned}2 \sin^2 x + 3 \sin x - 2 &= 0 \\ (2 \sin x - 1)(\sin x + 2) &= 0 \\ \text{So } 2 \sin x &= 1 \text{ or } \sin x = -2\end{aligned}$$

$$\sin x = \frac{1}{2} \quad \text{No solution}$$

$$x = \frac{\pi}{6} \text{ or } \pi - \frac{\pi}{6}$$

$$\therefore x = \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$$

Remember $-1 \leq \sin x \leq 1$ for all values of x .

→ PRACTICE

1 Solve $2 \sin^2 x - \sin x = 1$ for $0 \leq x \leq 2\pi$.

2 Solve $\frac{1}{\operatorname{cosec}\theta - \cot\theta} - \frac{1}{\operatorname{cosec}\theta + \cot\theta} = 2$ for $0 \leq \theta \leq 2\pi$.

3 Find all values of θ ($0 \leq \theta \leq 2\pi$) for which $\sin^2\theta = \operatorname{cosec}\theta$.

4 a Show that $\sec x \tan x + \sec^2 x = \frac{1}{1 - \sin x}$.

b Hence solve $\sec x \tan x + \sec^2 x = \frac{\sqrt{2}}{\sqrt{2} - 1}$ for $0 \leq x \leq \frac{\pi}{2}$.

5 Solve $2 \sin x = -3 \cot x$ for $0 \leq x \leq 2\pi$.

6* Solve $2 \cos^2(3x) - 3 \cos(3x) + 1 = 0$ for $0 \leq x \leq \pi$.

Answers → p. 50

9 Problem solving

→ Trig functions involving expressions of the form $k f(a(x + b)) + c$ can be used to model practical phenomena of a periodic nature.

→ When a particle oscillates about a central position its displacement can be modelled by a function such as $y = k \sin(a(x + b)) + c$ or $y = k \cos(a(x + b)) + c$. Some examples are the swinging of a pendulum or the bobbing of a cork in water.

→ Equations or problems may arise from finding the length of an arc ($l = r\theta$) or the area of a sector $\left(A = \frac{1}{2}r^2\theta\right)$ of a circle.

EXAMPLE 1

The displacement from the origin, x cm at time t seconds, of a particle is given by $x = 4 \sin 2t$.

- a What is the amplitude?
- b What is the period?
- c Briefly describe the motion of the particle.

a Amplitude = 4 cm

b Frequency = 2

$$\begin{aligned}\text{Period} &= \frac{2\pi}{2} \\ &= \pi\end{aligned}$$

c When $t = 0$, $x = 0$.

So the particle is initially at the origin.

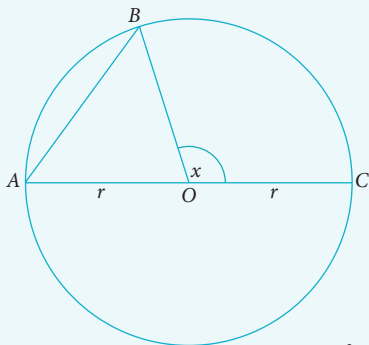
When $t = \frac{\pi}{4}$, $x = 4$.

After $\frac{\pi}{4}$ seconds the particle will be 4 cm to the right of the origin. It will then return, passing through the origin after $\frac{\pi}{2}$ seconds and reaching a position 4 cm to the left of the origin after $\frac{3\pi}{4}$ seconds. It will turn again reaching the origin after π seconds, completing its first oscillation and will continue on following the same path over and over again.

Positive values of x are to the right of the origin and negative values to the left. The particle is moving backwards and forwards in a straight line. It is not tracing out a sine curve.

EXAMPLE 2

O is the centre of a circle of radius r units. AC is a diameter of the circle and B is a point on the circumference such that $\angle BOC = x$ radians.

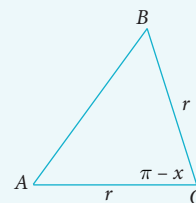


- a Show that the area of triangle ABO is $\frac{r^2 \sin x}{2}$ units².
- b If the area of sector BOC is twice that of $\triangle ABO$ show that $\sin x = \frac{x}{2}$.

- c By graphing $y = \sin x$ and $y = \frac{x}{2}$ on the same diagram, find an approximate value of x .

- a $OB = r$ units (radius of circle)
 $\angle AOB + \angle BOC = \pi$ (angles in st. line)
 $\therefore \angle AOB = (\pi - x)$ radians

$$\begin{aligned}\text{Area} &= \frac{1}{2} ab \sin C \\ &= \frac{1}{2} \times r \times r \times \sin(\pi - x) \\ &= \frac{1}{2} r^2 \sin(\pi - x)\end{aligned}$$



But $\sin(\pi - x) = \sin x$

So the area is $\frac{r^2 \sin x}{2}$ units²

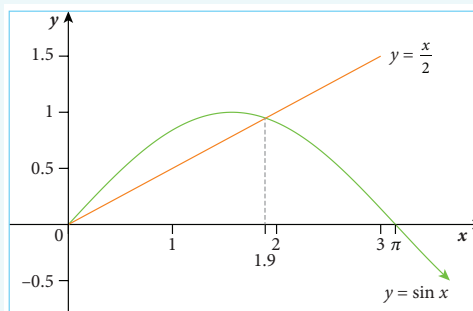
- b Area of sector = $\frac{1}{2} r^2 x$ units²

$$\text{So } 2 \left(\frac{r^2 \sin x}{2} \right) = \frac{1}{2} r^2 x$$

$$r^2 \sin x = \frac{r^2 x}{2}$$

$$\sin x = \frac{x}{2}$$

- c From the graph $x \approx 1.9$ radians.



So an angle of about 109° will mean that the area of the sector is twice that of the triangle.

EXAMPLE 3

On a certain day the depth of water at high tide, which is at 4:45 am, at the entrance to a particular bay is 2.35 m while low tide is at 11 am and the depth is 0.65 m.

- a Assuming that the action of the tide can be modelled by the equation $x = k \cos(at + b) + c$, where x is the depth of water in metres and t the time in hours after 4:45 am, show that

$$x = 0.85 \cos \frac{4\pi t}{25} + 1.5.$$

- b** A boat needs a depth of 2 m to cross the entrance to the bay. What is the latest time before 11 am that the boat can safely enter the bay?

- a** The tide oscillates between 2.35 m and 0.65 m.

$$\text{So } c = \frac{0.65 + 2.35}{2}$$

$$= 1.5$$

$$k = 2.35 - 1.5$$

$$= 0.85$$

The time between high and low tides is $6\frac{1}{4}$ hours.

$$\text{Period} = 2 \times 6.25 \text{ hours}$$

$$= 12.5 \text{ hours}$$

$$\text{So } \frac{2\pi}{a} = 12.5$$

$$a = \frac{2\pi}{12.5}$$

$$= \frac{4\pi}{25}$$

Tides can be quite accurately modelled with trig functions.

$$\text{Now } x = k \cos(at + b) + c$$

$$= 0.85 \cos\left(\frac{4\pi t}{25} + b\right) + 1.5$$

When it is high tide at 4:45 am, $t = 0$.

$$\text{So when } t = 0, \quad x = 2.35$$

$$2.35 = 0.85 \cos b + 1.5$$

$$0.85 = 0.85 \cos b$$

$$\cos b = 1$$

$$b = 0 \quad \left(0 \leq b \leq \frac{\pi}{2}\right)$$

$$\therefore x = 0.85 \cos \frac{4\pi t}{25} + 1.5$$

- b** When $x = 2$,

$$0.85 \cos \frac{4\pi t}{25} + 1.5 = 2$$

$$0.85 \cos \frac{4\pi t}{25} = 0.5$$

$$\cos \frac{4\pi t}{25} = \frac{0.5}{0.85}$$

$$\frac{4\pi t}{25} = \cos^{-1}\left(\frac{0.5}{0.85}\right)$$

$$t = \left(\cos^{-1}\left(\frac{0.5}{0.85}\right)\right) \div \frac{4\pi}{25}$$

$$\text{Time} = 1.87389308... \text{ h}$$

$$= 1 \text{ h } 52 \text{ min (nearest min)}$$

Now 1 h 52 min after 4:45 am is 6:37 am.

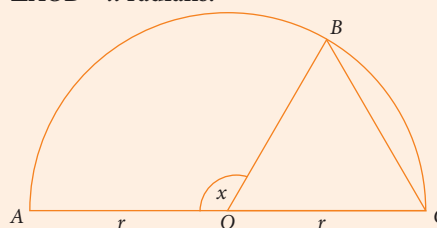
The latest time the boat can safely enter the bay before 11 am is 6:37 am.

→ PRACTICE

- 1** The displacement from the origin, x cm at time t seconds, of a particle is given by $x = 7 \cos 4t$.

- a** What is the amplitude?
b What is the period?
c Briefly describe the motion of the particle.

- 2** The diagram shows a semicircle of radius r units. B is a point on the circumference such that $\angle AOB = x$ radians.



- a** Show that the area of triangle OBC is

$$\frac{1}{2} r^2 \sin x \text{ units}^2.$$

- b** If the area of sector AOB is three times that of $\triangle OBC$ show that $3 \sin x = x$.

- c** By graphing $y = 3 \sin x$ and $y = x$ on the same diagram, find an approximate value of x .

- 3** On a certain day the depth of water at high tide, which is at 9:36 am, at the entrance to a particular river is 1.35 m while low tide is at 3:21 pm and the depth is 0.45 m.

- a** Assuming that the action of the tide can be modelled by the equation $x = k \cos(at + b) + c$, where x is the depth of water in metres and t the time in hours after 9:36 am, show that

$$x = 0.45 \cos \frac{4\pi t}{23} + 0.9.$$

- b** A boat needs a depth of 1.2 m to cross the entrance to the river. What is the last time before 3 pm that the boat can safely enter the bay?

- 4*** A particle is moving along the x -axis. It moves between positions $x = -4$ and $x = 4$ in 12.5 seconds. Assuming that the equation is of the form

$$x = k \sin(at + b) \text{ where } 0 \leq b \leq \frac{\pi}{2}, \text{ find the equation}$$

of the displacement at time t seconds if $x = 2$ when $t = 0$.

Answers → pp. 50–51

HSC EXAM-TYPE QUESTIONS

Total marks: 60

PART A

(Suggested time: 15 minutes)

(1 mark each)

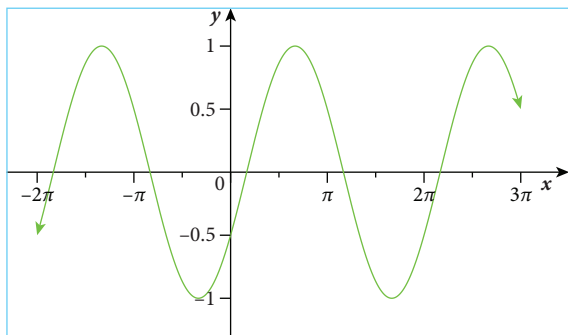
1 What is the amplitude of $y = \frac{1}{2} \cos \left(3x + \frac{\pi}{6} \right)$?

- A $\frac{1}{2}$ B 1 C 2 D 3

2 What is the period of the curve $y = \tan \pi x$?

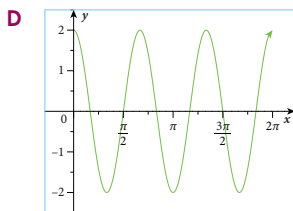
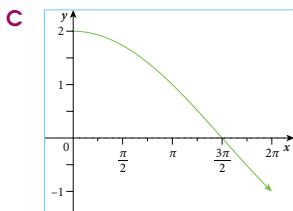
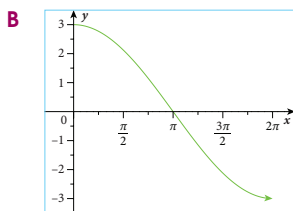
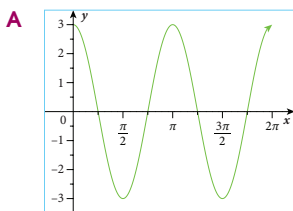
- A 2π B π C 2 D 1

3 What could be the equation of this curve?



- A $y = \sin \left(x + \frac{\pi}{6} \right)$ B $y = \sin \left(x - \frac{\pi}{6} \right)$
C $y = \cos \left(x + \frac{\pi}{6} \right)$ D $y = \cos \left(x - \frac{\pi}{6} \right)$

4 Which is the graph of $y = 3 \cos 2x$ ($0 \leq x \leq 2\pi$)?



5 Which is NOT correct?

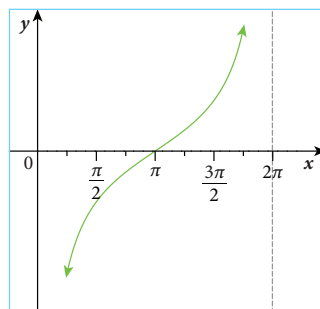
- A $\tan x = \frac{\sin x}{\cos x}$ B $\sin^2 x + \cos^2 x = 1$
C $\tan^2 x - \sec^2 x = 1$ D $\operatorname{cosec}^2 x - \cot^2 x = 1$

6 What is the period of $y = 3 - 2 \sin \left(4x - \frac{\pi}{3} \right)$?

- A $\frac{\pi}{4}$ B $\frac{\pi}{2}$ C 2π D 4π

7 What could be the equation of this curve?

- A $y = 2 \tan \left(\frac{x}{2} - \frac{\pi}{2} \right)$
B $y = 3 \tan \left(\frac{x}{2} - \pi \right)$
C $y = 4 \tan (2x - \pi)$
D $y = 5 \tan (2x + \pi)$



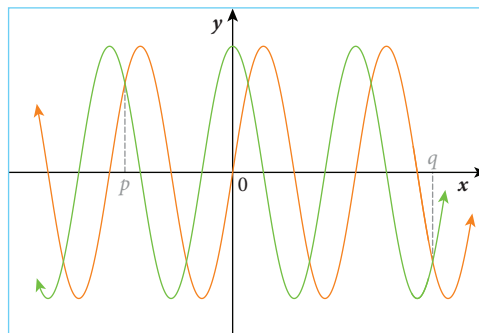
8 Which is NOT a solution to $\sqrt{2} \cos x + 1 = 0$?

- A $x = \frac{3\pi}{4}$ B $x = \frac{5\pi}{4}$ C $x = \frac{9\pi}{4}$ D $x = \frac{11\pi}{4}$

9 What is the range of $y = 7 - 4 \sin \frac{x}{2}$?

- A $-4 \leq x \leq 4$ B $-1 \leq x \leq 9$
C $5 \leq x \leq 9$ D $3 \leq x \leq 11$

10 The graph didn't get labelled but shows $y = \sin x$ and $y = \cos x$.



What are the values of p and q ?

- A $p = -\frac{5\pi}{4}$ and $q = \frac{13\pi}{4}$ B $p = -\frac{7\pi}{4}$ and $q = \frac{13\pi}{4}$
C $p = -\frac{5\pi}{4}$ and $q = \frac{21\pi}{4}$ D $p = -\frac{7\pi}{4}$ and $q = \frac{21\pi}{4}$

PART B

(Suggested time: 90 minutes)

Show all working.

11 Find a value of x for which $\tan\left(2x - \frac{\pi}{6}\right) = \sqrt{3}$. (2 marks)

12 A particle is moving about the origin, O . The displacement from O , x metres at time t seconds ($t \geq 0$) is given by $x = 4 \cos\left(2t + \frac{\pi}{3}\right)$.

- a What is the initial position of the particle? (1 mark)
- b When is the particle first at the origin? (2 marks)

13 Consider $y = 3 \sin\left(2x + \frac{\pi}{4}\right) + 1$.

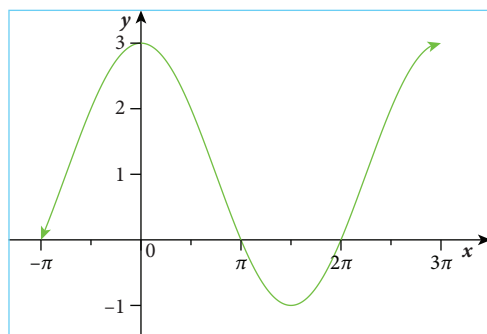
- a What is the amplitude? (1 mark)
- b What is the period? (1 mark)
- c What is the range? (1 mark)
- d What is the phase shift? (2 marks)
- e Sketch the curve. (2 marks)

14 a On the same diagram sketch $y = \sin 2x$ and $y = 2 \cos x$ for $0 \leq x \leq 2\pi$. (3 marks)

b Write down the solutions of $2 \cos x - \sin 2x = 0$ for $0 \leq x \leq 2\pi$. (1 mark)

c How many solutions will $2 \cos x - \sin 2x = 0$ have between $x = -4\pi$ and $x = 4\pi$? Justify your answer. (2 marks)

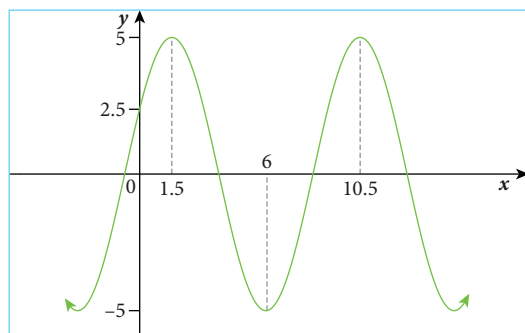
15 a Determine the equation of the cosine curve shown in the diagram. (3 marks)



b If the curve was a sine curve find its equation. (2 marks)

16 Solve $\sin^2 x = \sin x \cos x$ for $0 \leq x \leq 2\pi$. (3 marks)

17 The graph shows $y = k \sin(ax + b)$.



a What is the value of k ? (1 mark)

b Given $0 \leq b \leq \frac{\pi}{2}$, find b in terms of π . (1 mark)

c Find a in terms of π . (1 mark)

18 Find all values of x ($0 \leq x \leq \pi$) for which $\sqrt{2} \cos\left(2x - \frac{\pi}{3}\right) = 1$. (3 marks)

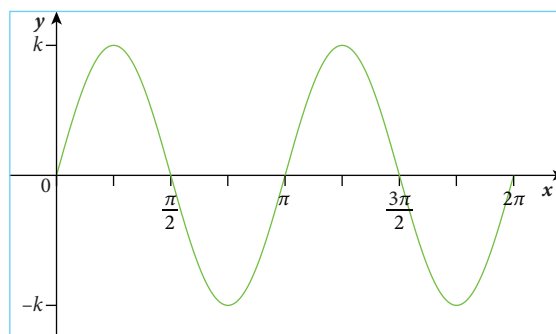
19 a Show that $\tan x = 2 \sin x$ when $x = \frac{\pi}{3}$. (1 mark)

b Sketch $y = \tan x$ and $y = 2 \sin x$ on the same diagram for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. (3 marks)

c Write down all solutions of $\tan x = 2 \sin x$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. (1 mark)

d When is $\tan x \geq 2 \sin x$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$? (3 marks)

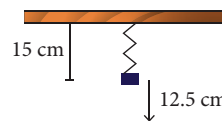
20 The graph shows $y = k \sin ax$ for $0 \leq x \leq 2\pi$.



Copy and show $y = k \sin\left(ax - \frac{\pi}{2}\right)$ on the same diagram, clearly showing any intercepts on the x -axis. (3 marks)

21 Solve $\sin^2 x + 2 \cos x + 2 = 0$ for $-2\pi \leq x \leq 2\pi$. (3 marks)

22 A weight is attached to the end of a spring that is hanging from a beam. At rest, the centre of the weight is 15 cm from the beam. The weight is pulled down 12.5 cm and released and oscillates about its rest position with period π .



Assume that the motion can be modelled by the equation $x = k \cos(at + b) + c$, $0 \leq b \leq \frac{\pi}{2}$, where x cm is the distance from the beam at time t seconds. Find the values of k , a , b and c . (4 marks)

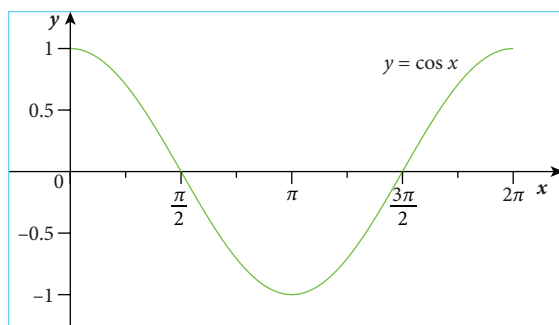
PRACTICE

1 Review of trig functions ⇨ pp. 33–34

1 $108^\circ = \frac{108\pi}{180} = \frac{3\pi}{5}$

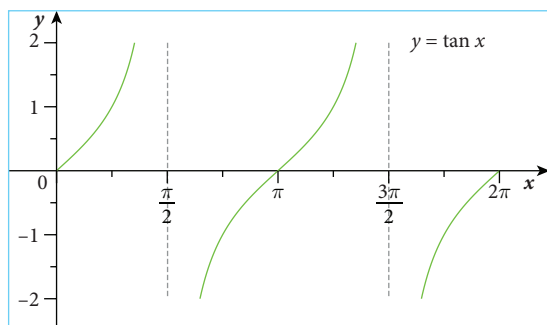
2 $\sin \frac{3\pi}{8} = 0.923\,879\,532\dots$
 $= 0.9239$ (4 d.p.)

3 a



b Range: $-1 \leq y \leq 1$

4 a

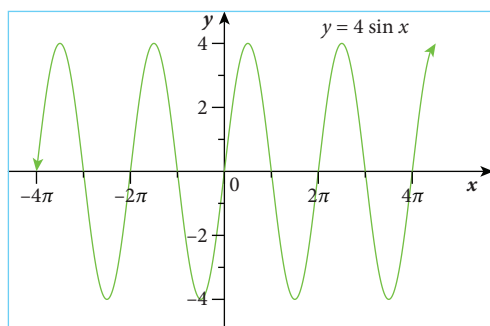


b Range: all real y

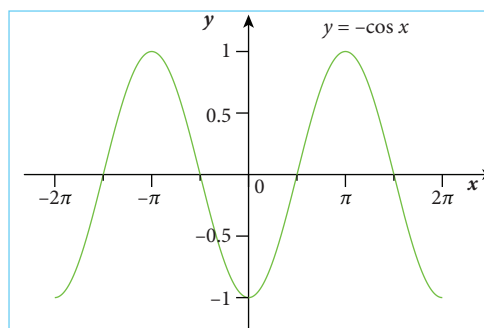
c $y = 1$ is the value when $x = \frac{\pi}{4}$. So it is the value of the curve for each x -value halfway between each value when $y = 0$ and the following asymptote.

2 Amplitude and $y = kf(x)$ ⇨ p. 34

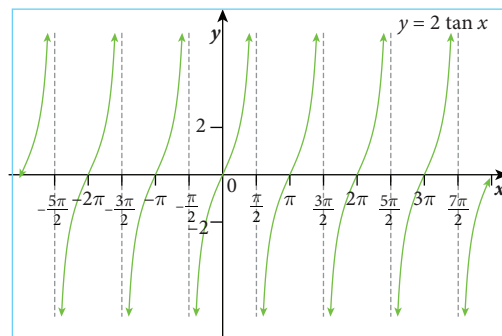
1



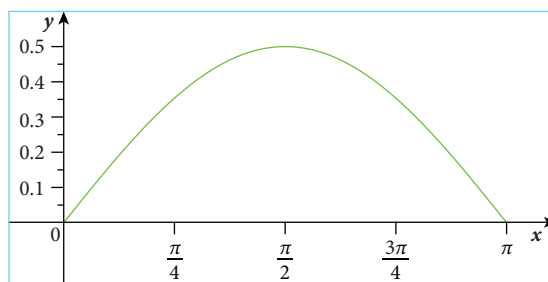
2



3



4



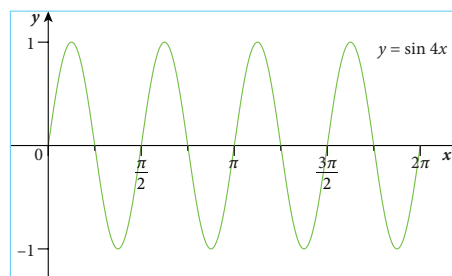
3 Period and $y = f(ax)$ ⇨ p. 35

1 $y = -2 \cos \frac{x}{3}$

Period = $\frac{2\pi}{\frac{1}{3}} = 6\pi$

2 $y = \sin 4x$ for $0 \leq x \leq 2\pi$.

Period = $\frac{2\pi}{4} = \frac{\pi}{2}$



3 $y = \tan ax$

Period = $\frac{3\pi}{2}$

$\therefore \frac{\pi}{a} = \frac{3\pi}{2}$

$3a = 2$

$a = \frac{2}{3}$

4 $y = k \sin ax$

Amplitude = 1.5

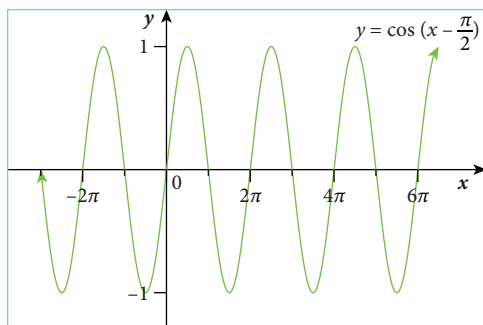
So $k = 1.5$

The curve completes 1.5 oscillations in 2π .

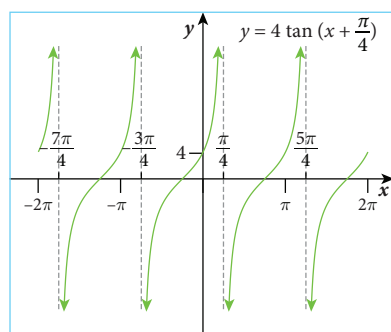
So $a = 1.5$ or $\frac{3}{2}$.

4 Phase and $y = f(x + b) \Rightarrow$ pp. 36–37

1



2



3 $y = k \cos(ax + b)$

The graph has amplitude 2, so $k = 2$. ($k > 0$)

The graph completes 3 cycles from 0 to 2π .

So $a = 3$.

When $x = 0$, $y = \sqrt{2}$

So $\sqrt{2} = 2 \cos b$

$\cos b = \frac{\sqrt{2}}{2}$

$= \frac{1}{\sqrt{2}}$

$b = \frac{\pi}{4} \quad \left(0 \leq b \leq \frac{\pi}{2}\right)$

So $k = 2$, $a = 3$ and $b = \frac{\pi}{4}$.

4 $y = -2 \sin\left(3x + \frac{\pi}{6}\right)$

Amplitude = 2

Period = $\frac{2\pi}{3}$

$y = -2 \sin\left(3x + \frac{\pi}{6}\right)$

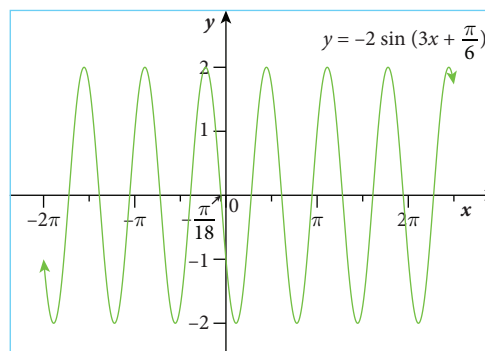
$= -2 \sin 3\left(x + \frac{\pi}{18}\right)$

So the curve moves $\frac{\pi}{18}$ units to the left.

When $x = 0$,

$y = -2 \sin\left(\frac{\pi}{6}\right)$

$= -1$



5 $y = 4 \cos\left(\frac{x}{2} - \frac{\pi}{3}\right)$, $0 \leq x \leq 2\pi$.

Amplitude = 4

Period = $\frac{2\pi}{\frac{1}{2}} = 4\pi$

$y = 4 \cos\left(\frac{x}{2} - \frac{\pi}{3}\right)$

$= 4 \cos \frac{1}{2}\left(x - \frac{2\pi}{3}\right)$

So the curve moves $\frac{2\pi}{3}$ units to the right.

When $x = 0$,

$y = 4 \cos\left(-\frac{\pi}{3}\right)$

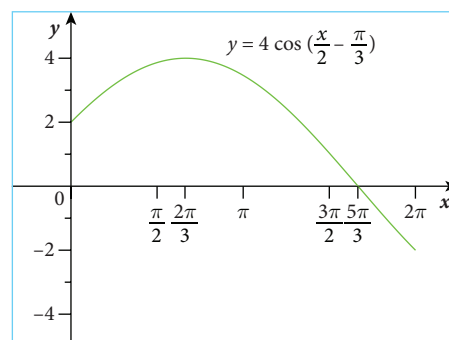
$= 2$

When $x = 2\pi$,

$y = 4 \cos\left(\pi - \frac{\pi}{3}\right)$

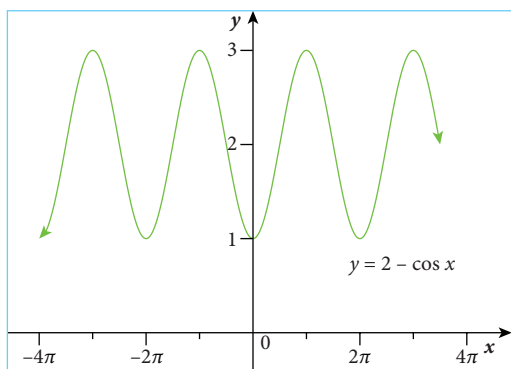
$= 4 \times -\frac{1}{2}$

$= -2$



5 Vertical shift and $y = f(x) + c \Rightarrow$ p. 37

1



2 $y = 2 \sin\left(\frac{x+\pi}{4}\right) + 3$

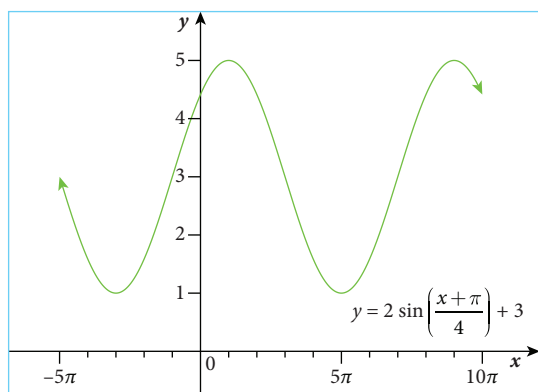
a Amplitude = 2

b Range: $1 \leq y \leq 5$

c Period = $\frac{2\pi}{\frac{1}{4}}$
 $= 8\pi$

d $2 \sin\left(\frac{x+\pi}{4}\right) + 3 = 2 \sin \frac{1}{4}(x+\pi) + 3$

So the phase shift is π units to the left.



3 $y = 3 \sin(4x + 1) + 2$

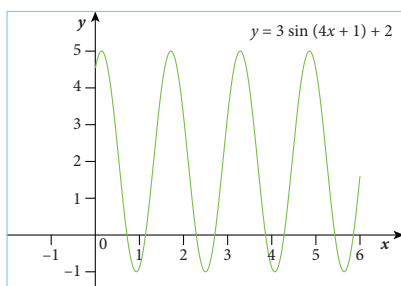
$= 3 \sin 4\left(x + \frac{1}{4}\right) + 2$

Amplitude = 3

Range: $-1 \leq x \leq 5$

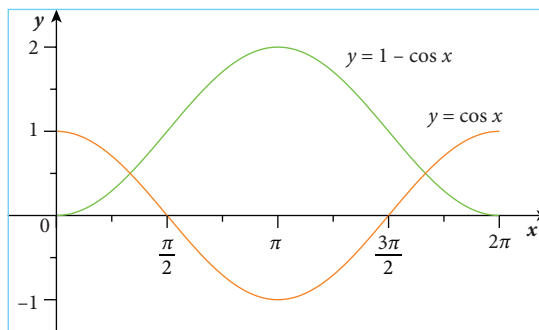
Period = $\frac{2\pi}{4}$
 $= \frac{\pi}{2}$
 ≈ 1.57

Phase shift $\frac{1}{4}$ unit to the left.



6 Solving equations graphically $\Rightarrow$ pp. 38–39

1 a

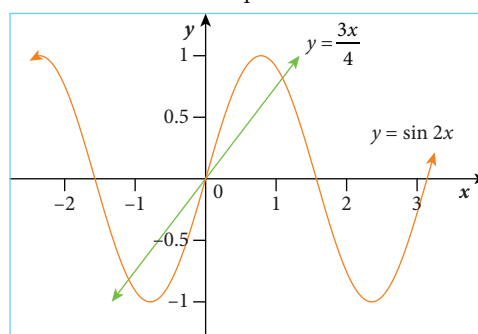


b $2 \cos x = 1$

$\therefore \cos x = \frac{1}{2} \Rightarrow \cos x = 1 - \cos x$

From the graph $x = \frac{\pi}{3}$ or $\frac{5\pi}{3}$.

2 From the graph $\sin 2x = \frac{3x}{4}$ when $x = 0$ and when $x \approx \pm 1.1$.

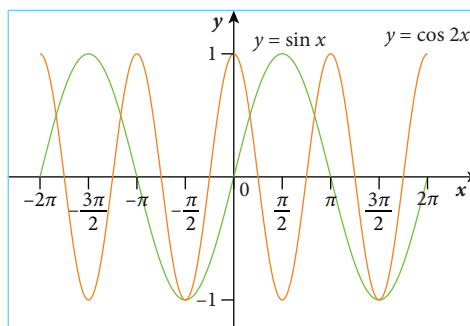


3 a $\sin \frac{\pi}{6} = \frac{1}{2}$

$\cos 2\left(\frac{\pi}{6}\right) = \cos \frac{\pi}{3}$
 $= \frac{1}{2}$
 $= \sin \frac{\pi}{6}$

$\therefore \frac{\pi}{6}$ is a solution of $\cos 2x = \sin x$.

b



c From the diagram, $\cos 2x = \sin x$ when:

$x = -\frac{11\pi}{6}, -\frac{7\pi}{6}, -\frac{\pi}{2}, \frac{\pi}{6}, \pi, \frac{5\pi}{6}$ and $\frac{3\pi}{2}$.

4 $y = 2 \cos x$ and $y = \tan \frac{3x}{4}$

a 2

b No solutions

- c They are both symmetrical curves but $y = \cos x$ is an even function symmetrical about the y -axis while $y = \tan x$ is an odd function with point symmetry about the origin.

d $2 \cos \frac{\pi}{3} = 2 \times \frac{1}{2}$
 $= 1$

When $x = \frac{\pi}{3}$,

$$\begin{aligned}\tan \frac{3x}{4} &= \tan \frac{3 \times \frac{\pi}{3}}{4} \\ &= \tan \frac{\pi}{4} \\ &= 1 \\ &= 2 \cos \frac{\pi}{3}\end{aligned}$$

$\therefore x = \frac{\pi}{3}$ is a solution of $2 \cos x = \tan \frac{3x}{4}$.

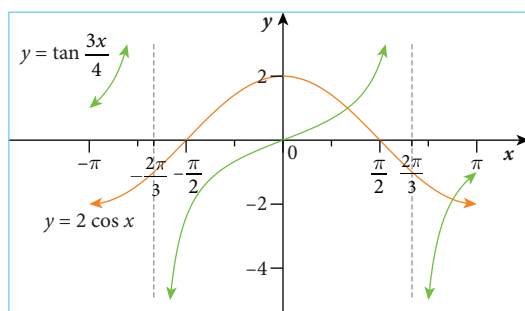
- e $y = \tan x$ has period π and asymptotes at (among others)

$$x = \pm \frac{\pi}{2}$$

$y = \tan \frac{3x}{4}$ has period $\frac{\pi}{3}$ or $\frac{4\pi}{3}$.

The asymptotes include $x = \pm \frac{2\pi}{3}$.

So $2 \cos x \geq \tan \frac{3x}{4} \left(-\pi \leq x \leq \frac{\pi}{2} \right)$ when $-\frac{2\pi}{3} < x \leq \frac{\pi}{3}$.



7 Solving trigonometric equations algebraically ⇨ p. 40

1 $\tan x = 5$

Now $\tan^{-1}5 = 1.373\,400\,76\dots$

$\tan$ is positive in the first and third quadrants.

$\therefore x = 1.373\,400\,76\dots$ or $x = \pi + 1.373\,400\,76\dots$

$x = 1.373\,400\,76\dots$ or $x = 4.514\,993\,42\dots$

$x = 1.373$ or $x = 4.515$ (3 d.p.)

2 $2 \sin^2 \theta - 1 = 0$

$$2 \sin^2 \theta = 1$$

$$\sin^2 \theta = \frac{1}{2}$$

$$\sin \theta = \pm \sqrt{\frac{1}{2}}$$

$$= \pm \frac{1}{\sqrt{2}}$$

$$\theta = \frac{\pi}{4}, \pi - \frac{\pi}{4}, \pi + \frac{\pi}{4} \text{ or } 2\pi - \frac{\pi}{4}$$

$$= \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4} \text{ or } \frac{7\pi}{4}$$

3 $3x = 2 - 5 \sin x$

$$\therefore 5 \sin x = 2 - 3x$$

$$\text{Now } -1 \leq \sin x \leq 1$$

$$\therefore -5 \leq 5 \sin x \leq 5$$

$$\text{So } -5 \leq 2 - 3x \leq 5$$

$$-7 \leq -3x \leq 3$$

$$\frac{7}{3} \geq x \geq -1$$

So any solution to $3x = 2 - 5 \sin x$ lies between $x = -1$ and $x = 2\frac{1}{3}$.

4 $10 \sin \left(2t + \frac{\pi}{4} \right) - 5 = 0$

$$0 \leq t \leq 2\pi$$

$$0 \leq 2t \leq 4\pi$$

$$\frac{\pi}{4} \leq 2t + \frac{\pi}{4} \leq 4\pi + \frac{\pi}{4}$$

$$10 \sin \left(2t + \frac{\pi}{4} \right) = 5$$

$$\sin \left(2t + \frac{\pi}{4} \right) = \frac{1}{2}$$

Now $\sin \frac{\pi}{6} = \frac{1}{2}$

$$\text{So } 2t + \frac{\pi}{4} = \pi - \frac{\pi}{6} \text{ or } 2\pi + \frac{\pi}{6} \text{ or } 3\pi - \frac{\pi}{6} \text{ or } 4\pi + \frac{\pi}{6}$$

$$= \frac{5\pi}{6} \text{ or } \frac{13\pi}{6} \text{ or } \frac{17\pi}{6} \text{ or } \frac{25\pi}{6}$$

$$2t = \frac{7\pi}{12} \text{ or } \frac{23\pi}{12} \text{ or } \frac{31\pi}{12} \text{ or } \frac{47\pi}{12}$$

$$t = \frac{7\pi}{24} \text{ or } \frac{23\pi}{24} \text{ or } \frac{31\pi}{24} \text{ or } \frac{47\pi}{24}$$

5 $2 \cos \left(3x + \frac{\pi}{2} \right) = 1$

$$0 \leq x \leq \frac{\pi}{2}$$

$$0 \leq 3x \leq \frac{3\pi}{2}$$

$$\frac{\pi}{2} \leq 3x + \frac{\pi}{2} \leq 2\pi$$

$$2 \cos \left(3x + \frac{\pi}{2} \right) = 1$$

$$\cos \left(3x + \frac{\pi}{2} \right) = \frac{1}{2}$$

Now $\cos \frac{\pi}{3} = \frac{1}{2}$

$\cos$ is positive in the first and fourth quadrants.

But $\frac{\pi}{2} \leq 3x + \frac{\pi}{2} \leq 2\pi$

$$\text{So } 3x + \frac{\pi}{2} = 2\pi - \frac{\pi}{3}$$

$$= \frac{5\pi}{3}$$

$$3x = \frac{7\pi}{6}$$

$$x = \frac{7\pi}{18}$$

8 Further trigonometric equations and expressions ⇨ p. 41

1 $2 \sin^2 x - \sin x = 1$

Let $u = \sin x$

$\therefore 2u^2 - u - 1 = 0$

$(2u + 1)(u - 1) = 0$

$2u + 1 = 0$ or $u - 1 = 0$

$u = -\frac{1}{2}$ or $u = 1$

So $\sin x = -\frac{1}{2}$ or $\sin x = 1$

$x = \pi + \frac{\pi}{6}$ or $2\pi - \frac{\pi}{6}$ or $x = \frac{\pi}{2}$

$\therefore x = \frac{\pi}{2}, \frac{7\pi}{6} \text{ or } \frac{11\pi}{6}$

2 $\frac{1}{\operatorname{cosec} \theta - \cot \theta} - \frac{1}{\operatorname{cosec} \theta + \cot \theta} = 2$

$$\begin{aligned} \frac{1}{\operatorname{cosec} \theta - \cot \theta} - \frac{1}{\operatorname{cosec} \theta + \cot \theta} &= \frac{\operatorname{cosec} \theta + \cot \theta - (\operatorname{cosec} \theta - \cot \theta)}{(\operatorname{cosec} \theta - \cot \theta)(\operatorname{cosec} \theta + \cot \theta)} \\ &= \frac{\operatorname{cosec} \theta + \cot \theta - \operatorname{cosec} \theta + \cot \theta}{\operatorname{cosec}^2 \theta - \cot^2 \theta} \\ &= \frac{2 \cot \theta}{1} \end{aligned}$$

So $2 \cot \theta = 2$

$\cot \theta = 1$

$\therefore \tan \theta = 1$

$\tan \frac{\pi}{4} = 1$

$\theta = \frac{\pi}{4}$ or $\pi + \frac{\pi}{4}$

$\therefore \theta = \frac{\pi}{4}$ or $\theta = \frac{5\pi}{4}$

3 $\sin^2 \theta = \operatorname{cosec} \theta$

$= \frac{1}{\sin \theta}$

$\sin^3 \theta = 1$

$\sin \theta = 1$

$\therefore \theta = \frac{\pi}{2} \quad (0 \leq \theta \leq 2\pi)$

4 a $\sec x \tan x + \sec^2 x = \frac{1}{\cos x} \times \frac{\sin x}{\cos x} + \frac{1}{\cos^2 x}$

$$\begin{aligned} &= \frac{\sin x}{\cos^2 x} + \frac{1}{\cos^2 x} \\ &= \frac{\sin x + 1}{1 - \sin^2 x} \\ &= \frac{1 + \sin x}{(1 - \sin x)(1 + \sin x)} \\ &= \frac{1}{1 - \sin x} \end{aligned}$$

b $\sec x \tan x + \sec^2 x = \frac{\sqrt{2}}{\sqrt{2} - 1}$

$$\begin{aligned} \therefore \frac{1}{1 - \sin x} &= \frac{\sqrt{2}}{\sqrt{2} - 1} \\ 1 - \sin x &= \frac{\sqrt{2} - 1}{\sqrt{2}} \\ &= 1 - \frac{1}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} \sin x &= \frac{1}{\sqrt{2}} \\ \therefore x &= \frac{\pi}{4} \quad \left(0 \leq x \leq \frac{\pi}{2}\right) \end{aligned}$$

5 $2 \sin x = -3 \cot x$

$2 \sin x = \frac{-3 \cos x}{\sin x}$

$2 \sin^2 x = -3 \cos x$

$2(1 - \cos^2 x) = -3 \cos x$

$2 - 2 \cos^2 x = -3 \cos x$

$0 = 2 \cos^2 x - 3 \cos x - 2$

$0 = (2 \cos x + 1)(\cos x - 2)$

$2 \cos x = -1$ or $\cos x = 2$

$\cos x = -\frac{1}{2}$ No solution

Now $\cos \frac{\pi}{3} = \frac{1}{2}$

$\cos$ is negative in the second and third quadrants.

$x = \pi - \frac{\pi}{3}$ or $\pi + \frac{\pi}{3}$

$\therefore x = \frac{2\pi}{3}$ or $\frac{4\pi}{3}$

6 $2 \cos^2(3x) - 3 \cos(3x) + 1 = 0$

Let $u = \cos 3x$

$2u^2 - 3u + 1 = 0$

$(2u - 1)(u - 1) = 0$

$u = \frac{1}{2}$ or $u = 1$

So $\cos 3x = \frac{1}{2}$ or $\cos 3x = 1$

Now $0 \leq x \leq \pi$

So $0 \leq 3x \leq 3\pi$

$3x = \frac{\pi}{3}, \frac{5\pi}{3}$ or $\frac{7\pi}{3}$ or $3x = 0$ or 2π

$x = \frac{\pi}{9}, \frac{5\pi}{9}$ or $\frac{7\pi}{9}$ or $x = 0$ or $\frac{2\pi}{3}$

So $x = 0, \frac{\pi}{9}, \frac{5\pi}{9}, \frac{2\pi}{3}$ or $\frac{7\pi}{9}$.

9 Problem solving ⇨ p. 43

1 a $x = 7 \cos 4t$

Amplitude = 7

b Period = $\frac{2\pi}{4}$

$$= \frac{\pi}{2}$$

c When $t = 0$, $x = 7$

So the particle is initially 7 cm to the right of the origin.

After $\frac{\pi}{8}$ seconds the particle will be at the origin and will

continue through the origin reaching a position 7 cm to the left of the origin after $\frac{\pi}{4}$ seconds. It will turn passing through

the origin again after another $\frac{\pi}{8}$ seconds and after a total of $\frac{\pi}{2}$ seconds will return to its starting position. It will continue, following the same path over and over again.

2 a $OB = r$ units (radius of circle)
 $\angle AOB + \angle BOC = \pi$ (angles in st. line)
 $\therefore \angle BOC = (\pi - x)$ radians
Area = $\frac{1}{2} ab \sin C$
 $= \frac{1}{2} \times r \times r \times \sin(\pi - x)$
 $= \frac{1}{2} r^2 \sin(\pi - x)$

But $\sin(\pi - x) = \sin x$

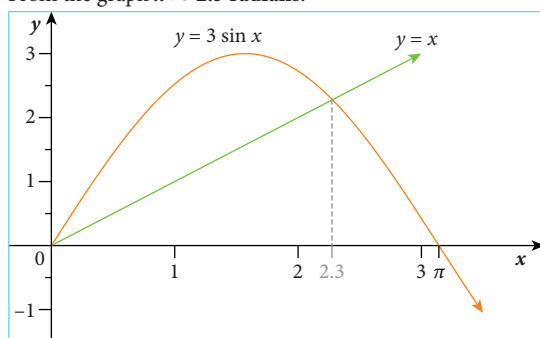
So the area of $\triangle OBC$ is $\frac{1}{2} r^2 \sin x$ units².

b Area of sector AOB is $\frac{1}{2} r^2 x$ units².

Now 3 times the area of $\triangle OBC$ equals the area of the sector.

So $3 \times \frac{1}{2} r^2 \sin x = \frac{1}{2} r^2 x$
 $3 \sin x = x$

c From the graph $x \approx 2.3$ radians.



3 a The tide oscillates between 1.35 m and 0.45 m.

So $c = \frac{0.45 + 1.35}{2} = 0.9$

$k = 1.35 - 0.9 = 0.45$

The time between high and low tides is $5\frac{3}{4}$ hours.

Period = 2×5.75 hours
 $= 11.5$ hours

So $\frac{2\pi}{a} = 11.5$

$a = \frac{2\pi}{11.5}$
 $= \frac{4\pi}{23}$

Now $x = k \cos(at + b) + c$

$= 0.45 \cos\left(\frac{4\pi t}{23} + b\right) + 0.9$

When it is high tide at 9:36 am, $t = 0$.

So when $t = 0$, $x = 1.35$

$1.35 = 0.45 \cos b + 0.9$

$0.45 = 0.45 \cos b$

$\cos b = 1$

$b = 0 \quad \left(0 \leq b \leq \frac{\pi}{2}\right)$

$\therefore x = 0.45 \cos \frac{4\pi t}{23} + 0.9$

b When $x = 1.2$,

$0.45 \cos \frac{4\pi t}{23} + 0.9 = 1.2$

$0.45 \cos \frac{4\pi t}{23} = 0.3$

$\cos \frac{4\pi t}{23} = \frac{0.3}{0.45}$

$\frac{4\pi t}{23} = \cos^{-1}\left(\frac{0.3}{0.45}\right)$

$t = \left(\cos^{-1}\left(\frac{0.3}{0.45}\right)\right) \div \frac{4\pi}{23}$

Time = 1.539 392 71... h

$= 1$ h 32 min (nearest min)

Now 1 h 32 min after 9:36 am is 11:08 am.

The latest time the boat can safely enter the bay before 3 pm is 11:08 am.

4 $x = k \sin(at + b)$

Amplitude = 4 units

$\therefore k = 4$

Period = (2×12.5) seconds

$= 25$ seconds

$\therefore \frac{2\pi}{a} = 25$

$a = \frac{2\pi}{25}$

$x = 4 \sin\left(\frac{2\pi t}{25} + b\right)$

When $t = 0$, $x = 2$

$2 = 4 \sin b$

$\sin b = \frac{1}{2}$

$b = \frac{\pi}{6} \quad \left(0 \leq b \leq \frac{\pi}{2}\right)$

$\therefore x = 4 \sin\left(\frac{2\pi t}{25} + \frac{\pi}{6}\right)$

HSC EXAM-TYPE QUESTIONS

Part A

1 A. $y = \frac{1}{2} \cos\left(3x + \frac{\pi}{6}\right)$

Amplitude = $\frac{1}{2}$

2 D. $y = \tan \pi x$

Period = $\frac{\pi}{\pi} = 1$

3 B. Sine curve with phase shift $\frac{\pi}{6}$ units to the right.

So $y = \sin\left(x - \frac{\pi}{6}\right)$

4 A. $y = 3 \cos 2x$

Amplitude = 3

Period = $\frac{2\pi}{2}$

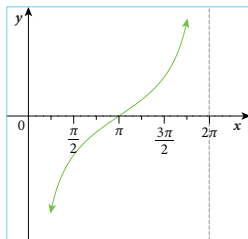
$= \pi$

5 C. $\tan^2 x + 1 = \sec^2 x$

So $\tan^2 x - \sec^2 x = -1$

6 B. $y = 3 - 2 \sin\left(4x - \frac{\pi}{3}\right)$
 Period = $\frac{2\pi}{4}$
 $= \frac{\pi}{2}$

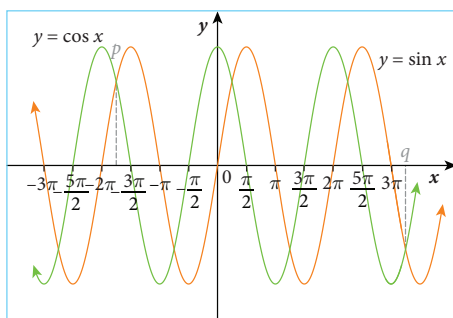
7 A. Period = 2π
 Phase shift π units to the right.
 So the equation is of the form
 $y = k \tan \frac{1}{2}(x - \pi)$.
 So $y = 2 \tan\left(\frac{x}{2} - \frac{\pi}{2}\right)$ is a
 possible equation.



8 C. $\sqrt{2} \cos x + 1 = 0$
 $\sqrt{2} \cos x = -1$
 $\cos x = -\frac{1}{\sqrt{2}}$
 $\therefore x = \pi - \frac{\pi}{4}, \pi + \frac{\pi}{4}, 3\pi - \frac{\pi}{4}, 3\pi + \frac{\pi}{4}, \dots$
 $= \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{11\pi}{4}, \dots$
 $x = \frac{9\pi}{4}$ is not a solution.

9 D. The range of $4 \sin \frac{x}{2}$ is $-4 \leq x \leq 4$.
 So $y = 7 - 4 \sin \frac{x}{2}$ ranges between $7 - (-4) = 11$ and $7 - 4 = 3$.
 Range: $3 \leq x \leq 11$

10 B. $p = -2\pi + \frac{\pi}{4}$ and $q = 3\pi + \frac{\pi}{4}$
 So $p = -\frac{7\pi}{4}$ and $q = \frac{13\pi}{4}$



Part B

11 $\tan\left(2x - \frac{\pi}{6}\right) = \sqrt{3}$
 Now $\tan \frac{\pi}{3} = \sqrt{3}$
 So $2x - \frac{\pi}{6} = \frac{\pi}{3}$ ✓
 $2x = \frac{\pi}{2}$
 $x = \frac{\pi}{4}$ ✓
 (Other answers are possible.)

12 a $x = 4 \cos\left(2t + \frac{\pi}{3}\right)$
 When $t = 0$,
 $x = 4 \cos \frac{\pi}{3}$
 $= 2$

The initial position is 2 m right of the origin. ✓

b When $x = 0$,
 $4 \cos\left(2t + \frac{\pi}{3}\right) = 0$
 $\cos\left(2t + \frac{\pi}{3}\right) = 0$

Now $t = \frac{\pi}{2}$ is the first value greater than 0 for which $\cos t = 0$.

So $2t + \frac{\pi}{3} = \frac{\pi}{2}$ ✓
 $2t = \frac{\pi}{6}$
 $t = \frac{\pi}{12}$

The particle is first at the origin after $\frac{\pi}{12}$ seconds. ✓

13 $y = 3 \sin\left(2x + \frac{\pi}{4}\right) + 1$

a Amplitude = 3 ✓

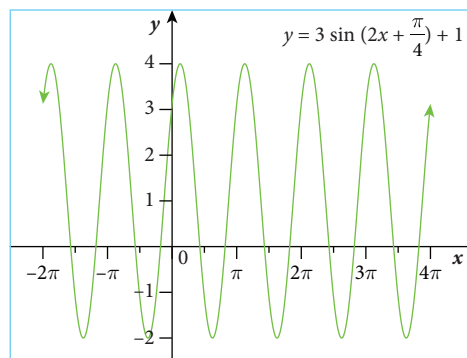
b Period = $\frac{2\pi}{2}$
 $= \pi$ ✓

c Range: $-2 \leq x \leq 4$ ✓

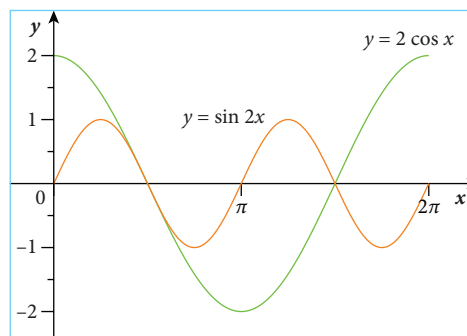
d $y = 3 \sin\left(2x + \frac{\pi}{4}\right) + 1$
 $= 3 \sin 2\left(x + \frac{\pi}{8}\right) + 1$

Phase shift is $\frac{\pi}{8}$ units to the left. ✓✓

e



14 a



✓✓✓

b $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$ ✓

c 8 solutions ✓

$y = 2 \cos x$ has period 2π and there will be 2 solutions in every complete oscillation of $y = 2 \cos x$. For $-4\pi \leq x \leq 4\pi$ there will be 4 complete oscillations. ✓

15 a Equation is of the form $y = k \cos ax + c$

Amplitude = 2

$\therefore k = 2$ ✓

Range is $-1 \leq y \leq 3$

So $c = 1$ ✓

Period = 3π

So $\frac{2\pi}{a} = 3\pi$

$a = \frac{2}{3}$ ✓

So the equation is $y = 2 \cos \frac{2x}{3} + 1$.

b If the curve was a sine curve then $y = 2 \sin \left(\frac{2x}{3} + b \right) + 1$. ✓

When $x = 0$, $y = 3$

$3 = 2 \sin b + 1$

$\sin b = 1$

$b = \frac{\pi}{2}$

$y = 2 \sin \left(\frac{2x}{3} + \frac{\pi}{2} \right) + 1$ ✓

16 $\sin^2 x = \sin x \cos x$

$\sin^2 x - \sin x \cos x = 0$

$\sin x (\sin x - \cos x) = 0$

$\sin x = 0$ or $\sin x = \cos x$ ✓

If $\sin x = 0$

$x = 0, \pi$ or 2π . ✓

If $\sin x = \cos x$

$\tan x = 1$

$x = \frac{\pi}{4}$ or $\frac{5\pi}{4}$ ✓

So $x = 0, \frac{\pi}{4}, \pi, \frac{5\pi}{4}$ or 2π .

17 $y = k \sin (ax + b)$

a $k = 5$ ✓

b $y = 5 \sin (ax + b)$

When $x = 0$, $y = 2.5$

$\therefore 2.5 = 5 \sin b$

$\sin b = \frac{1}{2}$

$b = \frac{\pi}{6}, 0 \leq b \leq \frac{\pi}{2}$ ✓

c Period = 9

So $9 = \frac{2\pi}{a}$

$a = \frac{2\pi}{9}$ ✓

18 $\sqrt{2} \cos \left(2x - \frac{\pi}{3} \right) = 1$

$0 \leq x \leq \pi$

$0 \leq 2x \leq 2\pi$

$\therefore -\frac{\pi}{3} \leq 2x - \frac{\pi}{3} \leq \frac{5\pi}{3}$ ✓

$\cos \left(2x - \frac{\pi}{3} \right) = \frac{1}{\sqrt{2}}$

Now $\cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$

$2x - \frac{\pi}{3} = -\frac{\pi}{4}$ or $\frac{\pi}{4}$ ✓

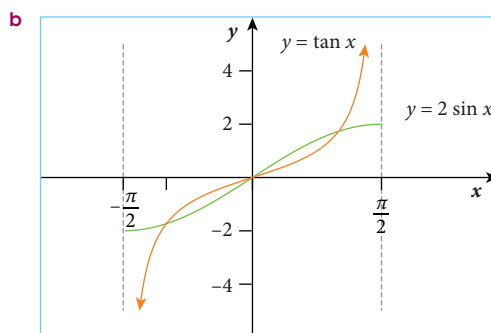
$2x = \frac{\pi}{12}$ or $\frac{7\pi}{12}$

$x = \frac{\pi}{24}$ or $\frac{7\pi}{24}$ ✓

19 a $\tan \frac{\pi}{3} = \sqrt{3}$

$2 \sin \frac{\pi}{3} = 2 \times \frac{\sqrt{3}}{2}$
 $= \sqrt{3}$

$\therefore \tan x = 2 \sin x$ when $x = \frac{\pi}{3}$. ✓



c From the graph $\tan x = 2 \sin x$ when $x = -\frac{\pi}{3}, 0$ and $\frac{\pi}{3}$. ✓

d $\tan x \geq 2 \sin x$ for $-\frac{\pi}{3} \leq x \leq 0$ and $\frac{\pi}{3} \leq x < \frac{\pi}{2}$. ✓✓✓

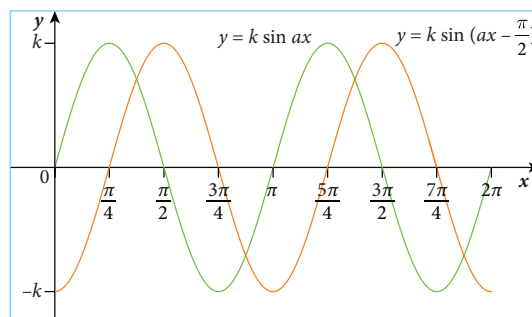
20 $y = k \sin ax$ has period π .

So $a = 2$

$y = k \sin \left(2x - \frac{\pi}{2} \right)$

$= k \sin 2 \left(x - \frac{\pi}{4} \right)$

Phase shift is $\frac{\pi}{4}$ units to the right. ✓



21 $\sin^2 x + 2 \cos x + 2 = 0$

$$1 - \cos^2 x + 2 \cos x + 2 = 0$$

$$-\cos^2 x + 2 \cos x + 3 = 0$$

$$\cos^2 x - 2 \cos x - 3 = 0 \quad \checkmark$$

$$(\cos x - 3)(\cos x + 1) = 0$$

$$\cos x = 3 \text{ or } \cos x = -1$$

$$\text{No solution} \quad x = \pi \text{ or } -\pi \quad \checkmark \checkmark$$

22 x cm is the distance from the beam.

The weight oscillates about $x = 15$.

So $c = 15$ $\checkmark$

The amplitude is 12.5 cm.

So $k = 12.5$ $\checkmark$

The period is π .

So $\frac{2\pi}{a} = \pi$

$a = 2$ $\checkmark$

Now $x = k \cos(at + b) + c$

$$x = 12.5 \cos(2t + b) + 15$$

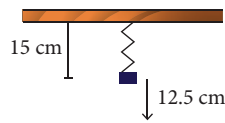
When $t = 0$, $x = 27.5$

$$27.5 = 12.5 \cos b + 15$$

$$12.5 = 12.5 \cos b$$

$$\cos b = 1$$

$\therefore b = 0 \quad \left(0 \leq b \leq \frac{\pi}{2}\right) \quad \checkmark$



SAMPLE HSC EXAMINATION 1

Try to complete these papers as if they are the real thing. These are the instructions you need to follow in the HSC Exam:

General instructions

- Reading time: 10 minutes
- Working time: 3 hours
- Write using black pen.
- NESA approved calculators may be used.
- A reference sheet is provided at the back of this book.
- For questions in Section II, show relevant mathematical reasoning and/or calculations.

Total marks: 100

Section I: 10 marks

Section II: 90 marks

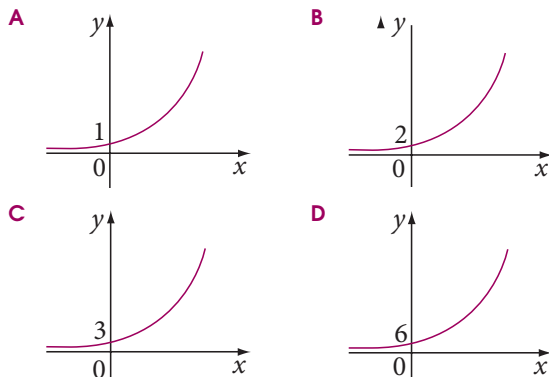
Section I: 10 Marks

Allow about 15 minutes for this section.

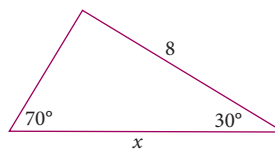
- Mary's z-score in a test was 1.4. If the mean was 69% and the standard deviation 5%, what was Mary's raw score?
A 77% B 76% C 75% D 74%
- $\log_4 8 - \log_4 2 =$
A 0 B 1 C 2 D 4
- Patrick wants to save some money for a special project. What is the minimum amount, to the nearest dollar, he could invest each quarter at 7.6% p.a. interest, compounded quarterly, to have \$8000 at the end of 3 years?
A \$500 B \$550 C \$600 D \$650

Future values of an annuity of \$1			
Period	Interest rate		
	1.9%	3.8%	7.6%
3	3.0574	3.1154	3.2338
4	4.1155	4.2338	4.4795
12	13.337	14.855	18.533
36	51.006	74.452	170.68
48	77.268	131.33	429.61

- Which diagram could show the graph of $y = 3(2^x)$?



- If $f(x) = x^3 - 7x^2 + 2x - 5$ then $f''(3) = ?$
A 4 B 2 C -3 D -11
- Which scatterplot shows weak negative correlation?
A B C D
- The first term of a geometric series is 3 and the common ratio is -2. What is the 10th term?
A -1536 B 1536 C -3072 D 3072
- Which expression will give the value of x ?



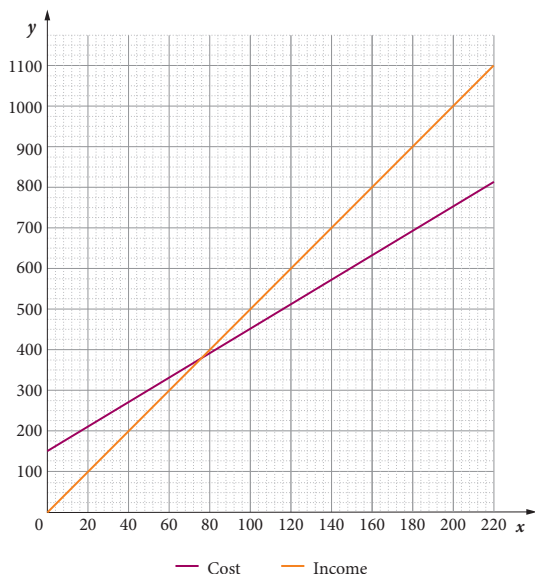
- A $\frac{8 \sin 30^\circ}{\sin 70^\circ}$ B $\frac{\sin 70^\circ}{8 \sin 30^\circ}$
- C $\frac{8 \sin 70^\circ}{\sin 30^\circ}$ D $\frac{8 \sin 80^\circ}{\sin 70^\circ}$
- Consider this set of scores:
4, 13, 15, 17, 18, 20, 24, 24, 27, 30, 44.
Using the method involving the interquartile range, which of the scores are outliers?
A none B 4 only C 44 only D 4 and 44
 - The exact value of $\int_0^{\frac{\pi}{3}} \sec^2 x \, dx = ?$
A $\frac{1}{\sqrt{2}}$ B $\frac{1}{\sqrt{3}}$ C $\frac{\sqrt{3}}{2}$ D $\sqrt{3}$

Section II: 90 marks

Allow about 2 hours and 45 minutes for this section.

Question 11 (4 marks)

Yvonne makes pies to sell at a stall each weekend. The graph shows the cost and income for different numbers of pies.



- How many pies must Yvonne sell to break even? (1 mark)
- How much profit will Yvonne make on a weekend when she sells 200 pies? (1 mark)
- What is the y -intercept of the cost line? What does that represent? (1 mark)
- What is the gradient of the income line? What does that represent? (1 mark)

Question 12 (4 marks)

Tim made a study of the arm length (from elbow to wrist) and leg length (from knee to ankle) of some students. He found that the arm lengths had mean $\bar{x} = 26$ cm and standard deviation 4.25 cm. The leg lengths had mean $\bar{y} = 45$ cm and standard deviation 5 cm and the correlation coefficient was 0.935.

- Briefly describe the correlation between the measurements. (1 mark)
- Tim found the equation of the least squares line of best fit to be $y = 1.1x + 16.4$. Using this equation, what would be the leg length of someone whose arm length is 20 cm? (1 mark)
- Tim's leg length was 56 cm. What would you predict his arm length to be? (1 mark)

- One of the students, Melanie, has an arm length of 39 cm. Would you describe this as being unusual? Discuss. (1 mark)

Question 13 (4 marks)

Differentiate with respect to x :

- $x^2 e^x$ (2 marks)
- $(1 - \cos x)^3$ (2 marks)

Question 14 (5 marks)

- Find $\int \frac{x}{x^2 + 1} dx$ (1 mark)
- Evaluate $\int_0^{\frac{\pi}{3}} \sin x dx$ (2 marks)
- Find the value of b ($b > 1$), if $\int_1^b x dx = 12$. (2 marks)

Question 15 (4 marks)

A teacher gave a test to her class where all the questions had to be answered as either true or false. She checked the answers and drew up the following table.

	Answered true	Answered false
Question true	104	78
Question false	75	193

- If 18 students did the test and each one answered every question, how many questions were on the test? (1 mark)
- What percentage of questions were answered correctly? (1 mark)
- What is the probability that a true question was answered true? (1 mark)
- What is the probability of having a correct answer when the question was answered as true? (1 mark)

Question 16 (2 marks)

Solve $12x - 3x^2 \geq 0$. (2 marks)

Question 17 (2 marks)

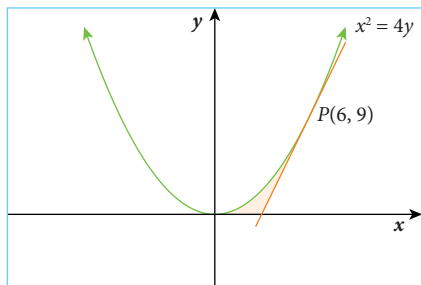
The gradient function of a curve is given

by $\frac{dy}{dx} = e^x - e^{-x}$. The curve passes through the origin.

Find the equation of the curve. (2 marks)

Question 18 (5 marks)

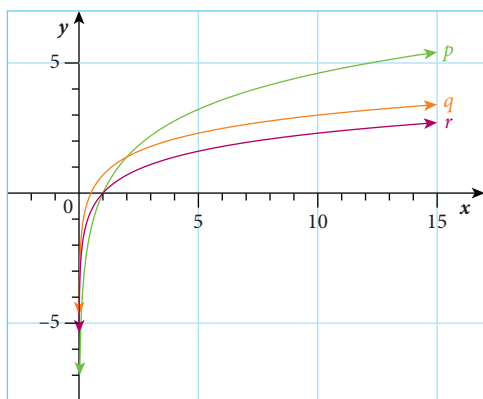
The diagram shows the parabola $x^2 = 4y$ and the tangent to the parabola at the point $P(6, 9)$.



- Find the equation of the tangent to the parabola at the point P . (2 marks)
- Find the shaded area. (3 marks)

Question 19 (3 marks)

The graphs of $y = f(x)$, $y = 2f(x)$ and $y = f(2x)$ where $f(x) = \ln x$ are shown below. Identify each curve and briefly comment on the similarities and differences between the curves. (3 marks)



Question 20 (5 marks)

A sample of 150 packets of toothpicks were examined and the number in each packet was recorded.

Number in packet	147	148	149	150	151	152	153	154	155	156	157
Number of packets	2	5	10	16	23	37	22	18	12	4	1

- Find the mean number of toothpicks per packet. (1 mark)
- Find the sample standard deviation. (Give the answer to two decimal places.) (1 mark)
- What percentage of packets held more than 155 toothpicks? (1 mark)

- The manufacturer of the toothpicks claims that the contents of each packet is 150. Assuming that the contents of the packets are normally distributed, with mean and standard deviation as calculated above, approximately what percentage of packets will have less than 150 toothpicks? (2 marks)

Question 21 (5 marks)

A random variable, X , has probability density function

$$\text{given by } f(x) = \begin{cases} \frac{9-x}{32} & 1 \leq x \leq 9 \\ 0 & \text{elsewhere} \end{cases}$$

- Verify that the properties of a probability density function are satisfied. (2 marks)
- Find $P(X > 5)$. (2 marks)
- Briefly explain the significance, if any, of the answer to part **b**. (1 mark)

Question 22 (2 marks)

If $\sin x \neq \pm 1$ show that

$$1 + \sin^2 x + \sin^4 x + \sin^6 x + \dots = \sec^2 x. \quad (2 \text{ marks})$$

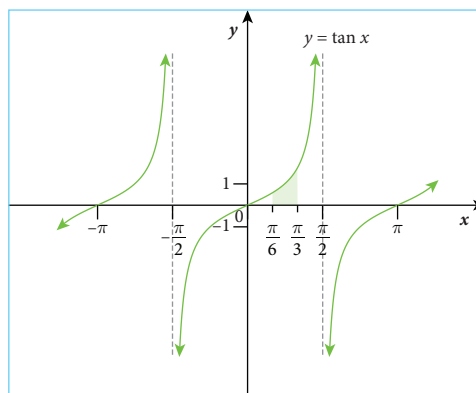
Question 23 (6 marks)

The velocity $v \text{ ms}^{-1}$ of a moving particle is given by $v = t^2 + t - 12$ ($t \geq 0$).

- When is the particle at rest? (1 mark)
- If the particle was initially 2 metres to the left of the origin, find its position after 6 seconds. (2 marks)
- Find the initial acceleration. (1 mark)
- Briefly describe the motion. (2 marks)

Question 24 (4 marks)

- Differentiate $\log_e(\cos x)$. (2 marks)
- Hence find the exact area shaded in the diagram. (2 marks)



Question 25 (5 marks)

Consider $y = 2 \sin \pi(x - 0.25) + 1$.

- a What is the period? (1 mark)
- b What is the range? (2 marks)
- c Sketch $y = 2 \sin \pi(x - 0.25) + 1$ for $0 \leq x \leq 4$. (2 marks)

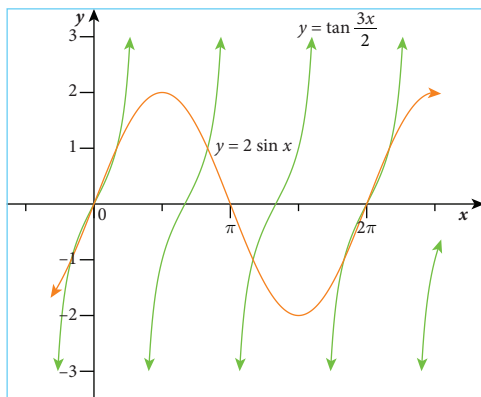
Question 26 (4 marks)

An orchard is trapezoidal in shape. There are 20 trees in the first row and 95 trees in the last row. The difference between the number of trees in successive rows is constant. There are a total of 1495 trees in the orchard.

- a Find the number of rows of trees. (2 marks)
- b In any row, how many more trees are there than in the previous row? (2 marks)

Question 27 (6 marks)

A software application was used to graph $y = 2 \sin x$ and $y = \tan \frac{3x}{2}$ on the same diagram.



- a How many solutions are there of $2 \sin x = \tan \frac{3x}{2}$ for $0 \leq x \leq 2\pi$? (1 mark)
- b Show that $\frac{\pi}{6}$ is a solution of the equation $2 \sin x = \tan \frac{3x}{2}$. (2 marks)
- c Solve $2 \sin x \geq \tan \frac{3x}{2}$ for $0 \leq x \leq \frac{\pi}{2}$. (3 marks)

Question 28 (7 marks)

Lachlan borrows \$370 000 and agrees to repay the principal plus interest over 25 years. The rate of reducible interest charged is 0.6% per month. Let A_n be the amount owing after n months and M be the amount of each monthly repayment.

- a Show that $A_3 = 370\,000(1.006)^3 - M(1 + 1.006 + 1.006^2)$. (2 marks)

- b Write down a similar expression for A_n and hence show that

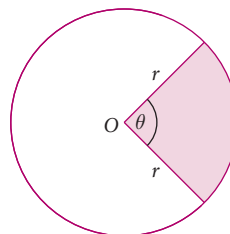
$$A_n = 370\,000(1.006)^n - M \left(\frac{1.006^n - 1}{0.006} \right). \quad (2 \text{ marks})$$

- c If the loan is to be repaid over 25 years find the amount of each monthly instalment. (2 marks)
- d Find the total amount of interest that Lachlan pays. (1 mark)

Question 29 (8 marks)

Consider the curve $y = 4 + 9x - 3x^2 - x^3$.

- a Where does the curve cut the y -axis? (1 mark)
- b Find the stationary points and determine their nature. (3 marks)
- c Sketch the curve showing all essential features. (4 marks)

Question 30 (5 marks)

The diagram shows a sector of a circle, radius r and the angle subtended at the centre θ . The area of the sector is 36 cm^2 .

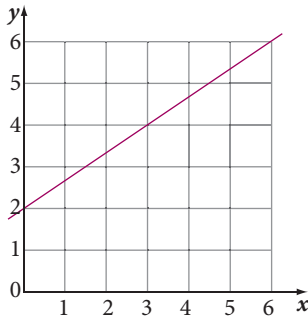
- a Show that $\theta = \frac{72}{r^2}$. (1 mark)
- b Hence show that the minimum perimeter of the sector will occur when the arc length of the sector is equal to the diameter of the circle. (4 marks)

SAMPLE HSC EXAMINATION 2

Section I: 10 Marks

Allow about 15 minutes for this section.

- 1 What is the equation of the line?



- A $y = \frac{3}{2}x + 2$ B $y = 2x + \frac{3}{2}$
C $y = \frac{2}{3}x + 2$ D $y = 2x + \frac{2}{3}$

- 2 A two-way table has been drawn up showing the results of a lie-detector test.

	Detected true	Detected false
True statement	84	16
False statement	36	124

What is the probability that a statement was correctly detected?

- A $\frac{21}{65}$ B $\frac{31}{65}$ C $\frac{4}{5}$ D $\frac{6}{13}$

- 3 A certain amount is to be invested at the end of each month into an account earning 9% p.a. interest, compounded monthly. At the end of 5 years the future value will be \$60 339. What is the present value (to the nearest dollar)?

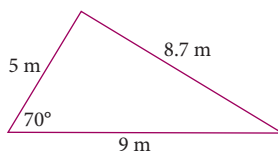
- A \$38 539 B \$39 216
C \$27 153 D \$33 186

- 4 In a normal distribution, the mean is 64 and the standard deviation 12. Approximately what percentage of scores lie between 40 and 76?

- A 68% B 81.5%
C 87.5% D 95%

- 5 Which is closest to the area of the triangle?

- A 21.1 m^2 B 7.7 m^2
C 20.4 m^2 D 36.8 m^2



- 6 Which is an anti-derivative of $\frac{1}{x^2}$?

- A $\frac{3}{x^3}$ B $\frac{1}{3x^3}$ C $\frac{1}{x}$ D $\frac{x-1}{x}$

- 7 $\frac{\sin x \cos x + \sin^2 x}{\sin x \cos x + \cos^2 x} =$

- A $\operatorname{cosec}^2 x$ B $\sec^2 x$
C $\cot x$ D $\tan x$

- 8 $f(x) = (x+1)^2$ and $g(x) = x^2 + 1$. Which function is even?

- A $f(x)$ only
B $g(x)$ only
C both $f(x)$ and $g(x)$
D neither $f(x)$ nor $g(x)$

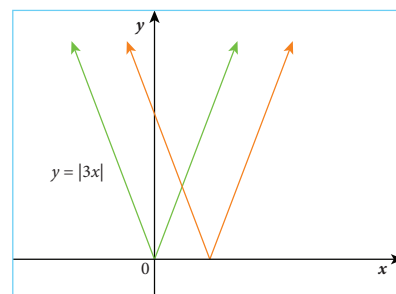
- 9 The probability density function of a random variable

$$\text{is given by } f(x) = \begin{cases} \frac{x}{8} & 0 \leq x \leq 4 \\ 0 & \text{elsewhere} \end{cases}$$

What is the median (correct to one decimal place)?

- A 2.2 B 2.4
C 2.6 D 2.8

- 10 The graph shows $y = |3x|$ and another function. What could be the equation of the second function?



- A $y = |3x + 2|$ B $y = |3x - 2|$
C $y = |3x| + 2$ D $y = |3x| - 2$

Section II: 90 marks

Allow about 2 hours and 45 minutes for this section.

Question 11 (4 marks)

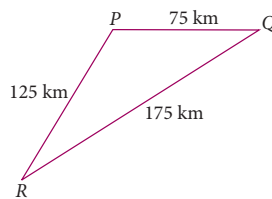
Class scores			
X		Y	
9 3	4	6	
7 7 2 0	5	1 4 8	
9 8 5 4 1	6	0 3 5 6	
8 6 5 3 2	7	1 2 4 7 8	
9 6 5 3	8	2 4 4 7 9 9	
4 1	9	0 3 5	

A back-to-back stem-and-leaf plot has been drawn showing the scores by students in two different classes, X and Y.

- What is the range of scores in class X? (1 mark)
- What is the median score in class Y? (1 mark)
- Compare the results, referring to the shape and patterns of the display. (2 marks)

Question 12 (3 marks)

In the diagram, P is 75 km due west of Q . R is 125 km from P and 175 km from Q .



- Use the cosine rule to find the size of $\angle QPR$. (2 marks)
- Find the true bearing of R from P . (1 mark)

Question 13 (8 marks)

Two different tradespeople charge different rates for their work. Each price is made up of a fixed charge and an hourly rate. The graph shows the cost for getting work done by each of the tradespeople for different hours.



- How much does Barry charge as a fixed rate? (1 mark)
- How much will it cost for Carrie to work for 10 hours? (1 mark)

- At what time are the two costs the same? (1 mark)
- If you wanted to spend about \$500, who would be the most economical person to hire? (1 mark)
- What is the gradient of Barry's line and what does it represent? (2 marks)
- Write an equation for the amount, \$ C , that Carrie charges per hour, h . (2 marks)

Question 14 (3 marks)

Differentiate:

- $\log_e(x^2 + 4)$ (1 mark)
- $e^x \sin x$ (2 marks)

Question 15 (6 marks)

Lily wants to travel overseas in two years time and estimates that she will need \$30 000.

- How much, to the nearest dollar, will Lily need to invest each month into an account earning 6% p.a. interest, compounded monthly? (2 marks)

Future values of an annuity of \$1					
Period	0.5%	1.5%	3%	5%	6%
2	2.0050	2.0150	2.0300	2.0500	2.0600
4	4.0301	4.0909	4.1836	4.3101	4.3746
8	8.1414	8.4328	8.8923	9.5491	9.8975
12	12.336	13.041	14.192	15.917	16.870
18	18.786	20.489	23.414	28.132	30.906
24	25.432	28.634	34.426	44.502	50.816

- Lily believes that this investment will mean she will have exactly half of \$30 000 in one year. Briefly explain why Lily is wrong and find the amount that she will have at the end of one year if she invests the amount found in part a. (2 marks)
- Lily's dad invests \$750 at the end of each quarter at 6% p.a. interest, compounded quarterly. How much will he have at the end of 2 years? (1 mark)
- Lily's dad presents this extra amount to Lily as she is about to leave on her trip. How much extra would Lily have needed to save each month to have the amount that she now has for her trip? (1 mark)

Question 16 (4 marks)

Evaluate:

- $\int_1^4 \frac{dx}{\sqrt{x}}$ (2 marks)
- $\int_0^{\frac{\pi}{6}} \sin 2x \, dx$ (2 marks)

Question 17 (2 marks)

Of 48 students enrolled in a particular course at university, all but 6 also study either physics or chemistry or both. 26 study physics and 30 study chemistry.

- a Draw a Venn diagram to illustrate this information. (1 mark)
- b What is the probability that a student who studies chemistry, chosen at random from that group, also studies physics? (1 mark)

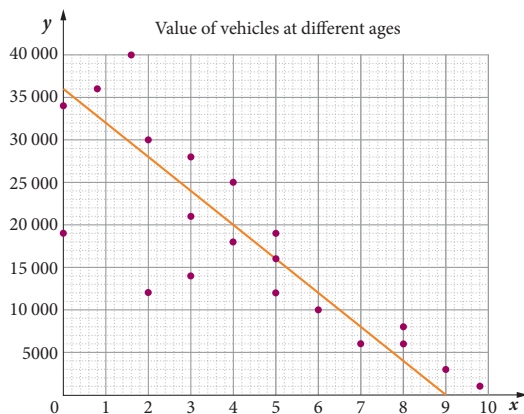
Question 18 (2 marks)

The sum of the infinite geometric series $1 + 3^x + 3^{2x} + 3^{3x} + \dots$ is 1.5. Find the value of x . (2 marks)

Question 19 (8 marks)

Corey collected some data on the value, in dollars, of vehicles of different ages, in years.

- a Corey used an online calculator and found Pearson's correlation coefficient to be -0.8168 for his data. What information can be found from the correlation coefficient? (1 mark)
- b Corey produced a scatterplot from his data and drew a line of fit on it. How many vehicles did Corey have data for? (1 mark)



- c What is the equation of Corey's line of fit? (2 marks)
- d What value does the line of fit suggest for a vehicle that is $6\frac{1}{2}$ years old? (1 mark)
- e What is the suggested age of a vehicle valued at \$22 000? (1 mark)
- f What restrictions must be placed on the equation of the line of fit? Give reasons for your answer. (2 marks)

Question 20 (4 marks)

- a Without the use of calculus sketch the graph of $y = 3 \sin 2x$. (2 marks)

- b On the same set of axes sketch $y = x$. (1 mark)

- c How many solutions has the equation $3 \sin 2x = x$? (1 mark)

Question 21 (4 marks)

The life in hours of a certain organism has the probability

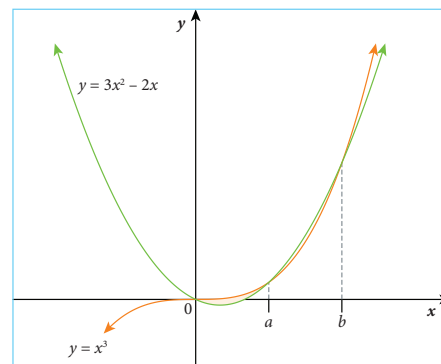
$$\text{density function } f(x) = \begin{cases} \frac{1}{50} e^{-\frac{x}{50}} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Giving each answer correct to three decimal places, find the probability that such an organism has a life-span of:

- a at most 24 hours (2 marks)
- b anywhere from 48 to 60 hours (1 mark)
- c at least 36 hours (1 mark)

Question 22 (5 marks)

The diagram shows the curves $y = x^3$ and $y = 3x^2 - 2x$.



- a The two curves intersect at the origin and at the points where $x = a$ and $x = b$ ($0 < a < b$). Find the value of a . (2 marks)
- b Show that the area shaded in the diagram can be found by evaluating the integral $\int_0^a (x^3 - 3x^2 + 2x) dx$. (2 marks)
- c Find the shaded area. (1 mark)

Question 23 (5 marks)

The acceleration (in ms^{-2}) of a moving particle

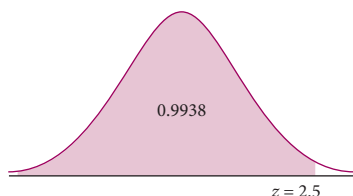
is $\frac{d^2x}{dt^2} = -2$. When $t = 0$, $\frac{dx}{dt} = 6$ and $x = 7$.

- a Find an expression for the velocity of the particle. (1 mark)
- b Find an expression for the displacement. (1 mark)
- c Find the time when the particle is at rest. (1 mark)
- d Find the distance travelled in the first 5 seconds. (2 marks)

Question 24 (6 marks)

In a national examination testing 58 000 students the results were found to be normally distributed. The exam was marked out of 80 and the mean was 52.5 and the standard deviation 8.2.

- a** Micky scored 73. Show that the z -score corresponding to his mark is 2.5. (1 mark)
- b** Micky found this diagram showing that the area beneath the bell curve, to the left of a z -score of 2.5, was 0.9938. About how many students did better than Micky in the exam? (2 marks)



- c** What is the probability, as a percentage, that a student chosen at random from those who did the exam scored more than 40%? (3 marks)

Question 25 (3 marks)

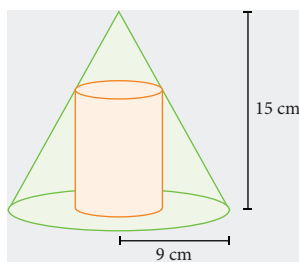
Sketch the graph of $y = \log_2 x$ and hence find any solutions of the equation $1 + \log_2 x = 2x$. (3 marks)

Question 26 (3 marks)

Kylie invests \$10 000 on 1 June each year into a superannuation fund, beginning in June 2005. What will be the value of her investment on 31 May 2035 if the fund pays 6.5% per annum interest? (3 marks)

Question 27 (6 marks)

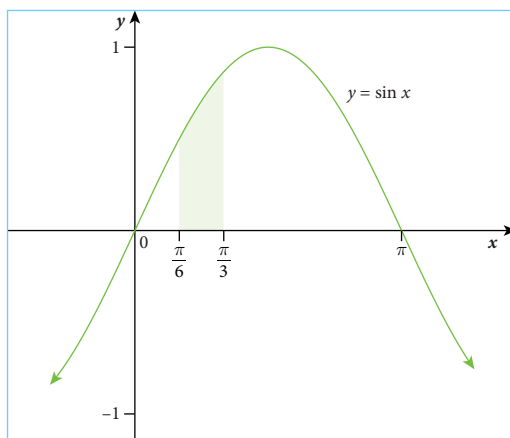
A cylinder is to be placed in a cone of height 15 cm and radius 9 cm as shown in the diagram. If the radius of the cylinder is r cm and the height is h cm:



- a** show that $h = 15 - \frac{5}{3}r$ (2 marks)
- b** show that the volume V cm³ of the cylinder is given by $V = 15\pi r^2 - \frac{5\pi r^3}{3}$ (1 mark)
- c** show that the maximum volume of the cylinder occurs when $r = 6$ (2 marks)
- d** show that the largest possible cylinder has $\frac{4}{9}$ of the volume of the cone (1 mark)

Question 28 (5 marks)

- a** Find the exact area shown in the diagram. (2 marks)



- b** Use one application of the trapezoidal rule to find an approximation for the area shown in the diagram. (Give the answer in terms of π and $\sqrt{3}$.) (2 marks)
- c** Use the answers to parts **a** and **b** to show that $\pi \approx 12(2 - \sqrt{3})$. (1 mark)

Question 29 (9 marks)

Consider the curve $y = 2xe^{-x}$.

- a** Where does the curve cross the y -axis? (1 mark)
- b** Show that $\frac{dy}{dx} = 2e^{-x}(1 - x)$. (1 mark)
- c** Show that $\frac{d^2y}{dx^2} = -2e^{-x}(2 - x)$. (1 mark)
- d** Find the turning point and determine its nature. (2 marks)
- e** Find the point where the curve changes concavity. (2 marks)
- f** Determine the behaviour of y as $x \rightarrow \infty$ and as $x \rightarrow -\infty$. (1 mark)
- g** Sketch the graph of $y = 2xe^{-x}$. (1 mark)

SAMPLE HSC EXAMINATION 1

Section I

1 B. $z = 1.4$, $\mu = 69$, $\sigma = 5$

$$z = \frac{x - \mu}{\sigma}$$

$$1.4 = \frac{x - 69}{5}$$

$$7 = x - 69$$

$$x = 76$$

Mary's score is 76%.

2 B. $\log_4 8 - \log_4 2 = \log_4 \left(\frac{8}{2} \right)$

$$= \log_4 4$$

$$= 1$$

3 C. $FV = \$8000$, $r = 1.9\%$, $n = 12$

$$\$8000 = a \times 13.337$$

$$a = \$8000 \div 13.337$$

$$= \$599.835\ 04\dots$$

$$= \$600 \text{ (nearest dollar)}$$

4 C. $y = 3(2^x)$

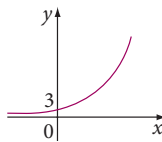
When $x = 0$,

$$y = 3(2^0)$$

$$= 3 \times 1$$

$$= 3$$

C is the correct graph.



5 A. $f(x) = x^3 - 7x^2 + 2x - 5$

$$f'(x) = 3x^2 - 14x + 2$$

$$f''(x) = 6x - 14$$

$$f''(3) = 6 \times 3 - 14$$

$$= 4$$

6 C. C shows weak negative correlation.



7 A. $a = 3$, $r = -2$, $n = 10$

$$T_n = ar^{n-1}$$

$$T_{10} = 3 \times (-2)^{10-1}$$

$$= -1536$$

8 D. Third angle $= 180^\circ - (70^\circ + 30^\circ)$

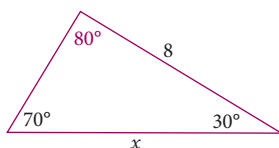
$$= 80^\circ$$

By the sine rule:

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

$$\frac{x}{\sin 80^\circ} = \frac{8}{\sin 70^\circ}$$

$$x = \frac{8 \sin 80^\circ}{\sin 70^\circ}$$



9 A. 4, 13, 15, 17, 18, 20, 24, 24, 27, 30, 44

Median = 20

$$Q_1 = 15 \text{ and } Q_3 = 27$$

$$IQR = 27 - 15$$

$$= 12$$

$$\text{So } Q_1 - 1.5 \times IQR = 15 - 1.5 \times 12$$

$$= -3$$

$$Q_3 + 1.5 \times IQR = 27 + 1.5 \times 12$$

$$= 45$$

There are no scores less than -3 or greater than 45, so there are no outliers.

10 D. $\int_0^{\pi/3} \sec^2 x \, dx = [\tan x]_0^{\pi/3}$

$$= \tan \frac{\pi}{3} - \tan 0$$

$$= \sqrt{3}$$

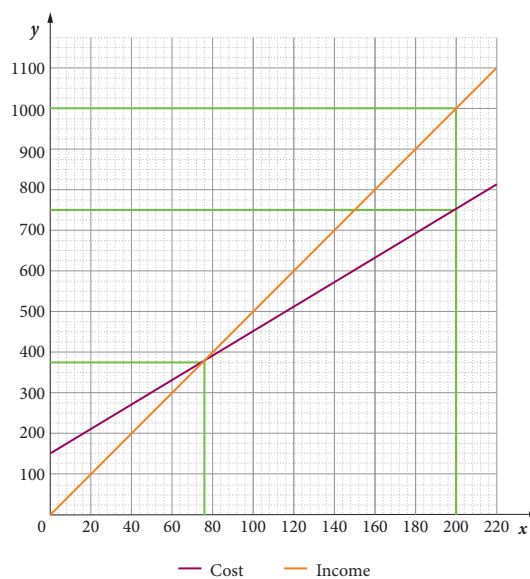
Section II

11 a From the graph, Yvonne must sell 75 pies to break even. ✓

b If Yvonne sells 200 pies the cost is \$750 and the income is \$1000.

$$\text{Profit} = \$1000 - \$750$$

$$= \$250 \quad \checkmark$$



c The y-intercept of the cost line is 150. That represents the amount of fixed costs that Yvonne must pay. ✓

d The gradient of the income line is $\frac{100}{20} = 5$. It represents the amount that Yvonne receives for each pie. So she sells the pies for \$5 each. ✓

- 12 a There is a **strong positive** correlation between the measurements. ✓
- b $y = 1.1x + 16.4$
When $x = 20$,
 $y = 1.1 \times 20 + 16.4$
 $= 38.4$
So, using the equation, a person whose arm length is 20 cm would have a leg length of **38.4 cm**. ✓
- c $y = 1.1x + 16.4$
When $y = 56$,
 $1.1x + 16.4 = 56$
 $1.1x = 39.6$
 $x = 36$
So a prediction for Tim's arm length is **36 cm**. ✓
- d The arm lengths have mean 26 cm and standard deviation 4.25 cm.
Now $26 + 3 \times 4.25 = 38.75$.
So Melanie's arm length is more than three standard deviations above the mean so it is **unusual**. ✓

- 13 a $y = x^2 e^x$
 $u = x^2 \quad v = e^x$
 $\frac{du}{dx} = 2x \quad \frac{dv}{dx} = e^x$
 $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$
 $= x^2 \times e^x + e^x \times 2x$ ✓
 $= x^2 e^x + 2xe^x$
 $= xe^x(x + 2)$ ✓
- b $y = (1 - \cos x)^3$
 $\frac{dy}{dx} = 3(1 - \cos x)^2 \times \frac{d}{dx}(1 - \cos x)$ ✓
 $= 3(1 - \cos x)^2 \times \sin x$
 $= 3 \sin x (1 - \cos x)^2$ ✓

14 a $\int \frac{x}{x^2 + 1} dx = \frac{1}{2} \int \frac{2x}{x^2 + 1} dx$
 $= \frac{1}{2} \log_e(x^2 + 1) + c$ ✓

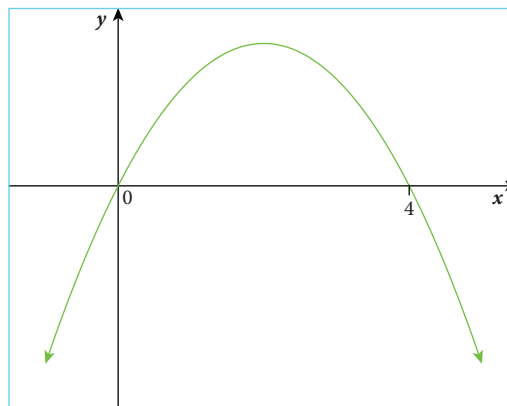
b $\int_0^{\frac{\pi}{3}} \sin x dx = [-\cos x]_0^{\frac{\pi}{3}}$ ✓
 $= -\cos \frac{\pi}{3} - (-\cos 0)$
 $= -\frac{1}{2} + 1$
 $= \frac{1}{2}$ ✓

c $\int_1^b x dx = 12$
Now $\int_1^b x dx = \left[\frac{x^2}{2} \right]_1^b$
 $= \frac{b^2}{2} - \frac{1}{2}$ ✓
So $\frac{b^2}{2} - \frac{1}{2} = 12$
 $b^2 - 1 = 24$
 $b^2 = 25$
 $b = 5 \quad (b > 0)$ ✓

	Answered true	Answered false
Question true	104	78
Question false	75	193

- a Total questions = $104 + 78 + 75 + 193$
 $= 450$
Questions per test = $450 \div 18$
 $= 25$ ✓
- b Questions answered correctly = $104 + 193$
 $= 297$
Percentage answered correctly = $\frac{297}{450} \times 100\%$
 $= 66\%$ ✓
- c True questions = $104 + 78$
 $= 182$
 $P(\text{answered true}) = \frac{104}{182}$
 $= \frac{4}{7}$ ✓
- d True answers = $104 + 75$
 $= 179$
 $P(\text{correct answer}) = \frac{104}{179}$ ✓

16 $12x - 3x^2 \geq 0$
Let $y = 12x - 3x^2$
 $= 3x(4 - x)$ ✓



From the graph, $y \geq 0$ when $0 \leq x \leq 4$.
 $\therefore 12x - 3x^2 \geq 0$ when $0 \leq x \leq 4$. ✓

- 17 $\frac{dy}{dx} = e^x - e^{-x}$
 $y = e^x - \frac{e^{-x}}{-1} + C$
 $y = e^x + e^{-x} + C$ ✓
The curve passes through the origin.
 $\therefore$ when $x = 0$, $y = 0$
 $0 = e^0 + e^{-0} + C$
 $0 = 1 + 1 + C$
 $C = -2$
The equation of the curve is $y = e^x + e^{-x} - 2$. ✓

18 a $x^2 = 4y$

$$y = \frac{x^2}{4}$$

$$\frac{dy}{dx} = \frac{2x}{4}$$

$$= \frac{x}{2}$$

When $x = 6$, $\frac{dy}{dx} = \frac{6}{2}$

So the gradient of the tangent is 3. ✓

$(6, 9)$, $m = 3$

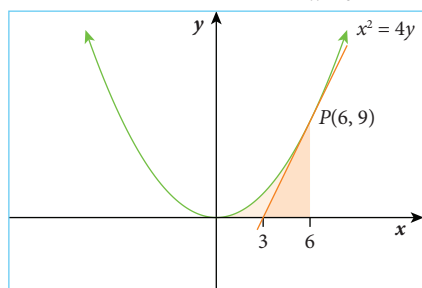
$$y - y_1 = m(x - x_1)$$

$$y - 9 = 3(x - 6)$$

$$y - 9 = 3x - 18$$

$$y = 3x - 9 \quad \checkmark$$

- b The tangent cuts the x -axis when $y = 0$,
i.e. $3x - 9 = 0$
 $3x = 9$
 $x = 3$



Area of triangle:

$$A = \frac{1}{2}bh$$

$$= \frac{1}{2} \times 3 \times 9$$

$$= 13.5 \quad \checkmark$$

Area under parabola:

$$A = \int_0^6 \frac{x^2}{4} dx$$

$$= \left[\frac{x^3}{12} \right]_0^6 \quad \checkmark$$

$$= \frac{6^3}{12} - \frac{0^3}{12}$$

$$= 18$$

Required area = $(18 - 13.5)$ units²

$$= 4.5 \text{ units}^2 \quad \checkmark$$

- 19 The curves are $y = \ln x$, $y = 2 \ln x$ and $y = \ln 2x$.

Now $y = \ln x$ passes through $(1, 0)$ and $(e, 1)$.

So r is the graph of $y = \ln x$.

$$\ln 2x = \ln 2 + \ln x$$

Now $\ln 2$ is a positive constant.

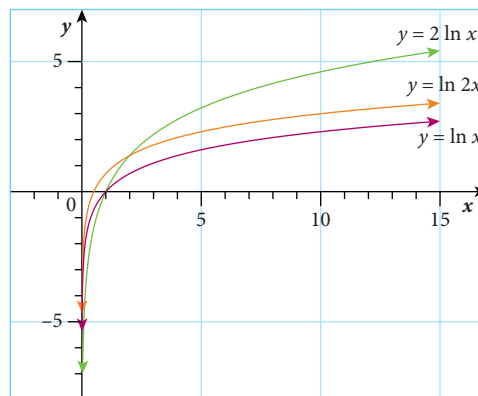
So $y = \ln 2x$ is the curve $y = \ln x$ translated up.

So q is the graph of $y = \ln 2x$. ✓

p is the graph of $y = 2 \ln x$. ✓

Curves q and r are the same shape just in different positions.

Curve p is a similar logarithmic shape to the others but has been stretched and pulled into a slightly different shape. ✓



- 20 a $\bar{x} = 152$ (by calculator) ✓

b $\sigma_{n-1} = 2.0033528 \dots$

$$= 2.00 \quad (2 \text{ d.p.}) \quad \checkmark$$

- c From the table five packets hold more than 155 toothpicks.

$$\text{Percentage of packets} = \frac{5}{150} \times 100\%$$

$$= 3\frac{1}{3}\% \quad \checkmark$$

- d 150 is one standard deviation below the mean. ✓

68% are within one standard deviation of the mean.

Percentage outside one standard deviation

$$= 100\% - 68\%$$

$$= 32\%$$

Half of these are below 150.

So about 16% of packets will have fewer than 150 toothpicks. ✓

21 a $f(x) = \begin{cases} \frac{9-x}{32} & 1 \leq x \leq 9 \\ 0 & \text{elsewhere} \end{cases}$

If $x \leq 9$ then $9 - x \geq 0$ and so $\frac{9-x}{32} \geq 0$.

$\therefore x \geq 0$ for all values of x . ✓

$$\int_{-\infty}^{\infty} f(x) dx = \int_1^9 \frac{9-x}{32} dx$$

$$= \left[\frac{9x}{32} - \frac{x^2}{64} \right]_1^9$$

$$= \frac{9 \times 9}{32} - \frac{9^2}{64} - \left(\frac{9 \times 1}{32} - \frac{1^2}{64} \right)$$

$$= 1 \quad \checkmark$$

So the properties of a probability density function are satisfied.

b $P(X > 5) = \int_5^9 \frac{9-x}{32} dx$

$$= \left[\frac{9x}{32} - \frac{x^2}{64} \right]_5^9 \quad \checkmark$$

$$= \frac{9 \times 9}{32} - \frac{9^2}{64} - \left(\frac{9 \times 5}{32} - \frac{5^2}{64} \right)$$

$$= \frac{1}{4} \quad \checkmark$$

- c 5 is the upper quartile. ✓

22 $1 + \sin^2 x + \sin^4 x + \sin^6 x + \dots$

Infinite geometric series:

$$a = 1, \quad r = \sin^2 x$$

Now $-1 \leq \sin x \leq 1$ for all values of x .

But $\sin x \neq \pm 1$

$$\therefore -1 < r < 1$$

So a limiting sum exists. ✓

$$S = \frac{a}{1-r}$$

$$= \frac{1}{1-\sin^2 x}$$

$$= \frac{1}{\cos^2 x}$$

$$= \sec^2 x$$

$$\therefore 1 + \sin^2 x + \sin^4 x + \sin^6 x + \dots = \sec^2 x \quad \checkmark$$

23 $v = t^2 + t - 12 \quad (t \geq 0)$

a The particle is at rest when $v = 0$,

$$\text{i.e. } t^2 + t - 12 = 0$$

$$(t+4)(t-3) = 0$$

$$t = -4 \quad \text{or} \quad t = 3$$

$$\text{But } t \geq 0 \quad \therefore t = 3$$

The particle is at rest after 3 seconds. ✓

b $v = t^2 + t - 12$

$$x = \frac{t^3}{3} + \frac{t^2}{2} - 12t + C$$

When $t = 0$, $x = -2$

$$-2 = \frac{0^3}{3} + \frac{0^2}{2} - 12 \times 0 + C$$

$$C = -2$$

$$\therefore x = \frac{t^3}{3} + \frac{t^2}{2} - 12t - 2 \quad \checkmark$$

When $t = 6$,

$$x = \frac{6^3}{3} + \frac{6^2}{2} - 12 \times 6 - 2$$

$$= 16$$

After 6 seconds the particle is 16 m to the right of O. ✓

c $a = 2t + 1$

When $t = 0$,

$$a = 2 \times 0 + 1$$

$$= 1$$

The initial acceleration is 1 ms^{-2} . ✓

d The particle is initially 2 m to the left of the origin moving left (the velocity is negative) and slowing down (the acceleration is positive, opposite the velocity). ✓

It comes to rest after 3 seconds and then moves right at increasing speed reaching a position 16 m to the right of the origin after 6 seconds. ✓

24 a $y = \log_e(\cos x)$

$$\frac{dy}{dx} = \frac{1}{\cos x} \times \frac{d}{dx}(\cos x) \quad \checkmark$$

$$= \frac{1}{\cos x} \times -\sin x$$

$$= \frac{-\sin x}{\cos x}$$

$$= -\tan x \quad \checkmark$$

b $A = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \tan x \, dx$

$$= \left[-\log_e(\cos x) \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}} \quad \checkmark$$

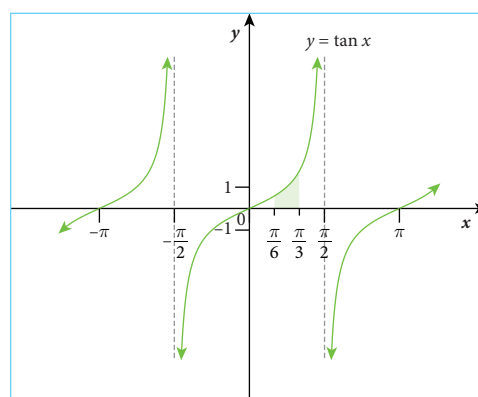
$$= -\log_e\left(\cos \frac{\pi}{3}\right) - \left(-\log_e\left(\cos \frac{\pi}{6}\right)\right)$$

$$= -\log_e \frac{1}{2} + \log_e \frac{\sqrt{3}}{2}$$

$$= \log_e \left(\frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} \right)$$

$$= \log_e \sqrt{3}$$

$$= \frac{1}{2} \log_e 3 \quad \checkmark$$



25 a $y = 2 \sin \pi(x - 0.25) + 1$

$$= 2 \sin \left(\pi x - \frac{\pi}{4} \right) + 1$$

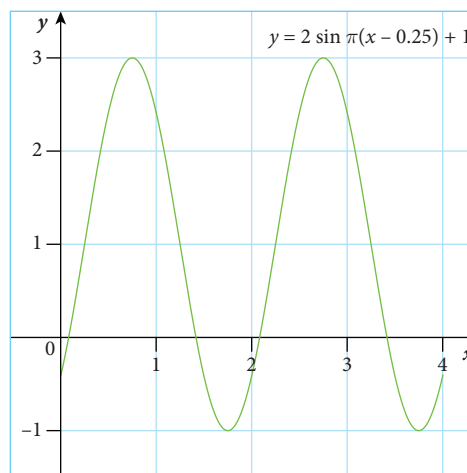
$$\text{Period} = \frac{2\pi}{\pi}$$

$$= 2 \quad \checkmark$$

b Amplitude = 2 ✓

Range: $-1 \leq y \leq 3$ ✓

c Phase shift = 0.25 units to the right (at the centre of the range) ✓



26 a The difference between successive rows is constant. So it is an arithmetic sequence.

$$a = 20, \quad l = 95, \quad S_n = 1495$$

$$S_n = \frac{n}{2}(a + l)$$

$$1495 = \frac{n}{2}(20 + 95) \quad \checkmark$$

$$2990 = n \times 115$$

$$n = 26$$

$\therefore$ there are 26 rows of trees. $\checkmark$

b $T_n = a + (n - 1)d$

$$95 = 20 + (26 - 1) \times d \quad \checkmark$$

$$75 = 25d$$

$$d = 3$$

There are three more trees in every row. $\checkmark$

27 a There are six points of intersection so there are six solutions. $\checkmark$

b When $x = \frac{\pi}{6}$,

$$2 \sin x = 2 \sin \frac{\pi}{6}$$

$$= 2 \times \frac{1}{2}$$

$$= 1$$

$$\tan \frac{3x}{2} = \tan \left(\frac{3 \times \frac{\pi}{6}}{2} \right)$$

$$= \tan \frac{\pi}{4}$$

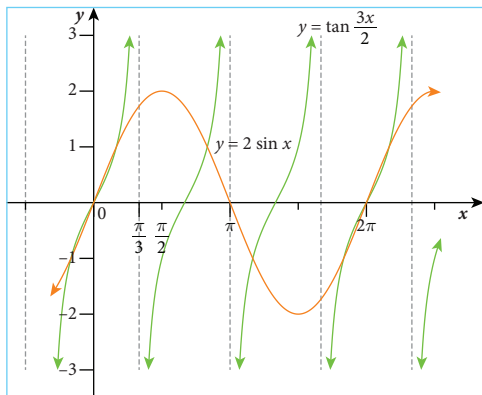
$$= 1 \quad \checkmark$$

So $2 \sin x = \tan \frac{3x}{2}$ when $x = \frac{\pi}{6}$.

c From the graph $2 \sin x \geq \tan \frac{3x}{2}$ when $0 \leq x \leq \frac{\pi}{6}$. $\checkmark$

Now, $y = \tan \frac{3x}{2}$ has an asymptote when $\frac{3x}{2} = \frac{\pi}{2}$

$$\text{i.e. } x = \frac{\pi}{3} \quad \checkmark$$



So, for $0 \leq x \leq \frac{\pi}{6}$, $2 \sin x \geq \tan \frac{3x}{2}$ when $0 \leq x \leq \frac{\pi}{6}$

and $\frac{\pi}{3} < x \leq \frac{\pi}{2}$. $\checkmark$

28 a $P = 370\,000$, $r = 0.6$

$$A_1 = 370\,000(1.006) - M$$

$$A_2 = A_1(1.006) - M$$

$$= [370\,000(1.006) - M](1.006) - M$$

$$= 370\,000(1.006)^2 - M(1.006) - M$$

$$= 370\,000(1.006)^2 - M(1 + 1.006) \quad \checkmark$$

$$A_3 = A_2(1.006) - M$$

$$= [370\,000(1.006)^2 - M(1 + 1.006)](1.006) - M$$

$$= 370\,000(1.006)^3 - M(1.006 + 1.006^2) - M$$

$$= 370\,000(1.006)^3 - M(1 + 1.006 + 1.006^2) \quad \checkmark$$

b $A_n = 370\,000(1.006)^n - M(1 + 1.006 + \dots + 1.006^{n-1}) \quad \checkmark$

Geometric series: $a = 1$, $r = 1.006$, $n = n$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$= \frac{1(1.006^n - 1)}{1.006 - 1}$$

$$= \frac{1.006^n - 1}{0.006}$$

$$\therefore A_n = 370\,000(1.006)^n - M \left(\frac{1.006^n - 1}{0.006} \right) \quad \checkmark$$

c 25 years = 300 months

$$A_{300} = 370\,000(1.006)^{300} - M \left(\frac{1.006^{300} - 1}{0.006} \right)$$

If the loan is repaid after 25 years $A_{300} = 0$.

$$0 = 370\,000(1.006)^{300} - M \left(\frac{1.006^{300} - 1}{0.006} \right) \quad \checkmark$$

$$M \left(\frac{1.006^{300} - 1}{0.006} \right) = 370\,000(1.006)^{300}$$

$$M = \frac{370\,000(1.006)^{300}(0.006)}{1.006^{300} - 1}$$

$$= 2662.478\,15\dots$$

$$= 2662.48 \quad (2 \text{ d.p.})$$

The amount of each monthly instalment would be **\$2662.48**. $\checkmark$

d Total repaid = $\$2662.48 \times 300$

$$= \$798\,744$$

$$\text{Total interest} = \$798\,744 - \$370\,000$$

$$= \$428\,744 \quad \checkmark$$

29 a $y = 4 + 9x - 3x^2 - x^3$

When $x = 0$,

$$y = 4 + 9 \times 0 - 3 \times 0^2 - 0^3$$

$$= 4$$

The curve cuts the y -axis at $(0, 4)$. $\checkmark$

b $\frac{dy}{dx} = 9 - 6x - 3x^2$

Stationary points occur when $\frac{dy}{dx} = 0$

$$\text{i.e. } 9 - 6x - 3x^2 = 0$$

$$3 - 2x - x^2 = 0$$

$$(3 + x)(1 - x) = 0$$

$$x = -3 \quad \text{or} \quad x = 1 \quad \checkmark$$

$$\frac{d^2y}{dx^2} = -6 - 6x$$

$$\text{When } x = -3, \frac{d^2y}{dx^2} = -6 - 6 \times (-3)$$

$$= 12 \quad (> 0)$$

$\therefore$ there is a minimum when $x = -3$.

$$\text{When } x = 1, \frac{d^2y}{dx^2} = -6 - 6 \times 1 \\ = -12 \quad (< 0)$$

$\therefore$ there is a maximum when $x = 1$. ✓

$$\text{When } x = -3, y = 4 + 9(-3) - 3(-3)^2 - (-3)^3 \\ = -23$$

$$\text{When } x = 1, y = 4 + 9 \times 1 - 3 \times 1^2 - 1^3 \\ = 9$$

There is a minimum at $(-3, -23)$ and a maximum at $(1, 9)$. ✓

c If $\frac{d^2y}{dx^2} = 0$

$$-6 - 6x = 0$$

$$-6x = 6$$

$$x = -1 \quad \checkmark$$

$$\text{If } x < -1, \frac{d^2y}{dx^2} > 0.$$

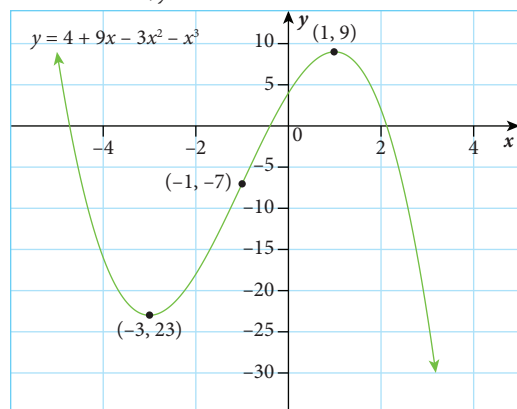
$$\text{If } x > -1, \frac{d^2y}{dx^2} < 0.$$

$$\text{When } x = -1, y = 4 + 9(-1) - 3(-1)^2 - (-1)^3 \\ = -7$$

There is a point of inflection at $(-1, -7)$. ✓

When $x \rightarrow \infty, y \rightarrow -\infty$

When $x \rightarrow -\infty, y \rightarrow \infty$ ✓



30 a $A = \frac{1}{2}r^2\theta$

$$\frac{1}{2}r^2\theta = 36$$

$$r^2\theta = 72$$

$$\theta = \frac{72}{r^2} \quad \checkmark$$

b $P = r + r + r\theta$

$$= 2r + r\theta$$

$$= 2r + r \times \frac{72}{r^2}$$

$$= 2r + \frac{72}{r} \quad \checkmark$$

$$= 2r + 72r^{-1}$$

$$\text{Now } \frac{dP}{dr} = 2 - 72r^{-2} \quad \checkmark$$

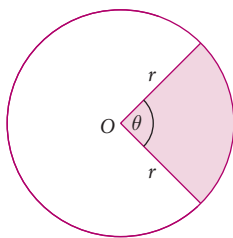
A maximum or minimum will occur when $\frac{dP}{dr} = 0$,

$$\text{i.e. } 2 - \frac{72}{r^2} = 0$$

$$2r^2 = 72$$

$$r^2 = 36$$

$$r = 6 \quad (r > 0)$$



$$\frac{d^2P}{dr^2} = 144r^{-3}$$

$$\text{So } \frac{d^2P}{dr^2} > 0 \text{ when } r = 6.$$

$\therefore$ the minimum perimeter occurs when $r = 6$. ✓

The diameter of the circle will be 12 cm.

$$\text{Arc length} = r\theta$$

$$= \frac{72}{r}$$

$$= 12 \text{ when } r = 6$$

$\therefore$ the minimum perimeter will occur when both the diameter of the circle and the arc length of the sector are 12 cm. ✓

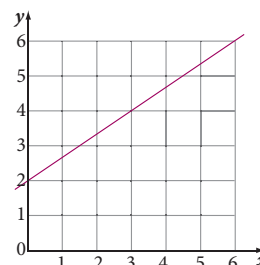
SAMPLE HSC EXAMINATION 2

Section I

1 C. Gradient = $\frac{2}{3}$

$$\text{Vertical intercept} = 2$$

$$\text{Equation is } y = \frac{2}{3}x + 2$$



2 C.

	Detected true	Detected false
True statement	84	16
False statement	36	124

$$\text{Number correctly detected} = 84 + 124 \\ = 208$$

$$\text{Total statements} = 208 + 16 + 36 \\ = 260$$

$$P(\text{correctly detected}) = \frac{208}{260} \\ = \frac{4}{5}$$

3 A. $FV = \$60\,339, r = 0.0075, n = 60$

$$PV = \frac{FV}{(1 + r)^n} \\ = \frac{\$60\,339}{(1.0075)^{60}} \\ = \$38\,539 \text{ (nearest dollar)}$$

4 B. 40 is two standard deviations below the mean.

76 is one standard deviation above the mean.

68% of scores lie within one standard deviation of the mean.

95% of scores lie within two standard deviations.

95% - 68% = 27% lie between one and two standard deviations.

$$\text{Percentage} = 68\% + \frac{1}{2} \text{ of } 27\% \\ = 81.5\%$$

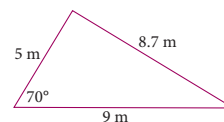
5 A. $A = \frac{1}{2}ab\sin C$

$$= \frac{1}{2} \times 5 \times 9 \times \sin 70^\circ$$

$$= 21.143\,0839 \dots$$

$$= 21.1 \text{ (1 d.p.)}$$

The area is 21.1 m².



$$6 \text{ D. } f(x) = \frac{1}{x^2}$$

$$= x^{-2}$$

$$F(x) = \frac{x^{-1}}{-1} + c$$

$$= -\frac{1}{x} + c$$

$$\text{Now } \frac{x-1}{x} = 1 - \frac{1}{x}.$$

So $\frac{x-1}{x}$ is an anti-derivative of $\frac{1}{x^2}$.

$$7 \text{ D. } \frac{\sin x \cos x + \sin^2 x}{\sin x \cos x + \cos^2 x} = \frac{\sin x (\cos x + \sin x)}{\cos x (\sin x + \cos x)}$$

$$= \frac{\sin x}{\cos x}$$

$$= \tan x$$

$$8 \text{ B. } f(x) = (x+1)^2 = x^2 + 2x + 1$$

$$f(-x) = ((-x)+1)^2$$

$$= x^2 - 2x + 1$$

$$\neq f(x)$$

So $f(x)$ is not even.

$$g(x) = x^2 + 1$$

$$g(-x) = (-x)^2 + 1$$

$$= x^2 + 1$$

$$= g(x)$$

So $g(x)$ is even.

$$9 \text{ D. } f(x) = \begin{cases} \frac{x}{8} & 0 \leq x \leq 4 \\ 0 & \text{elsewhere} \end{cases}$$

$$\text{Now } \int_0^M \frac{x}{8} dx = 0.5$$

$$\int_0^M \frac{x}{8} dx = \left[\frac{x^2}{16} \right]_0^M$$

$$= \frac{M^2}{16} - 0$$

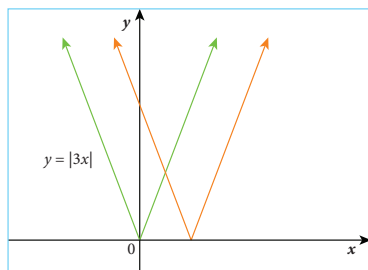
$$\text{So } \frac{M^2}{16} = 0.5$$

$$M^2 = 8$$

$$M = 2.82842712... \quad (M > 0)$$

$$= 2.8 \quad (1 \text{ d.p.})$$

10 B. The graph has moved to the right so the equation must be of the form $y = |3x - k|$.



Section II

Class scores

X	Y
93	46
7720	5148
98541	60356
86532	712478
9653	8244799
41	9035

11 a X: range = $94 - 43$

$$= 51 \quad \checkmark$$

b Y: median = $\frac{74+77}{2}$

$$= 75.5 \quad \checkmark$$

c Class Y did better than class X with more higher scores and a higher median (75.5 compared to 70.5). $\checkmark$

The results in class X are symmetrical but those in class Y are negatively skewed. $\checkmark$

$$12 \text{ a } \cos P = \frac{q^2 + r^2 - p^2}{2qr}$$

$$= \frac{125^2 + 75^2 - 175^2}{2 \times 125 \times 75} \quad \checkmark$$

$$= -0.5$$

$$P = 120^\circ \quad \checkmark$$

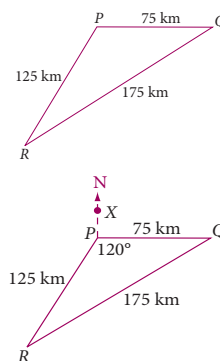
b Let X be a point due north of P.

P is due west of Q. $\angle XPQ = 90^\circ$

$$\text{Reflex angle } XPR = 90^\circ + 120^\circ$$

$$= 210^\circ$$

The bearing of R from P is 210° . $\checkmark$



13 a Barry charges \$200 as a fixed rate. $\checkmark$

b It will cost \$575 for Carrie to work for 10 hours. $\checkmark$

c The two costs are the same for 5 hours work. $\checkmark$

d Barry would work slightly longer than Carrie for \$500 so Barry would be the most economical. $\checkmark$

$$\text{e Gradient} = \frac{375-200}{5}$$

$$= 35 \quad \checkmark$$

This represents Barry's hourly rate (\$35/h). $\checkmark$

f y-intercept = 175

$$\text{Gradient} = \frac{375-175}{5}$$

$$= 40 \quad \checkmark$$

$$\text{So } C = 40h + 175. \quad \checkmark$$

14 a $y = \log_e(x^2 + 4)$

$$\frac{dy}{dx} = \frac{1}{x^2 + 4} \times \frac{d}{dx}(x^2 + 4)$$

$$= \frac{2x}{x^2 + 4} \quad \checkmark$$

b $y = e^x \sin x$

$$\frac{dy}{dx} = e^x \times \frac{d}{dx}(\sin x) + \sin x \times \frac{d}{dx}(e^x) \quad \checkmark$$

$$= e^x \times \cos x + \sin x \times e^x$$

$$= e^x(\cos x + \sin x) \quad \checkmark$$

$$15 \text{ a } FV = \$30\,000 \quad r = 6\% \div 12 = 0.5\% \quad n = 2 \times 12 = 24$$

$$\text{Factor} = 25.432 \quad \checkmark$$

$$\$30\,000 = a \times 25.432$$

$$a = \$30\,000 \div 25.432$$

$$= \$1179.61623\dots$$

$$= \$1180 \quad (\text{nearest dollar}) \quad \checkmark$$

- b Because the interest is compound interest, the amount of interest earned each month increases as the amount of the investment increases. It is not the same over the whole term of the loan. $\checkmark$

In 12 months the future value is given by:

$$FV = \$1180 \times 12.336$$

$$= \$14\,556.48 \quad \checkmark$$

- c Interest rate = $6\% \div 4 = 1.5\%$

$$\text{Period} = 2 \times 4 = 8$$

$$\text{Interest rate factor} = 8.4328$$

$$FV = \$750 \times 8.4328$$

$$= \$6324.60 \quad \checkmark$$

- d Extra amount = $\$6324.60 \div 25.432 = \$248.686693\dots = \$249 \quad (\text{nearest dollar})$

Lily would need to have saved an extra **\$249** each month. $\checkmark$

$$16 \text{ a } \int_1^4 \frac{dx}{\sqrt{x}} = \int_1^4 x^{-\frac{1}{2}} dx$$

$$= \left[\frac{x^{\frac{1}{2}}}{\frac{1}{2}} \right]_1^4 \quad \checkmark$$

$$= \left[2\sqrt{x} \right]_1^4$$

$$= 2\sqrt{4} - 2\sqrt{1}$$

$$= 2 \quad \checkmark$$

$$b \int_0^{\frac{\pi}{6}} \sin 2x \, dx = \left[-\frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{6}} \quad \checkmark$$

$$= -\frac{1}{2} \cos \left(2 \times \frac{\pi}{6} \right) - \left(-\frac{1}{2} \cos(2 \times 0) \right)$$

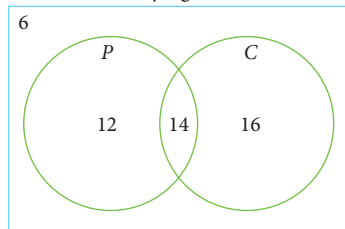
$$= -\frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times 1$$

$$= \frac{1}{4} \quad \checkmark$$

- 17 a Number studying physics or chemistry or both = $48 - 6 = 42$.

$$\text{Now } 26 + 30 = 56.$$

$$\text{So number studying both} = 56 - 42 = 14.$$



b $P(\text{chemistry student also studies physics}) = \frac{14}{30} = \frac{7}{15} \quad \checkmark$

$$18 \quad 1 + 3^x + 3^{2x} + 3^{3x} + \dots$$

Infinite geometric series:

$$a = 1, \quad r = 3^x, \quad S = 1.5$$

$$S = \frac{a}{1-r}$$

$$1.5 = \frac{1}{1-3^x} \quad \checkmark$$

$$1 - 3^x = \frac{1}{1.5}$$

$$= \frac{2}{3}$$

$$3^x = \frac{1}{3}$$

$$= 3^{-1}$$

$$\therefore x = -1 \quad \checkmark$$

- 19 a A correlation coefficient of -0.8168 means there is strong negative correlation. Generally as the age increases, the value of vehicles decrease. $\checkmark$

- b 20 $\checkmark$

$$c \text{ y-intercept} = 36\,000$$

$$\text{Gradient} = \frac{-36\,000}{9}$$

$$= -4000 \quad \checkmark$$

$$\text{Equation is } y = -4000x + 36\,000 \quad \checkmark$$

- d \$10 000 $\checkmark$

$$e \quad 3\frac{1}{2} \text{ years} \quad \checkmark$$

- f $x \geq 0$ because a vehicle cannot have a negative age. $\checkmark$

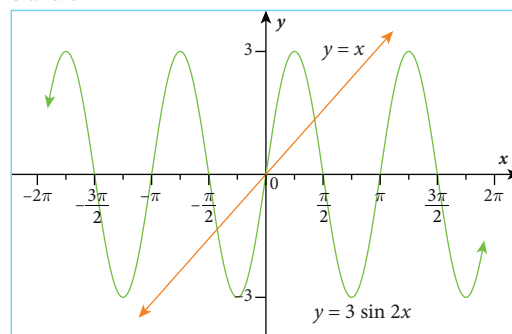
$$x \leq 9 \text{ because a vehicle cannot have a negative value. } \checkmark$$

- 20 a $y = 3 \sin 2x$

$$\text{Amplitude} = 3$$

$$\text{Period} = \frac{2\pi}{2} = \pi \quad \checkmark$$

a and b



- c From the graph $y = x$ and $y = 3 \sin 2x$ have three points of intersection.

$$\therefore 3 \sin 2x = x \text{ has three solutions. } \checkmark$$

$$21 \text{ a } f(x) = \begin{cases} \frac{1}{50} e^{-\frac{x}{50}} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$P(X \leq 24) = \int_0^{24} \frac{1}{50} e^{-\frac{x}{50}} dx$$

$$= \left[-e^{-\frac{x}{50}} \right]_0^{24} \quad \checkmark$$

$$= -e^{-\frac{24}{50}} - (-e^0)$$

$$= 1 - e^{-\frac{24}{50}}$$

$$= 0.381216608\dots$$

$$= 0.381 \quad (3 \text{ d.p.}) \quad \checkmark$$

$$\text{b } P(48 < X < 60) = \int_{48}^{60} \frac{1}{50} e^{-\frac{x}{50}} dx$$

$$= -e^{-\frac{60}{50}} - \left(-e^{-\frac{48}{50}} \right)$$

$$= 0.0816986740\dots$$

$$= 0.082 \quad (3 \text{ d.p.}) \quad \checkmark$$

$$\text{c } P(X \geq 36) = 1 - P(X < 36)$$

$$= 1 - \left(1 - e^{-\frac{36}{50}} \right)$$

$$= 0.48675225\dots$$

$$= 0.487 \quad (3 \text{ d.p.}) \quad \checkmark$$

22 a At points of intersection:

$$x^3 = 3x^2 - 2x$$

$$x^3 - 3x^2 + 2x = 0$$

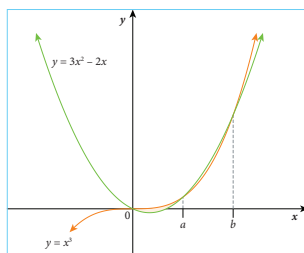
$$x(x^2 - 3x + 2) = 0$$

$$x(x-1)(x-2) = 0 \quad \checkmark$$

$$x = 0 \text{ or } x = 1 \text{ or } x = 2$$

$$\text{Now } 0 < a < b$$

$$\therefore a = 1 \quad \checkmark$$



b Let $x = c$ be the point where the curve cuts x -axis ($0 < c < a$).

$$\text{Area} = \int_0^a x^3 dx - \int_c^a (3x^2 - 2x) dx + \left| \int_0^c (3x^2 - 2x) dx \right| \quad \checkmark$$

$$\text{But } \int_0^c (3x^2 - 2x) dx < 0.$$

$$\text{So } \left| \int_0^c (3x^2 - 2x) dx \right| = - \int_0^c (3x^2 - 2x) dx.$$

$$\text{Area} = \int_0^a x^3 dx - \int_c^a (3x^2 - 2x) dx - \int_0^c (3x^2 - 2x) dx$$

$$= \int_0^a (x^3 - (3x^2 - 2x)) dx$$

$$= \int_0^a (x^3 - 3x^2 + 2x) dx \quad \checkmark$$

$$\text{c } A = \int_0^1 (x^3 - 3x^2 + 2x) dx$$

$$= \left[\frac{x^4}{4} - x^3 + x^2 \right]_0^1$$

$$= \frac{1^4}{4} - 1^3 + 1^2 - \left(\frac{0^4}{4} - 0^3 + 0^2 \right)$$

$$= \frac{1}{4}$$

The area is $\frac{1}{4}$ units². $\checkmark$

23 a $\frac{d^2x}{dt^2} = -2$

$$\frac{dx}{dt} = -2t + C_1$$

$$\text{When } t = 0, \frac{dx}{dt} = 6$$

$$\therefore 6 = -2 \times 0 + C_1$$

$$C_1 = 6$$

$$\therefore \frac{dx}{dt} = -2t + 6 \quad \checkmark$$

$$\text{b } x = -t^2 + 6t + C_2$$

$$\text{When } t = 0, x = 7$$

$$\therefore 7 = -0^2 + 6 \times 0 + C_2$$

$$C_2 = 7$$

$$\therefore x = -t^2 + 6t + 7 \quad \checkmark$$

$$\text{c Particle is at rest when } \frac{dx}{dt} = 0$$

$$\text{i.e. } -2t + 6 = 0$$

$$2t = 6$$

$$t = 3$$

The particle is at rest after 3 seconds. $\checkmark$

d When $t = 0$, $x = 7$

The particle is at rest at $t = 3$ so the maximum (since $\frac{d^2x}{dt^2} < 0$) displacement occurs at $t = 3$.

$$\text{When } t = 3, x = -3^2 + 6 \times 3 + 7$$

$$= 16 \quad \checkmark$$

$$\text{When } t = 5, x = -5^2 + 6 \times 5 + 7$$

$$= 12$$

$$\text{Total distance} = (16 - 7) + (16 - 12) \text{ m}$$

$$= 13 \text{ m} \quad \checkmark$$

24 a $x = 73, \mu = 52.5, \sigma = 8.2$

$$z = \frac{x - \mu}{\sigma}$$

$$= \frac{73 - 52.5}{8.2}$$

$$= 2.5 \quad \checkmark$$

b Mickey did better than 99.38% of students.

$$\text{Now } 0.9938 \times 58\,000 = 57\,640.4.$$

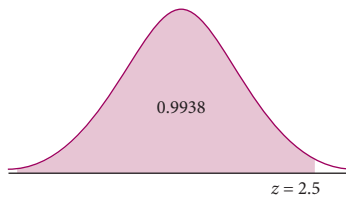
So Micky did better than about 57 640 students. $\checkmark$

Number who did better than Micky

$$\approx 58\,000 - 57\,640$$

$$= 360$$

So about 360 students did better than Micky. $\checkmark$



c 40% of 80 = 32

So a score of 32 in the test is a score of 40%. $\checkmark$

$$x = 32, \mu = 52.5, \sigma = 8.2$$

$$z = \frac{x - \mu}{\sigma}$$

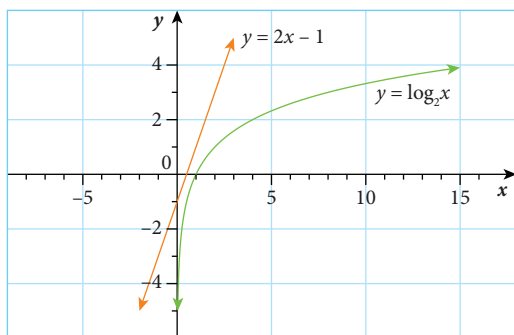
$$= \frac{32 - 52.5}{8.2}$$

$$= -2.5 \quad \checkmark$$

Using the symmetry of the curve, the area to the right of $z = -2.5$ is 0.9938.

$P(\text{student did better than 40\%}) = 99.38\% \quad \checkmark$

25 $1 + \log_2 x = 2x$
 $\therefore \log_2 x = 2x - 1$



From the graph there are no points of intersection.

So, there are no solutions of the equation $1 + \log_2 x = 2x$. ✓

26 Last amount invested earns interest for 1 year. $A_1 = \$10\,000(1.065)$

First investment earns interest for 30 years: $A_{30} = \$10\,000(1.065)^{30}$

Total investment

$= \$[10\,000(1.065) + 10\,000(1.065)^2 + \dots + 10\,000(1.065)^{30}]$ ✓

This is a geometric series:

$a = 10\,000(1.065), \quad r = 1.065, \quad n = 30$

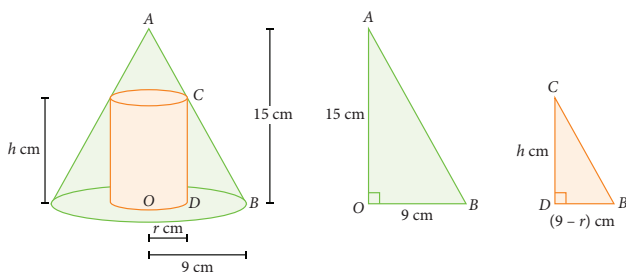
$S_n = \frac{a(r^n - 1)}{r - 1}$

$= \frac{10\,000(1.065)(1.065^{30} - 1)}{1.065 - 1}$ ✓

$= 919\,892.302\dots$

The total investment will be worth \$919 892 to the nearest dollar. ✓

27 a



In $\triangle AOB, \triangle CDB$

$\angle AOB = \angle CDB$ (both 90°)

$\angle ABO = \angle CBD$ (common angle)

$\therefore$ Triangles AOB and CDB are similar. ✓

$\therefore \frac{h}{15} = \frac{9-r}{9}$ (corresponding sides, similar triangles)

$\frac{h}{15} = 1 - \frac{r}{9}$

$h = 15 - \frac{5}{3}r$ ✓

b $V = \pi r^2 h$

$= \pi r^2 \left(15 - \frac{5}{3}r\right)$

$= 15\pi r^2 - \frac{5\pi r^3}{3}$ ✓

c $\frac{dV}{dr} = 30\pi r - 5\pi r^2$

When $\frac{dV}{dr} = 0$

$30\pi r - 5\pi r^2 = 0$

$5\pi r(6 - r) = 0$

$r = 0$ or $r = 6$ ✓

$\frac{d^2V}{dr^2} = 30\pi - 10\pi r$

$\frac{d^2V}{dr^2} < 0$ when $r = 6$

$\therefore$ the maximum volume occurs when $r = 6$. ✓

d $V = 15\pi r^2 - \frac{5\pi r^3}{3}$

When $r = 6$,

$V = 15 \times \pi \times 6^2 - \frac{5\pi \times 6^3}{3}$

$= 180\pi$

The maximum possible volume of the cylinder is $180\pi \text{ cm}^3$.

Volume of cone: $V = \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \pi \times 9^2 \times 15$

$= 405\pi$

The volume of the cone is $405\pi \text{ cm}^3$.

Ratio of volumes $= \frac{180\pi}{405\pi}$

$= \frac{4}{9}$ ✓

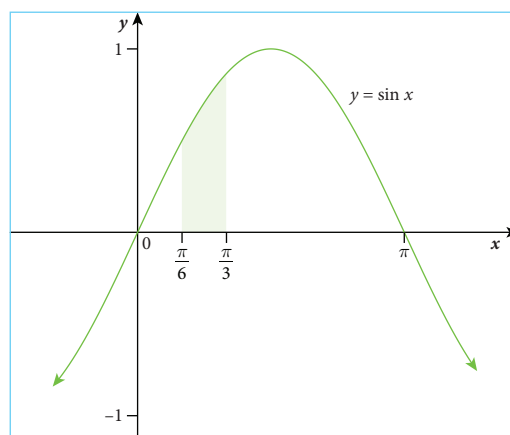
28 a $A = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin x \, dx$

$= [-\cos x]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$ ✓

$= -\cos \frac{\pi}{3} - \left(-\cos \frac{\pi}{6}\right)$

$= -\frac{1}{2} + \frac{\sqrt{3}}{2}$

The exact area is $\left(\frac{\sqrt{3}-1}{2}\right) \text{ units}^2$. ✓



$$\begin{aligned} \text{b } \int_a^b f(x) dx &\approx \frac{b-a}{2} [f(a) + f(b)] \\ \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin x dx &\approx \frac{\frac{\pi}{3} - \frac{\pi}{6}}{2} \left[\sin \frac{\pi}{6} + \sin \frac{\pi}{3} \right] \quad \checkmark \\ &= \frac{\pi}{12} \left(\frac{1}{2} + \frac{\sqrt{3}}{2} \right) \\ &= \frac{\pi}{24} (1 + \sqrt{3}) \quad \checkmark \end{aligned}$$

Using one application of the trapezoidal rule the area is approximately $\frac{\pi}{24}(1 + \sqrt{3})$ units².

$$\begin{aligned} \text{c } \frac{\pi}{24}(1 + \sqrt{3}) &\approx -\frac{1}{2} + \frac{\sqrt{3}}{2} \\ \pi(1 + \sqrt{3}) &\approx -12(1 - \sqrt{3}) \\ \pi &\approx \frac{-12(1 - \sqrt{3})}{1 + \sqrt{3}} \\ &= \frac{-12(1 - \sqrt{3})}{1 + \sqrt{3}} \times \frac{1 - \sqrt{3}}{1 - \sqrt{3}} \\ &= \frac{-12(1 - 2\sqrt{3} + 3)}{1 - 3} \\ &= 6(4 - 2\sqrt{3}) \end{aligned}$$

$$\therefore \pi \approx 12(2 - \sqrt{3}) \quad \checkmark$$

29 a $y = 2xe^{-x}$

Curve crosses y -axis when $x = 0$

$$\begin{aligned} y &= 2 \times 0 \times e^{-0} \\ &= 0 \end{aligned}$$

The curve cuts the y -axis at the origin. $\checkmark$

$$\begin{aligned} \text{b } \frac{dy}{dx} &= 2x \times \frac{d}{dx}(e^{-x}) + e^{-x} \times \frac{d}{dx}(2x) \\ &= 2x \times -e^{-x} + e^{-x} \times 2 \\ &= 2e^{-x}(-x + 1) \\ &= 2e^{-x}(1 - x) \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{c } \frac{d^2y}{dx^2} &= 2e^{-x} \times \frac{d}{dx}(1 - x) + (1 - x) \times \frac{d}{dx}(2e^{-x}) \\ &= 2e^{-x} \times -1 + (1 - x) \times -2e^{-x} \\ &= -2e^{-x}(1 + 1 - x) \\ &= -2e^{-x}(2 - x) \quad \checkmark \end{aligned}$$

d At turning point $\frac{dy}{dx} = 0$

$$\text{i.e. } 2e^{-x}(1 - x) = 0$$

$$2e^{-x} = 0 \quad \text{or} \quad 1 - x = 0$$

$$\text{no solution} \quad x = 1 \quad \checkmark$$

When $x = 1$,

$$\frac{d^2y}{dx^2} = -2e^{-1}(2 - 1) \quad (< 0)$$

$\therefore$ there is a maximum when $x = 1$.

$$\begin{aligned} \text{When } x = 1, \quad y &= 2 \times 1 \times e^{-1} \\ &= 2e^{-1} \end{aligned}$$

The turning point is a maximum at $(1, 2e^{-1})$. $\checkmark$

e If the curve changes concavity $\frac{d^2y}{dx^2} = 0$

$$\text{i.e. } 2e^{-x}(2 - x) = 0$$

$$2e^{-x} = 0 \quad \text{or} \quad 2 - x = 0$$

$$\text{no solution} \quad x = 2$$

$$\text{If } x < 2, \quad \frac{d^2y}{dx^2} < 0$$

$$\text{If } x > 2, \quad \frac{d^2y}{dx^2} > 0$$

$\therefore$ the curve changes concavity when $x = 2$. $\checkmark$

$$\begin{aligned} \text{When } x = 2, \quad y &= 2 \times 2 \times e^{-2} \\ &= 4e^{-2} \end{aligned}$$

The curve changes concavity at $(2, 4e^{-2})$. $\checkmark$

f As $x \rightarrow \infty$ $y \rightarrow 0$

$$\text{As } x \rightarrow -\infty \quad y \rightarrow -\infty \quad \checkmark$$

g

